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Standard Guide for Laboratory Informatics¹

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1. Scope

1.1 This guide helps describe the laboratory informatics landscape and covers issues commonly encountered at all stages in the life cycle of laboratory informatics from inception to retirement. It explains the evolution of laboratory informatics tools used in today's laboratories such as laboratory information management systems (LIMS), laboratory execution systems (LES), laboratory information systems (LIS), electronic laboratory notebooks (ELN), scientific data management systems (SDMS), and chromatography data systems (CDS). It also covers the relationship (interactions) between these tools and the external systems in a given organization. The guide discusses supporting laboratory informatics tools and a wide variety of the issues commonly encountered at different stages in the life cycle. The subsections that follow describe the scope of this document in specific areas.

1.2 *High-Level Purpose*—The purpose of this guide includes: (1) educating new users on laboratory informatics tools; (2) providing a standard terminology that can be used by different vendors and end users; (3) establishing minimum requirements for laboratory informatics; (4) providing guidance for the specification, evaluation, cost justification, implementation, project management, training, and documentation of the systems; and (5) providing a functional requirements checklist for laboratory informatics systems that can be adopted within the laboratory and integrated with existing systems.

1.3 *Laboratory Informatics Definition*—Laboratory informatics is the specialized application of information technology aimed at optimizing laboratory operations. It is a collection of informatics tools utilized within laboratory environments to collect, store, process, analyze, report, and archive data and information from the laboratory and its supporting processes. Laboratory informatics includes the effective use of critical data management systems, the electronic delivery of results to customers, and the use and integration of supporting systems (for example, training and policy management). Examples of

primary laboratory informatics tools include laboratory information management systems (LIMS), laboratory execution systems (LES), laboratory information systems (LIS), electronic laboratory notebooks (ELN), scientific data management systems (SDMS), and chromatography data systems (CDS).

1.4 *Scope Considerations when Selecting and Implementing Laboratory Informatics Solutions*—Many laboratories have determined that they need to deploy multiple laboratory informatics systems to automate their laboratory processes and manage their data. Selection of an informatics solution requires a detailed analysis of the laboratory's requirements and should not be a simple product category decision. Information technology (IT) representatives and **subject matter experts (SMEs) who understand the needs of the laboratory need to be involved in the selection and implementation of a laboratory informatics system to ensure that the needs of the laboratory are met and IT can support it. Customers (internal and external) of laboratory information should also be included in the laboratory informatics solution design to ensure full electronic integration between systems.**

1.5 The scope of this guide covers a wide range of laboratory types, industries, and sizes. Examples of laboratory types and industries include:

1.5.1 General Laboratories:

1.5.1.1 Standards (ASTM, IEEE, ISO) and

1.5.1.2 Government (EPA, FDA, JPL, NASA, NRC, USDA, USGS, FERC).

1.5.2 Environmental:

1.5.2.1 Environmental monitoring.

1.5.3 Life Science Laboratories:

1.5.3.1 Biotechnology and

1.5.3.2 Diagnostic.

1.5.4 Healthcare and Medical:

1.5.4.1 Bionomics/genomics,

1.5.4.2 Medical devices,

1.5.4.3 Pharmaceutical,

1.5.4.4 Veterinary,

1.5.4.5 Public health, and

1.5.4.6 Hospital.

1.5.5 Heavy Industry Laboratories:

1.5.5.1 Energy and resources,

1.5.5.2 Manufacturing and construction,

1.5.5.3 Materials and chemicals, and

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1.5.5.4 Transportation and shipping.

1.5.6 *Food and Beverage Laboratories:*

1.5.6.1 Agriculture,

1.5.6.2 Beverages,

1.5.6.3 Food, and

1.5.6.4 Food service and hospitality.

1.5.7 *Public Sector Laboratories:*

1.5.7.1 Law enforcement/forensic,

1.5.7.2 State and local government,

1.5.7.3 Education and nonprofits, and

1.5.7.4 Public utilities (water, electric, waste treatment).

1.6 *Integration*—The scope of integration covered in this guide includes communication and meaningful data exchange between different laboratory informatics tools and other external systems (document management, chromatography data systems, laboratory instruments, spectroscopy data systems, enterprise resource planning (ERP), manufacturing execution systems (MES), investigations/deviations and CAPA management systems), and other integrated business systems (for example, clinical or hospital environments) provide significant business benefits to any laboratory and is discussed at a high level in this guide.

1.7 *Life-Cycle Phases*—The scope of this guide is intended to provide an understanding of laboratory informatics tools' life cycle from project initiation point to retirement and decommissioning. This guide was designed to help newer audiences in understanding the complexity in the relationships between different laboratory informatics tools and how to plan and manage the implementation project, while seasoned users may use the different life cycles to maintain existing laboratory informatics tools. Integrating additional informatics tools to existing ones in today's evolving laboratory environment adds constraints that need to be considered. The life-cycle discussion includes both the laboratory informatics solution life cycle as well as the project life cycle.

1.7.1 The product life cycle encompasses a specific laboratory informatics system and the expected useful life of that system before it needs to be replaced or upgraded.

1.7.2 The project life cycle encompasses the activities to acquire, implement, operate, and eventually retire a specific laboratory informatics system.

1.8 *Audience*—This guide has been created with the needs of the following stakeholders in mind: (1) end users of laboratory informatics tools, (2) implementers of laboratory informatics tools, (3) quality personnel, (4) information technology personnel, (5) laboratory informatics tools vendors, (6) instrument vendors, (7) individuals who approve laboratory informatics tools funding, (8) laboratory informatics applications support specialists, and (9) software test/validation specialists. Information contained in this guide will benefit a broad audience of people who work in or interact with a laboratory. New users can use this guide to understand the purpose and functions of the wide variety of laboratory informatics tools as well as the interactions between these tools with external systems. The guide can also help prospective users in understanding terminology, configurations, features, design, benefits, and costs of these different laboratory informatics tools.

Individuals who are purchasing specific tools may also use this guide to identify functions that are recommended for specific laboratory environments. Research and development staff of different commercial laboratory informatics systems vendors may use the guide as a tool to evaluate, identify, and potentially improve the capabilities of their products. The vendors' sales staff may use the guide to represent functions of their laboratory informatics products to prospective customers in more generic and product-neutral terms.

1.9 *Out of Scope*—This guide does not attempt to define the boundaries of laboratory informatics, as they continue to evolve and blur between the different types of tools; rather, it focuses on the functionality that is provided by laboratory informatics as a whole.

1.10 *This international standard was developed in accordance with internationally recognized principles on standardization established in the Decision on Principles for the Development of International Standards, Guides and Recommendations issued by the World Trade Organization Technical Barriers to Trade (TBT) Committee.*

2. Referenced Documents

2.1 ASTM Standards:²

E1394 Specification for Transferring Information Between Clinical Instruments and Computer Systems (Withdrawn 2002)³

E1947 Specification for Analytical Data Interchange Protocol for Chromatographic Data

E1948 Guide for Analytical Data Interchange Protocol for Chromatographic Data

E2077 Specification for Analytical Data Interchange Protocol for Mass Spectrometric Data

E2078 Guide for Analytical Data Interchange Protocol for Mass Spectrometric Data

E2369 Specification for Continuity of Care Record (CCR)

2.2 CDISC Standard:⁴

SEND Standard for Exchange of Nonclinical Data;

2.3 CIDX Standard:⁵

CIDX Chemistry Industry Data eXchange

2.4 EPA Standard:⁶

ESAR Environmental Sampling, Analysis and Results Data Standard

2.5 FDA Regulation:⁷

21 CFR Part 11 Electronic Records; Electronic Signatures, 62 FR 13464

² For referenced ASTM standards, visit the ASTM website, www.astm.org, or contact ASTM Customer Service at service@astm.org. For *Annual Book of ASTM Standards* volume information, refer to the standard's Document Summary page on the ASTM website.

³ The last approved version of this historical standard is referenced on www.astm.org.

⁴ Available from Clinical Data Interchange Standards Consortium (CDISC), 401 West 15th Street, Suite 800, Austin, TX 78701, <https://www.cdisc.org/>.

⁵ The CIDX Chem eStandards are available at OAGi, P.O. Box 4897, Marietta, GA 30061-4897, <http://www.oagi.org/>.

⁶ Available from United States Environmental Protection Agency (EPA), 1200 Pennsylvania Ave., NW, Washington, DC 20460, <http://www.epa.gov/>.

⁷ Available from U.S. Food and Drug Administration (FDA), 10903 New Hampshire Ave., Silver Spring, MD 20993, <http://www.fda.gov>.

FDA Data Integrity and Compliance with CGMP: Guidance for Industry

2.6 HL7 Standards:

Health Level Seven⁸ Standards

2.7 ICH Standard:⁹

ICH Quality Guideline Q9 Quality Risk Management

2.8 IEEE Standards:¹⁰

IEEE 1012 Standard for System, Software, and Hardware Verification and Validation

IEEE 1028 Standard for Software Reviews and Audits

2.9 ISA Standard:¹¹

ANSI/ISA-95.00.06 Enterprise-Control System Integration--Part 6: Messaging Service Model

2.10 ISO Standards:¹²

ISO/IEC/IEEE 12207 Systems and software engineering -- Software life cycle processes

ISO/IEC 27000 Information technology -- Security techniques -- Information security management systems -- Overview and vocabulary

ISO/HL7 27932 Data Exchange Standards -- HL7 Clinical Document Architecture, Release 2

ISO/IEEE 11073-10101 Health informatics -- Point-of-care medical device communication -- Part 10101: Nomenclature

ISO/IEC/IEEE 26511 Systems and software engineering -- Requirements for managers of user documentation

ISO/IEC/IEEE 26512 Systems and software engineering -- Requirements for acquirers and suppliers of information

ISO/IEC/IEEE 29119-4 Systems and software engineering -- Software testing -- Part 4: Test techniques

ISO/IEC/IEEE 29119-5 Systems and software engineering -- Software testing -- Part 5: Keyword-Driven Testing

ISO/IEC/IEEE 29148 Systems and software engineering -- Life cycle processes -- Requirements engineering

2.11 ISPE GAMP¹³ Guides:

ISPE GAMP 5: A Risk-based Approach to Compliant GxP Computerized Systems

ISPE GAMP Guide: Records & Data Integrity

2.12 MHRA Standard:¹⁴

MHRA GxP Data Integrity Definitions and Guidance for Industry

2.13 NCPDP Standard:¹⁵

Batch Transaction Format

2.14 NIST Standard:¹⁶

NIST Cybersecurity Framework

2.15 NRC Standards:¹⁷

10 CFR Appendix B to Part 50 Quality Assurance Criteria for Nuclear Power Plants and Fuel Reprocessing Plants, 72 FR 49505

10 CFR Appendix E to Part 50 Emergency Planning and Preparedness for Production and Utilization Facilities, 80 FR 74980

10 CFR Appendix K to Part 50 ECCS Evaluation Models, 65 FR 34921

2.16 PIC/S Standard:¹⁸

PIC/S Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments

2.17 Regenstrief Institute Standard:

LOINC¹⁹ Logical Observation Identifiers Names and Codes

2.18 SNOMED International Standard:

SNOMED-CT Systematized Nomenclature of Medicine-Clinical Terms²⁰

2.19 WHO Standard:²¹

WHO Technical Report Series, No. 996, Annex 5 Guidance on good data and record management practices

3. Terminology

3.1 Definitions—This guide defines the majority of terminology used in the field of laboratory informatics. Users of this guide should request a terminology list from each vendor with a cross reference to the terms used in this guide.

3.2 Definitions of Terms Specific to This Standard:

3.2.1 artificial intelligence, AI, *n*—behavior by machines or computers versus the natural intelligence of humans and animals.

3.2.1.1 Discussion—In the computer science arena, any device that perceives its environment and takes action to maximize success in achieving a goal is exhibiting AI. Machine learning is an application of artificial intelligence that provides systems the ability to automatically learn and improve from experience without being explicitly programmed.

3.2.2 chromatography data system, CDS, *n*—computer system used to acquire, analyze, store, and report information from chromatographic instruments.

⁸ Registered trademark of and available from Health Level Seven (HL7) International, 3300 Washtenaw Avenue, Suite 227, Ann Arbor, MI 48104, <http://www.hl7.org/>.

⁹ Available from International Conference on Harmonisation of Technical Requirements for Registration of Pharmaceuticals for Human Use (ICH), ICH Secretariat, 9, chemin des Mines, P.O. Box 195, 1211 Geneva 20, Switzerland, <http://www.ich.org>.

¹⁰ Available from Institute of Electrical and Electronics Engineers, Inc. (IEEE), 445 Hoes Ln., Piscataway, NJ 08854-4141, <http://www.ieee.org>.

¹¹ Available from The International Society of Automation (ISA), 67 T. W. Alexander Drive, P. O. Box 12277, Research Triangle Park, NC 27709, <http://www.isa.org>.

¹² Available from International Organization for Standardization (ISO), ISO Central Secretariat, BIBC II, Chemin de Blandonnet 8, CP 401, 1214 Vernier, Geneva, Switzerland, <http://www.iso.org>.

¹³ Registered trademark of and available from International Society for Pharmaceutical Engineering (ISPE), 600 N. Westshore Blvd., Suite 900, Tampa, FL 33609, <http://www.ispe.org>.

¹⁴ Available from Medicines and Healthcare products Regulatory Agency, Her Majesty's Government, 10 South Colonnade, London E14 4PU, United Kingdom, <https://www.gov.uk/>.

¹⁵ Available from National Council for Prescription Drug Programs (NCPDP), 9240 East Raintree Drive, Scottsdale, AZ 85260-7518, <https://www.ncdp.org/>.

¹⁶ Available from National Institute of Standards and Technology (NIST), 100 Bureau Dr., Stop 1070, Gaithersburg, MD 20899-1070, <http://www.nist.gov>.

¹⁷ Available from U. S. Nuclear Regulatory Commission (NRC), 11555 Rockville Pk., Rockville, MD 20852, <http://www.nrc.gov>.

¹⁸ Available from PIC/S Secretariat, 14 rue du Roveray CH, 1207 Geneva, Switzerland, <https://picscheme.org/>.

¹⁹ Registered trademark of and available from The Regenstrief Institute, Inc, 410 West 10th Street, Suite 2000, Indianapolis, IN 46202-3012, <http://loinc.org>.

²⁰ Available from SNOMED International, One Kingdom St., Paddington Central, London W2 6BD, United Kingdom, <https://www.snomed.org/>.

²¹ Available from World Health Organization (WHO), Avenue Appia 20, 1202 Geneva, Switzerland, <http://www.who.int>.

3.2.3 *cloud computing*, *v*—term generally used to refer to software applications that are delivered as a software service through remote hosting using the public internet (public cloud) or within the users' network environment (private cloud).

3.2.3.1 *Discussion*—Essentially, the difference between cloud computing and traditional application deployment is that the application's users may not be responsible for the installation and maintenance of the computing infrastructure and application software.

3.2.4 *corrective and preventative action*, *CAPA*, *n*—CAPA applications are used to collect information, analyze information, identify and investigate product and quality problems, and take appropriate and effective corrective or preventive (or both) action to prevent their recurrence.

3.2.4.1 *Discussion*—Verifying or validating corrective and preventive actions, communicating corrective and preventive action activities to responsible people, providing relevant information for management review, and documenting these activities are essential in dealing effectively with product and quality problems, preventing their recurrence, and preventing or minimizing device failures.²²

3.2.5 *cybersecurity*, *n*—set of technologies, practices, and processes used to protect computers, networks, programs, and data from attack, damage, exploitation, and unauthorized access.

3.2.6 *data exchange standardization*, *n*—as defined by the International Organization for Standardization (ISO) in ISO/HL7 27932, the process of agreeing on standards, which represent the common language that allows the exchange of data between disparate data systems.

3.2.6.1 *Discussion*—The goals of standardization are to achieve comparability, compatibility, and interoperability between independent systems, to ensure compatibility of data for comparative statistical purposes, and to reduce duplication of effort and redundancies. A data standard often includes data elements, data element definitions, and such agreements as formats, message structures, and vocabulary. In the context of this paper, a standard is a specification or requirement and is not synonymous with a policy, procedure, guideline, framework, technique, or best practice. Adopting standards has the potential to improve interoperability and reduce costs by facilitating the ability of networked laboratories to coordinate activities during public health incidents where surge capacity may be required (for example, national response and readiness). Adopting standards may reduce the costs of laboratory informatics system implementation and vendor/developer support.

3.2.7 *data integrity*, *n*—extent to which data are attributable, complete, consistent, accurate, and reliable throughout the data life cycle.

3.2.8 *electronic document management system*, *EDMS*, *n*—computer system used to store, catalog, review/approve, retrieve, view, and print digital documents.

3.2.8.1 *Discussion*—Modern EDMS applications typically provide the ability to manage a document throughout its life

cycle, including document initiation, multiple review levels, version control, security, and archiving of historical versions of documents.

3.2.9 *electronic laboratory notebook*, *ELN*, *n*—software program designed to replace paper laboratory notebooks; an electronic system on which to create, store, retrieve, and share fully electronic records in ways that meet all legal, regulatory, technical, and scientific requirements.

3.2.9.1 *Discussion*—Laboratory notebooks, in general, are used by scientists, engineers, and technicians to document research, experiments, and procedures performed in a laboratory. A laboratory notebook is often maintained to be a legal document and may be used in a court of law as evidence. Similar to an inventor's notebook, the laboratory notebook is also often referred to in patent prosecution and intellectual property litigation.

3.2.10 *electronic signature*, *n*—electronic representation of a handwritten signature.

3.2.11 *enterprise resource planning*, *ERP*, *n*—computer system to integrate different types of data such as inventory levels, product orders, manufacturing capacity, inspection results, accounting data, and customer relationship management information from organizations within an enterprise (company), facilitating the flow of information between various business functions across a company as well as with outside business partners.

3.2.12 *good automated manufacturing practice forum*, *GAMP Forum*, *n*—volunteer group under the auspices of the International Society of Pharmaceutical Engineering (ISPE) for writing guidance for the validation of computerized systems used in the regulated portions of the pharmaceutical and allied industries and it is specifically designed to aid suppliers and users in the pharmaceutical industry.

3.2.13 *integration broker*, *n*—messaging application that can receive or extract data from a source system at the appropriate time, transform the data, and route the reformatted data to the target node.

3.2.13.1 *Discussion*—An integration broker application can also provide a repository for archiving, searching, and retrieving these messages.

3.2.14 *internet of things*, *IoT*, *n*—system of objects—computing devices, machines, objects, people, animals, and so forth—that can connect to a network and communicate among themselves, often without human intervention.

3.2.14.1 *Discussion*—An IoT device is an object operating within that system.

3.2.15 *laboratory execution system*, *LES*, *n*—computer system used in the laboratory at the analyst work level to aid in step enforcement for laboratory test method execution.

3.2.15.1 *Discussion*—LES focus on step execution of defined laboratory test methods. The LES is typically used in quality control laboratories that have defined test methods. The functionality of a LES and a laboratory information management system (LIMS) overlaps in the areas of result entry, instrument integration, and specification flagging. Deployment

²² For additional information, visit the FDA's CAPA page at <http://www.fda.gov/>.

options include LES and LIMS systems deployed as an integrated solution, LIMS-only, or LES-only (for limited functions).

3.2.16 *laboratory informatics, n*—term used to describe the specialized application of information technology aimed at optimizing laboratory operations.

3.2.16.1 *Discussion*—That technology includes informatics tools used within laboratory environments to collect, store, process, analyze, report, and archive data and information from the laboratory and supporting processes. Laboratory informatics includes the effective use of critical data management systems, the electronic delivery of results to customers, and the use and integration of supporting systems (for example, training and policy management). Examples of primary laboratory informatics tools include, LIMS, LES, CDS, ELN, laboratory information systems (LIS), and scientific data management systems (SDMS).

3.2.17 *laboratory informatics tools configuration, n*—refers to the process of changing the functions of any laboratory informatics tool to match the business processes used in a particular laboratory, and it does not involve the use of writing software code either via a recognized software language or a language provided by the informatics application supplier.

3.2.17.1 *Discussion*—This is a GAMP Category 4 software and is defined as “Configured software including, LIMS, SCADA, DCS, CDS, etc.” Such configuration typically involves using an interface provided by the vendor to enter information that describes the types of samples, analytical methods, specifications, and so forth, used in the laboratory. It may also involve the configuration of options and businesses rules within the tool.

3.2.18 *laboratory informatics tools customization, n*—refers to the process of changing the functions of any laboratory informatics tool to match the business processes used in a particular laboratory.

3.2.18.1 *Discussion*—This is different from the previously mentioned tools configuration in that customization involves writing software code either via a recognized software language or a language provided by the informatics application supplier. This is a GAMP 5 software category. Such customization typically involves adding tables, modifying table structures, and writing code or programs to alter the behavior of any laboratory informatics tool.

3.2.19 *laboratory information management system, LIMS, n*—(1) computer software and hardware that can acquire, analyze, report, and manage data and information in the laboratory; (2) computer software that is used in the laboratory for the management of samples, test results, laboratory users, instruments, standards, and other laboratory functions such as invoicing, plate management, product/material stability programs, and work flow automation; and (3) a class of application software which handles storing and managing of information generated by laboratory processes.

3.2.19.1 *Discussion*—These systems are used to manage laboratory processes, including master data definition, sample management and chain of custody, work assignment, instrument and equipment management, standard and reagent

management, scheduled sample collection and testing, result entry, capture of results from instruments, result review, reporting, trending, and business rule enforcement. These systems interface with laboratory instruments (for example, CDS and other information systems such as ERP, LIS, or manufacturing execution systems [MES]). A LIMS is a highly flexible application, which can be configured or customized to facilitate a wide variety of laboratory workflow models.

3.2.20 *laboratory information system, LIS, n*—class of application software that supports clinical laboratories by helping laboratory personnel manage the quality and integrity of test samples, departmental workflow functions, result review processes, reports of finalized results, interpretations, and diagnoses.

3.2.20.1 *Discussion*—These systems often interface with instruments and other information systems such as hospital information systems (HIS). A LIS is a highly configurable application and often includes laboratory-specific electronic medical records, direct clinician access via secure web connections, billing modules for laboratories performing commercial testing, sophisticated interface engines for routing orders and results to external systems, and on-board image archival systems for pathology images. Patient confidentiality and HIPAA requirements define unique security functionality for a LIS. The College of American Pathologists (CAP) publishes LIS product guides²³ that list current LIS in the market.

3.2.21 *lean laboratory, n*—set of management and organizational processes that enables efficient testing flow, leveled workloads, visual work assignment and tracking, and the elimination of waste.

3.2.21.1 *Discussion*—Lean laboratory designs yield productive, high-quality laboratory environments that are sometimes supported by laboratory informatics tools.

3.2.22 *mapping tools, n*—graphical data mapping, conversion, and integration applications that map data between any combination of XML, database, flat file, EDI, Excel (OOXML), XBRL, or web service, or both, then transforms data or autogenerates data integration code for the execution of recurrent conversions.

3.2.23 *master data, n*—represents the business objects which are agreed on and shared across the enterprise.

3.2.23.1 *Discussion*—It can include relatively static reference, transactional, unstructured, analytical, and hierarchical data, as well as associated metadata. Examples of master data include product specifications, test method steps (to capture intermediate and final results), laboratory calculations, instrument information, and standard and reagent information.

3.2.24 *metadata, n*—(1) data about data and (2) information that describes another set of data.

3.2.24.1 *Discussion*—Additional information about the data that provides context and meaning, including how, when, and by whom it was collected, and its relationship to the subject or test. Metadata in any laboratory informatics tool’s context

²³ For additional information, visit CAP’s product guide page at <http://www.captodayonline.com>.

typically includes all data that supports a test result that is recorded in this tool. For example, a pH test includes a pH result that can be supported by metadata, including what instrument was used, what the calibration date of the instrument was, what standard buffer solutions (reagents) were used to calibrate the pH probe sensor, the expiration dates for the standard solutions, and the temperature of the solution at time of measurement.

3.2.25 *sample registration, n*—process of recording incoming sample information in a given laboratory informatics tool.

3.2.26 *scientific data management system, SDMS, n*—computer system used to capture, centrally store, catalog, and manage data generated in a laboratory environment.

3.2.26.1 *Discussion*—These data are then available for re-use and integration with other laboratory informatics systems. An example of an SDMS is an electronic repository for reports from laboratory informatics systems. The SDMS may include raw data file storage and archiving of data. It may also provide e-signature functionality for review/approval.

3.2.27 *spectroscopic data systems, n*—computer systems used to collect, process, visualize, interpret, store, and report information from spectroscopic instruments.

4. Significance and Use

4.1 *Relevance*—This guide is intended to educate the intended audience on many aspects of laboratory informatics. Specifically, the guide may:

- 4.1.1 Help educate new users of laboratory informatics;
- 4.1.2 Help educate general audiences in laboratories and other organizations that use laboratory informatics;
- 4.1.3 Help educate instrument manufacturers and producers of other commonly interfaced systems;
- 4.1.4 Provide standard terminology that can be used by laboratory informatics vendors and end users;
- 4.1.5 Establish a minimum set of requirements for primary laboratory informatics functions;
- 4.1.6 Provide guidance on the tasks performed and documentation created in the specification, evaluation, cost justification, implementation, project management, training, and documentation of laboratory informatics; and
- 4.1.7 Provide high-level guidance for the integration of laboratory informatics and other software tools.

4.2 *How to be Used*—This guide is intended to be used by all stakeholders involved in any aspect of laboratory informatics implementation, use, or maintenance.

4.2.1 It is intended to be used throughout the laboratory informatics life cycle by individuals or groups responsible for laboratory informatics implementation and use, including specification, build/configuration, validation, use, upgrades, and retirement/decommissioning.

4.2.2 This guide also provides an example of a laboratory informatics functional requirements checklist that can be used to guide the purchase, upgrade, or development of a laboratory informatics system.

5. Elements of Laboratory Informatics

5.1 *Laboratory Informatics Elements Overview*—Laboratory informatics is used to help laboratory personnel

better collect, process, analyze, report, store, and share the data and information derived from the laboratory and its supporting processes. These processes are often an integral part of a laboratory's workflow and include activities such as registration of samples or experiments, or both, assignment of tests, entry of results, review and approval of results, and reporting. Laboratory informatics' scope encompasses multiple technical solutions or systems that are responsible for streamlining these and other laboratory processes. Laboratory informatics is not solely about software managing laboratory data; it has many elements, some of which integrate or cross over with business management and other third-party tools. Those elements are also becoming increasingly complex, both in functionality and interoperability. Outside of standard laboratory information management systems (LIMS) and laboratory information systems (LIS), elements such as field data capture systems, advanced analytics tools, and artificial intelligence continue to shape the field of laboratory informatics. The division between these and other system categories continues to soften as functionality continues to be added to each of them. LIMS were originally created to address laboratories' need to manage laboratory operations and data, provide traceability for all laboratory samples and equipment, and ensure that laboratory procedures are followed. Electronic laboratory notebooks (ELNs), on the other hand, were originally created to meet scientists' need to document their experimental design, execution, and conclusion in an electronic format instead of in a paper notebook. The scientific data management system (SDMS) was created to provide a repository of all scientific data files and results regardless of instrument type. The current definitions of each of these system categories are far more encompassing and continue to evolve as the boundaries between categories continue to blur. That blurring of laboratory informatics elements, as well as their potential integration with enterprise elements—both within organizations and with customers of laboratory information—are illustrated in Fig. 1. Laboratory informatics and all it encompasses is shown with the large yellow circle on the left, while the internal business systems that support laboratories are found associated with the blue circle on the right. Surrounding both is a bubble representing third-party interactions with both laboratory generated data and business data. The figure highlights the wide variety of crossover and interactions that can occur both within and external to laboratory informatics. Laboratory informatics applications are also taking on some of the functionality of internal business systems. (See 7.6–7.11 for more information on integration cases.)

5.1.1 *Core Systems*—These laboratory systems most often provide the outward face of laboratory informatics and include LIMS, LIS, laboratory execution systems (LES), ELN, SDMS, and chromatography data systems (CDS). Not all systems will necessarily appear in a laboratory together; some are more typical to certain laboratory types than others. However, they usually play a key role in a laboratory's research or analysis activities, or both, and represent the key software systems with which laboratory personnel and customers of the laboratory may interact. From research samples and clinical specimens to

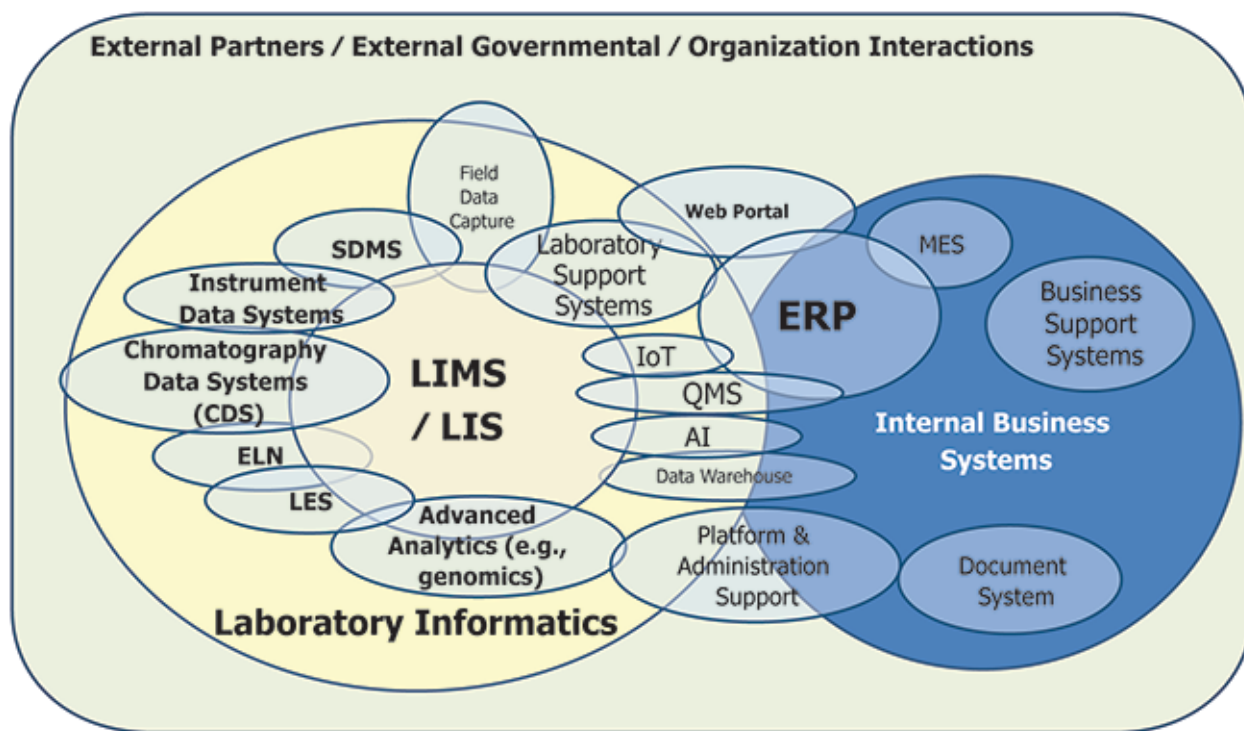


FIG. 1 Laboratory Informatics System Integration Concept Model

outlined experiments and raw instrument data, these core systems fill a vital role in the laboratory informatics sphere.

5.1.1.1 LIMS, LIS, and LES are alike in many regards in that they all act as core systems in a laboratory and handle data capture, analysis, review, storage, and reporting. These systems integrate in variable degrees with analytical instruments, automated tools, and other software systems, and they provide certain levels of regulated, industry-standard security for the data generated and transferred from its integrations. However, these systems also have fundamental differences that place them in specific use cases. A LIMS has been traditionally used to process and report on batches of samples from research, quality control, and manufacturing laboratories, all of which handle mostly anonymous, complex laboratory data. A LIS has normally been used in the clinical context of specimens and patients, and a LES is most often adopted in automated and regulated manufacturing environments where quality control, process control, test step execution, and instrument interface and calculation validation support laboratory testing.

5.1.1.2 An ELN largely serves as an electronic replacement for the traditional paper laboratory notebook associated with scientists and technicians in research-driven environments. The ELN may be tailored to the individual researcher, large collaborative research efforts, or both. It may also be designed to manage the activities related to a specific scientific discipline or application, or it may be cross-disciplinary, supporting data of all types. Traditionally used to document experiments and analysis, and act as intellectual property protection, the paper laboratory notebook has fallen out of favor with some laboratories that prefer the ELN's ability to improve search, support collaboration, and limit siloed data. Additional features of an

ELN include data import, content linking, preformatted and customizable templates, and messaging.

5.1.1.3 An SDMS is designed primarily to consolidate data and manage knowledge-based assets. The SDMS has typically excelled at handling unstructured files such as images, spreadsheets, raw instrument files, and PDFs. A set of agreed-upon rules in the system dictate how incoming data is processed and structured in the SDMS, acting as a gatekeeper for what is captured and how (including the application of metadata). An SDMS can be integrated with a LIMS, ELN, and so forth, to create a common repository for a laboratory's data, which can then be further associated with specific projects, experiments, or locations, or combinations thereof, or its data can simply be archived for long-term storage.

5.1.1.4 A CDS is designed for collecting, processing, and analyzing samples run on instruments managing chromatography techniques such as high-performance liquid chromatography (HPLC), ion chromatography (IC), gas chromatography (GC), size-exclusion chromatography (SEC), and affinity chromatography. The CDS typically consists of a combination of hardware and software connecting the instrument to the system and is computationally intensive, rapidly generating large data sets in laboratory environments. Complex algorithms and mathematical transformations of data can be performed within a CDS, which directly supports a wide range of chromatography instruments with bidirectional control of instrument settings (that is, temperature, pressure, and detector wavelength). The CDS typically provides sample handling (auto samplers), auto injection, mobile phase controls (temperature/pressure), detector control (wavelength), data collection (data points from one or more detectors), data analysis (for example, calibration

curves, peak detection, and integration), reporting, and audit trail support. The CDS can be deployed as a standalone system or in larger configurations supporting multiple instruments, sites, and geographic regions. The CDS is typically interfaced to other laboratory informatics tools (that is, LIMS, ELN, and SDMS) in which sample information is passed between the LIMS and the CDS, and the CDS test results (that is, peak areas) are passed back to the LIMS for final reporting. The CDS can also integrate with multiple analytical techniques in which data are passed between different instruments [such as a liquid chromatography-mass spectrometry (LC-MS) instrument]. The software is typically treated as a separate laboratory informatics element with its own IT infrastructure, including data acquisition modules (to attach to chromatography instruments) and administration, configuration, and security access controls.

5.1.1.5 Bioinformatics and genomic applications are used to collect and analyze biological data, including genomic data. Genomic instruments generate much larger amounts of data compared to traditional laboratory instruments. Genomic data is often analyzed using specialized, proprietary bioinformatics algorithms.

5.1.2 *Instrument Data Systems*—Some laboratory systems are purpose-made for specific instruments, with the CDS being a common case. Other examples include data acquisition systems (DAQ) for calibration equipment and digital oscilloscope software. These systems excel at “talking” with a specific instrument, capturing the data, and providing custom analysis tools related to the captured data. In the case of chromatography, a CDS will often both control a chromatography or spectroscopy instrument and provide a visual report of the chromatogram with contextual meaning. DAQ for calibration instruments may include additional functionality such as real-time data visualization and certificate generation.

5.1.3 *Advanced Analytics Tools*—This element represents a broad category of advanced analytical tools used in laboratories. Examples include the scientific field of genomics in which advanced tools are used to study complete genomes (genetic material within an organism). High-throughput genome analyzer instruments automate process steps to provide high-volume parallel operations that combine DNA molecules, primers, polymerase amplification, imaging, and computational functions to yield low-cost DNA sequencing outputs. Advanced tools are also used to automate DNA sequence assembly (reconstruction of an original DNA sequence). Advanced annotation tools/instruments are used to annotate DNA sequences by identifying portions of the genome that do and do not code for proteins, including supporting biological information.

5.1.4 *Field Data Capture Systems*—Field data capture and analysis is important to several industries, including many which do their own laboratory testing. Along with its base functionality, these systems are also often capable of integrating supervisory control and data acquisition (SCADA)-related infrastructure data (for example, from a wastewater or oil and gas stream) with manually collected field worker data to improve test data analysis and decision support. They also

typically allow native mobile apps to integrate, improving data analysis and reducing manual data input time.

5.1.5 *Laboratory Support Systems*—These systems are typically ancillary to the core systems, filling in functionality gaps that the core systems do not provide. Quality management software is one such example, used to formulate quality policy and objectives, standard operating procedures, and the required records used for the quality certification process. Middleware that can handle auto-validation of samples represents another example. The following may all be considered part of this category:

5.1.5.1 Artificial intelligence (AI) tools and algorithms are being used by researchers to inspect data better and make discoveries in the laboratory, while laboratory tools such as freezers, incubators, and air-cleaning systems are becoming “smarter” with added sensors that can feed data to one or more software systems for monitoring.

5.1.5.2 Batch and lot management tools assist with the creation, genealogy, review, and disposition of samples en masse, in the process assigning the same properties and process tracking for increased efficiency.

5.1.5.3 Capacity planning and laboratory scheduling improves the efficiency of laboratory workflow, allowing laboratory personnel to use better the time and resources of available personnel and equipment. Such tools take into account scheduled instrument maintenance and planned time off of researchers.

5.1.5.4 Compliance management, whether embedded in a core system or installed as a support system, helps keep laboratories of all types on track with complying with government regulations. Aside from the typical audit trails and electronic signatures, compliance management tools also may assist with risk assessments, business governance, and contractual obligation management.

5.1.5.5 Data integrity is a core competency and expectation of laboratory informatics solutions. Data integrity includes the maintenance and assurance of the accuracy and consistency of data over its entire life cycle. Data integrity is a critical design requirement that touches every element of laboratory informatics that stores, processes, or retrieves laboratory data. (See Section 10 for more on data integrity.)

5.1.5.6 Human resource management encompasses the activities associated with managing the data surrounding laboratory personnel. This includes the management of analyst training records, qualifications, certifications, and performance.

5.1.5.7 Instrument and equipment management tools help to determine on- or offline status, assist with calibration management, update service and preventative maintenance schedules, and present the qualification status of instruments for proper safety and functionality.

5.1.5.8 Instrument data capture and control can provide some of the greatest benefits to both laboratory efficiency and data quality and integrity. Traditionally, laboratory informatics tools have allowed direct data capture from simple instruments, indirect data capture through instrument data file parsing, and, less frequently, bi-directional control of simple and file-based instruments. Although it is common for laboratory informatics

solutions to have integrated instrument integration tools, there are also standalone instrument integration systems that may act as a data capture hub for the laboratory by connecting many instruments to a single application. These hub systems may then be interfaced to laboratory informatics solutions to provide data from multiple instruments via a single system-to-system interface. Instrument integration may also be available as a service, hosted in the cloud or on-premises, and newer instrumentation and devices are available that take advantage of the internet of things (IoT) web connection capabilities that share their data.

5.1.5.9 Inventory management involves the handling of controlled substances, reagents, and retention samples, and it may also handle stability analysis.

5.1.5.10 Investigation management is important to researchers and manufacturers in the pharmaceutical, medical device, and biologics industries as part of a regulated process that strives to ensure the safety and compliance of the components, containers, in-process materials, and finished products of those industries. This includes the management of the FDA's out-of-specification (OOS) requirements,²⁴ internal out-of-trend (OOT) requirements, process validation efforts, and any corrective and preventative action (CAPA) efforts related to investigations.

5.1.5.11 Process improvement is generally thought of as a series of discussions and changes towards making the laboratory leaner, often through the concept of continuous improvement. However, visual project management “dashboards” that collate performance data and visually present it may be integrated with other laboratory informatics tools.

5.1.5.12 Scheduled event management is enhanced through the use of event management and planning software or functionality. In particular, laboratories with shifts or rotations that move laboratory personnel around in various sections benefit from tools that assist with routine and nonroutine work.

5.1.5.13 Scientific data management does not necessarily need to be limited to an SDMS. Other core, instrument data, and standalone laboratory systems are designed to manage a wide variety of instrument raw data files, documents, and experiments for the purpose of improved data homogenization, sharing, and mining.

5.1.5.14 Standard and reagent management encompasses verifying the storage, quantity, on-order, and expiry status of a wide variety of inventoried additives, chemicals, and calibration standards.

5.1.5.15 Statistical trending and control charting is part of the data analysis and visualization efforts of a software system. These tools allow laboratory personnel to improve decision making and process development, among other things.

5.1.5.16 Systems integration functionality allows laboratory informatics systems, business management systems [for example, enterprise resource planning (ERP)], manufacturing systems [for example, manufacturing execution system (MES)], process management tools (for example, SCADA),

document management systems, statistical analysis tools, and more to connect with each other and limit the amount of “siloed” or isolated data in an organization.

5.1.6 *Artificial Intelligence*—The application of AI in clinical, research, and industrial laboratories is still in relative infancy, though laboratory personnel and researchers alike are increasingly finding ways to integrate AI into laboratory workflow. With AI's ability to “learn” from user interactions, data feeds, and more to improve or make insights into laboratory processes and data, the technology will likely continue to find ways into the laboratory. Machine learning tools that analyze structured data and natural language processing tools that extract information from unstructured data are already finding their way into some clinical laboratory workflows.

5.1.7 *Platform and Administration Support*—Platform and administration support tools not only help laboratory informatics end users develop and maintain important documentation, they also allow systems administrators and IT personnel to provide a better, more secure experience. These tools may be built into a core system or act as standalone components of an interconnected web of systems. Functionality among these tools includes document management (for standard operating procedures, specifications, training material, and so forth), configuration management (for handling all aspects of master data, electronic signatures, and so forth), system validation and commission (for requirements, specifications, test scripts, regression and stress tests, and so forth), system administration, and electronic security and privacy management.

5.1.8 *Data Warehouse*—An SDMS is a type of data warehouse in which a series of agreed-upon rules dictate how the incoming data is processed and structured. But other forms of data warehousing exist as well, which both clinical and nonclinical laboratories alike are using it viably. The late-binding or “data lake” approach to data storage represents a more modern architecture that takes in data of various structural states, tags it with appropriate metadata, and flattens it, leaving it alone otherwise until queried or used, or both. These data storage systems may be hosted locally or offered as a service over the cloud.

5.1.9 *Web Portals*—These tools allow third parties to access, in a controlled fashion, one or more laboratory informatics data repositories or data management systems. They are often part of an existing system such as a LIMS but may also be standalone with a connection to one or more databases. The level of access granted to individuals or groups are controlled by the web portal's administrative tools.

5.2 *Laboratory Informatics Systems Evolution*—Laboratory informatics systems have evolved over time, with developers adding and expanding functions as capabilities and needs change. LIMS and instrument data systems such as CDS began by performing simple laboratory functions. Over time, additional software tools entered the laboratory, and existing software products added functionality. Fig. 2 shows a timeline for the development of software products designed to meet the needs of the laboratory community. The expanding breadth of tools available illustrates the increased functionality and complexity of laboratory informatics solutions. The laboratory

²⁴ See the Food and Drug Administration's “Guidance for Industry: Investigating Out-of-Specification (OOS) Test Results for Pharmaceutical Production,” available via the FDA at <https://www.fda.gov/>.

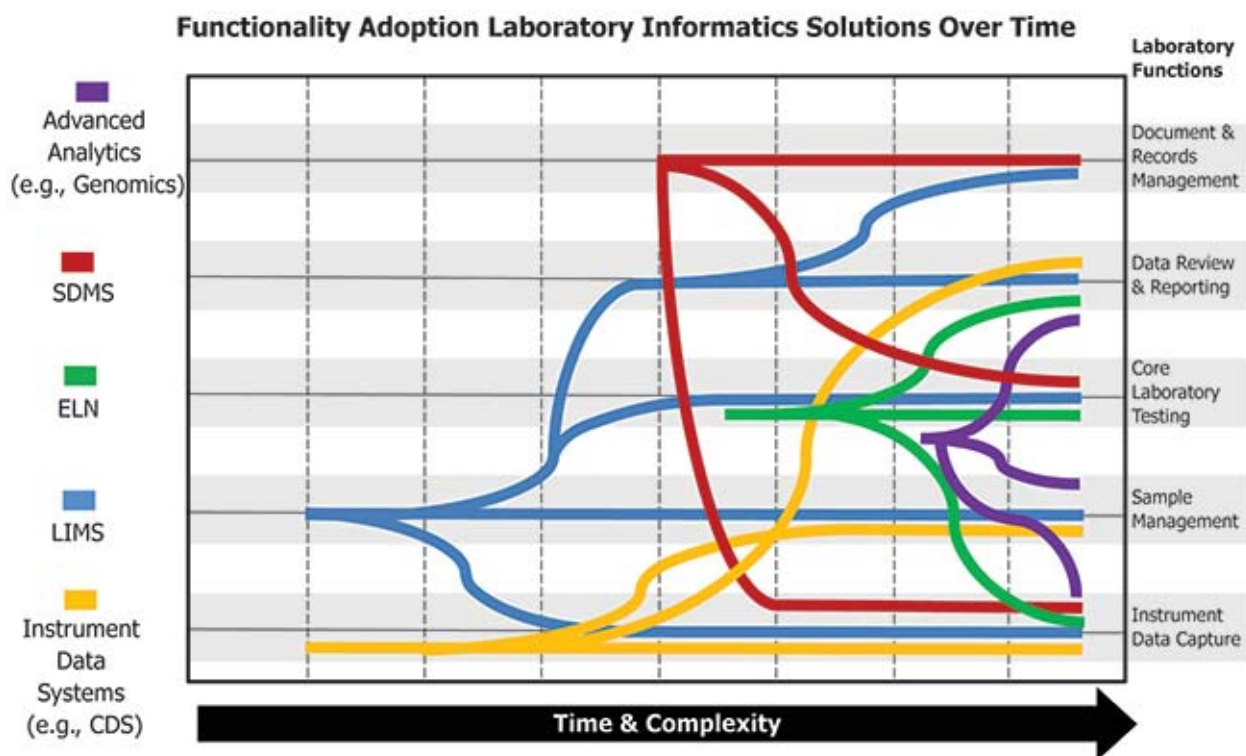


FIG. 2 Laboratory Informatics Systems Evolution

informatics solutions illustrated in this figure are examples, however, and do not imply that these are the only tools available.

5.3 Laboratory Informatics Functions—Laboratory informatics systems contain a rich set of functions. In Fig. 3, those functions using the primary functional categories that can be found in laboratory informatics systems are illustrated. At the center of these laboratory informatics functions is core laboratory testing (items labeled as C-x), including sample/experiment registration, sample management, testing, result review, sample approval, and reporting. Core laboratory testing is then supported by extended functions (items labeled as E-x), including planning and scheduling, instrument management, reagent and standard management, material disposition, experiment approvals, reporting, and trending. Finally, those core and supporting functions are propped up by laboratory informatics platform and administration support functions (items labeled as S-x and D-x), including master data management. Note that the labels assigned to these various sections directly tie into the items found in the requirements checklist of Appendix X1.

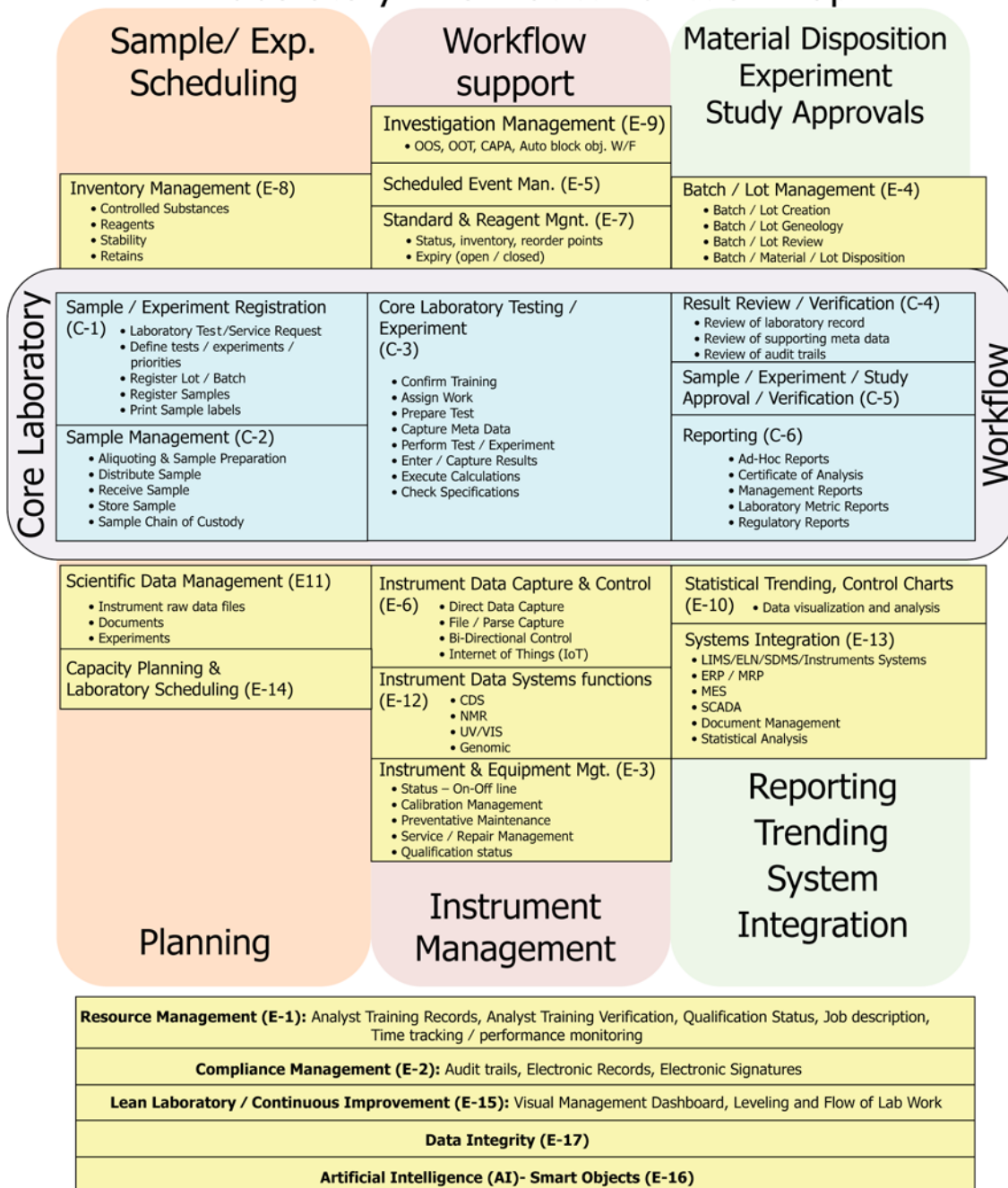
5.4 Laboratory Informatics Functions for Specific Industries—Laboratory informatics encompasses many technical solutions such as LIMS, LIS, ELN, and CDS, which aim to improve laboratory operations. However, not all laboratory workflows are the same. Laboratories in specific industries may require additional functionality to meet their workflow requirements. Some laboratory informatics systems vendors may design their solutions to meet the needs of many laboratory types, while others may design them for a particular industry segment with particular functionality needs. An environmental laboratory, for example, may require tracking of

sample containers, processing of samples in batches with control samples, instrument integration, multiple levels of review, and specific reporting requirements. In Fig. 4, some of the additional functions that may be required to address the needs of specific laboratory types are illustrated. The functions illustrated are over and above the basic laboratory workflow and are by no means an exhaustive list; these are merely examples of possible additional functionalities.

5.5 Laboratory Informatics Implementation and Technological Considerations—As with most any other software system, implementing laboratory informatics software in a laboratory requires at least a minimum set of plans and considerations. What operating systems are supported? What instruments shall be connected? Does the system need to be customized frequently? These types of questions quickly multiply as system complexity increases, operational needs expand, and integration needs become more complex. This line of questioning can become so complicated that functional requirements checklists are required to make sense of how a laboratory informatics solution fits into organizational workflow. As such, gaining a clearer understanding of a laboratory informatics solution's life cycle, as well as the technological elements that go into its development and implementation, is vital to its successful operation in the laboratory.

5.5.1 The implementation life cycle of laboratory informatics systems is not unlike other software systems, though it comes with its own intricacies, including regulatory and “best practice” considerations specific to clinical, research, and other types of laboratories. The laboratory informatics system life cycle is described in more detail in Section 8.

Laboratory Informatics Function Map



Platform & Administration Support

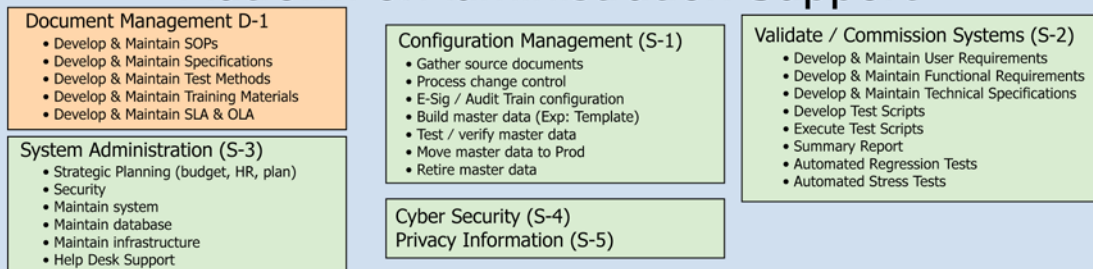


FIG. 3 Laboratory Informatics Functions

Additional Functions Required by Laboratory Types		Laboratory Functional Area					
		Sample / Experiment Management	Lab Testing	Review, Verification Approval	System Integration	Reporting and Trending	Other Functions
Laboratory Type	General Laboratories	Chain of custody	Batch Processing of Samples including QA		ELN, SDMS, instrument integration	Electronic Reporting of QC Results	
	Environmental	Container management, Chain of custody, regulatory compliance - significant auxiliary data	Batch Processing of Samples including QA	Multiple levels of review and approval	GIS integration	Electronic Reporting of QC Results by batch	Method QA/QC, Calculation validation, electronic data delivery
	Public Health Sector- (clinical microbiology and chemical)	Chain of custody, electronic test ordering, Emergency response, public health surveillance	Batch Processing of Samples including QA, agent regulations	Multiple levels of review and approval, CLIA certification	Emphasis on surveillance at multiple state and federal levels	Electronic Reporting multiple data formats, vocabulary, and secure transport	Clinical and non-clinical, sample centric and patient centric
	Life Sciences	Controlled substances Stability samples	Uniformity Calculations Product Specifications	Multiple levels of review and approval	ERP integration	SPC and Process variability analysis	Instrument maintenance and calibration tracking
	Food and Beverage	Lot geneology	Product Specifications		MES and ERP integration	SPC and Process variability analysis	Exception Reporting
	Heavy Industry	Automatic scheduling, Lot Management	Product Specifications		Process Control / PM integration	SPC and Process variability analysis	
	R&D	Experiment sharing	Method versioning		ELN, SDMS, instrument integration	Experiment Conclusions, Technical Reports	Expanded Search capability
	Medical Laboratory	Embedded LIS Standalone LIS Niche LIS Patient centric			interface engines to external systems	Unique functions Diagnostic and Surveillance	electronic medical records; direct clinician access, billing modules

FIG. 4 Laboratory Informatics Functions for Specific Industries

5.5.2 The technological elements of laboratory informatics systems and their design and deployment are worth noting. Technologies such as cloud computing and application programming interfaces (API) are continuing to shape system development and implementation across a wide variety of laboratory types. Key factors such as hardware infrastructure, application platform, and integration needs further drive decisions of how a system should be developed and deployed, thus influencing the use of technological elements towards those goals. The following may all be considered technological

elements relevant to laboratory informatics systems' development and deployment:

5.5.2.1 APIs allow software components to communicate clearly with each other, improving the way laboratory informatics applications are built and used. They not only allow core developers to simplify programming but also allow end users who need to customize aspects of how the software is used (and outputs data) a simpler way to do that. Some software vendors may ship their laboratory informatics offering with an API or software development kit (SDK) so the end user

can simplify how the software exchanges data with or calls background services from other systems in the laboratory, as well as customize the software in other ways.

5.5.2.2 Artificial intelligence (AI) and its application in the laboratory continue to grow with the use of smart objects, machine learning applied to laboratory data, and monitoring of laboratory processes and resources. See Section 11 for additional details on this subject.

5.5.2.3 Authentication mechanisms and protocols such as single sign-on (SSO) and OAuth are important to the security of a laboratory informatics application, and the decision of which to use is not taken lightly. These mechanisms and protocols help define how information is securely stored, accessed, and propagated. In some cases, regulations such as the Health Insurance Portability and Accountability Act (HIPAA) may further dictate selection, particularly in cases in which electronic-protected health information (ePHI) is involved.

5.5.2.4 Business intelligence (BI) and big data technology components increasingly find their way into or tangentially connected to laboratory informatics. This includes data warehousing, data mining, data analytics, data structuring, and web portal components that aid in the processing, managing, visualization, and dissemination of all sorts of data.

5.5.2.5 Cloud computing and storage offers a virtualized alternative to how laboratory informatics applications and data are leveraged. Whether because of a lack of information technology resources and internal expertise or a desire to reduce operating costs, informatics vendors and businesses alike are turning software into a service, accessible via the internet or over a private network. This technology has benefits and drawbacks, though it has also become more refined over the years. (See 7.1 for more.)

5.5.2.6 Data migration tools represent another important component of laboratory informatics deployment; many systems are not put into operation with a blank database. These tools are often standalone, meaning they do not become a permanent part of the overall informatics infrastructure, and they also may have data cleaning and transformation functionality to ensure data is in the right form when it arrives at its destination.

5.5.2.7 IoT devices have made their way to laboratories, promising greater productivity and insights. Collections of networkable sensors and devices give laboratories the option to interconnect, monitor, and make insights into various data streams, whether located in-house or thousands of miles away. However, implementing IoT in the laboratory comes with its own considerations, including communication methods, security, and quality of data. (See 7.2 for more.)

5.5.2.8 Software or technology stacks are essentially a collection of software subsystems or components that allow for the development and operation of a complete laboratory informatics solution. These subsystems and components are often chosen together frequently because of a solid history of integrating well, as well as the availability of abundant documentation to support the implementation of the stack. A

bioinformatics application, for example, may be built on a LAMP stack that includes Linux, Apache HTTP Server, MySQL, and Python.

5.6 *Laboratory Informatics: Standardization, Authentication, and Integration*—Several trends in laboratory informatics development, implementation, and use have slowly emerged, with a reasonable chance of impacting how laboratories manage data and its security. The concepts of data standardization, blockchain-based authentication, and end-to-end data integration are briefly described here.

5.6.1 Laboratory Informatics data standardization efforts have been active across multiple years and in different analytical techniques and industry segments. The adoption of data standards covering different laboratory instruments, different computer architectures, databases, and operating systems across different industry segments and laboratory types has been a slow tedious process. The demand and requirements for improved data-sharing mechanisms have expanded, driving further discussion and work efforts concerning electronic data formats. Data sources across a single scientific enterprise may number in the tens, making data sharing more difficult across hundreds of other such enterprises, many of which may include older or proprietary data sources that are not easily used by other entities. The durability of analytical data over long time periods is a growing concern. Data sources may be unorthodox too, such as from social media, focus groups, or regulatory filings in addition to traditional laboratory instrument data outputs. Emphasis on better managing, sharing, and mining this data has grown considerably, such that funding agencies, data organizations, standard organizations (like ASTM) and academic publishers are beginning to provide recommendations and requirements for shared laboratory data and how it is formatted, stored, and publicly accessed. Standardized electronic formats are important to laboratories of all types, especially public health and pharmaceutical laboratories. Groups such as Observational Health Data Sciences and Informatics are promoting data standardization efforts and data models such as the Observational Medical Outcomes Partnership (OMOP) Common Data Model²⁵ in recognition of this growing need. The OMOP data model attempts to transform, characterize, and analyze a wide variety of varying data sources associated with health care delivery and research “to ensure that research methods can be systematically applied to produce meaningfully comparable and reproducible results.”²⁶ Ongoing ASTM efforts supporting laboratory data standards include the Analytical Information Markup Language (AnIML),²⁷ an open XML-based format for analytical and biological data. The format supports data from many analytical and laboratory process techniques, enabling the creation of generic tools that can handle data from any instrument. Industry-specific consortiums (and their associated test method

²⁵ Available from Observational Health Data Sciences and Informatics (OHDSI), 622 W. 168th St., PH-20, New York, NY 10032, <https://www.ohdsi.org/>.

²⁶ See the OMOP Common Data Model documentation, housed at GitHub, <https://github.com>.

²⁷ Analytical Information Markup Language (AnIML), an emerging ASTM XML standard for chemistry data sanctioned by ASTM Subcommittee E13.15. Additional information available at <https://www.animl.org/>.

standards) have also made ongoing efforts to advance the management of laboratory data, though some require a license fee/commercial approach to generate data using standard formats.

5.6.2 Blockchain-based authentication and identity management was developed in the late 2000s, though it is in its infancy in relation to laboratory and medical data systems. The technology acts as a distributed ledger that enables multiple computer servers to access, reconcile, and lend integrity to a data record. Used initially with the management of digital currency records, some in the medical and laboratory communities are proposing the use of blockchains to improve data portability and security, particularly in the scope of electronic health records (EHR) and LIS. As data integrity, sharing, and portability take on greater importance for laboratories, blockchain technology offers a solution by storing, distributing, and allowing access to records such that all participants can interact without pre-existing trust (because of cryptographic techniques) while maintaining clear audit trails.

5.6.3 End-to-end data integration in laboratory informatics is more a concept than a reality, yet years of informatics software development have seen the functionality of what was previously found in several systems make its way into a single system. The functionality, for example, of a LIS, traditionally reserved for clinical use, has found its way into what was traditionally a more research-based LIMS. This convergence of system functionality continues, with enterprise resource planning functions finding their way into laboratory informatics applications as well. Most commonly we find separate systems integrated via APIs and data transfer protocols; however, differences in data formats between systems and unclear data standards make merging laboratory and business data (to mine it effectively) more challenging.

6. Laboratory Informatics Workflow and Sample Life Cycle

6.1 *End-to-End Business Integration*—Laboratory informatics solutions are central to the broader business that it supports, as illustrated in Fig. 5. Independent drivers create demand that feed into the laboratory environment. These factors are typically industry specific and may include clinical protocols, research projects, sales, production planning, stability and environmental monitoring, and investigations. Auxiliary and concurrent workflows also exist that are driven by information flowing in or out of the active laboratory informatics workflow. Testing status, intermediate/in-process results distribution, investigation requirements, instrument calibration and maintenance, standard and reagent management, and document management (for example, SOPs, change control, and validation) are parallel processes that interface with the informatics solution. Upon completion of laboratory testing, a series of broad workflows within the larger business may be initiated that provide feedback to the internal business or external customers. Data analytics and trending, performance metrics, dispositioning of samples, and data archiving are internally focused, while certificates of analysis (COAs), sales and distribution, sample storage, and external communication distributions may be delivered externally for use by customers.

6.2 *Laboratory Informatics Workflow Introduction*—The laboratory informatics workflow model shown in Fig. 6 provides a generic representation of the workflow in a typical analytical laboratory. Workflow steps shown are aligned with the system functions described elsewhere in this guide. The purpose of the workflow diagram is to depict the laboratory

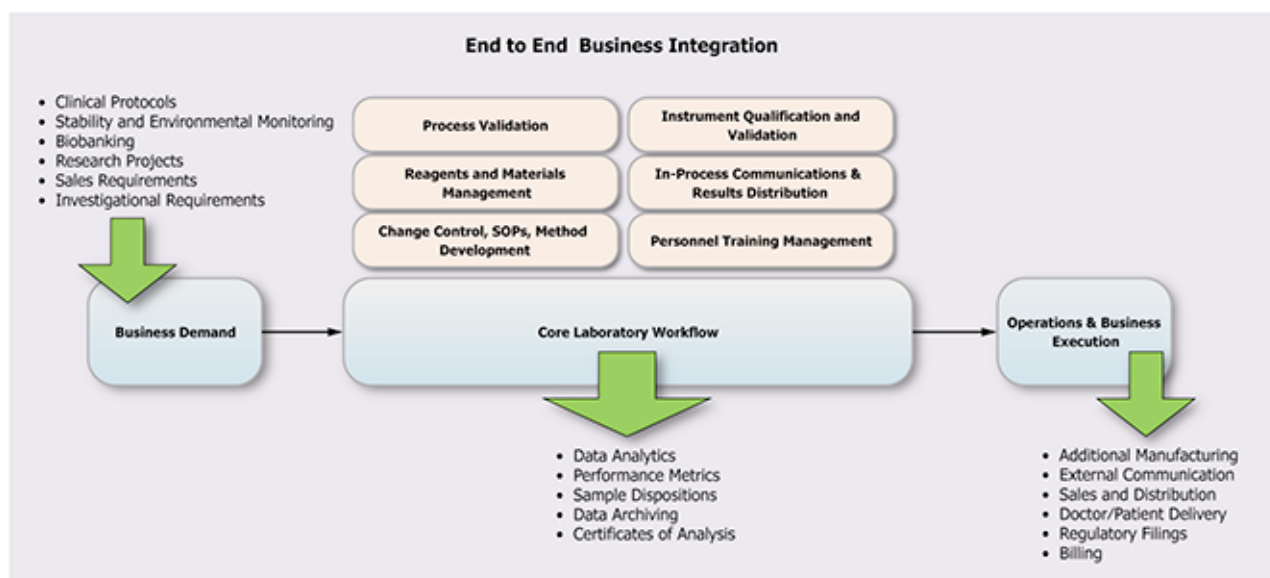


FIG. 5 End-to-End Business Integration

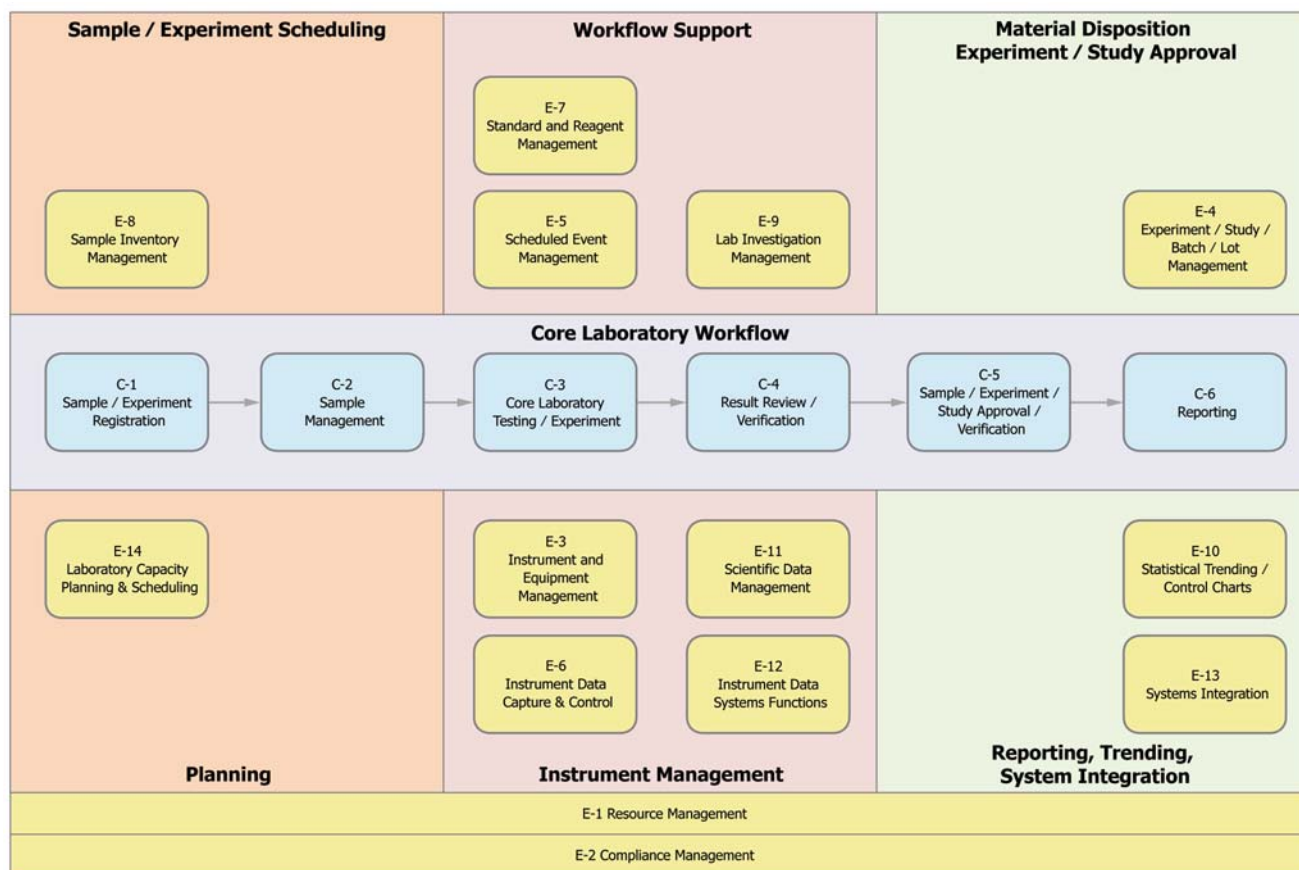


FIG. 6 Laboratory Informatics Workflow Model

informatics functions and interaction points with typical laboratory work processes. Specific laboratory workflow requirements may vary widely from one laboratory to another, as well as from one industry to another. For example, in a research and development (R&D) organization, the laboratory process typically starts with experimental design. Information input for experimental design may come from literature or from computational models based on tools such as computational chemistry and bioinformatics. However, before implementing a laboratory informatics solution, care should be taken to define and document completely the unique requirements and workflow model for the laboratory in question. This should include clearly delineating the functional boundaries of the laboratory informatics tool(s) being implemented. To achieve a successful implementation and use of a laboratory informatics solution, it shall be properly configured to support the target workflow and requirements. In addition, there is normally a significant amount of static or master data required to be imported into the system before use. Once configured, the laboratory informatics solution is able to support the laboratory workflow and sample life-cycle processes.

6.2.1 Laboratory Informatics Workflow Model—Defining the correct workflow model for the laboratory is essential to a successful laboratory informatics implementation and deployment. The laboratory informatics workflow model should support the business processes, which vary significantly in the level of flexibility they need to accommodate from industry to

industry. At one end of the spectrum, discovery research requires a highly flexible process to support constantly changing directions and methodologies. In these laboratories, only high-level policies exist in data governance. There are usually limited SOPs on how scientists conduct their research. The data model shall be able to capture different ways of describing a data entity. In the middle of the spectrum, many laboratories opt for data models that are procedure-centric (that is, test methods are defined from approved external procedures and SOPs) in which the requestor selects the appropriate tests based on knowledge of which procedures are appropriate for the sample in question. This model relies on the experience of the user and has great flexibility for the R&D laboratory or laboratories in which a wide variety of samples are submitted. At the other end of the spectrum, the business process for quality control laboratories is sample or product specific. That is, a suite of “approved” tests are bundled together and typically always applied to one sample type or product. This model removes the dependency upon the requestor to select the appropriate tests when submitting the sample for analysis and improve compliance to testing protocols.

6.2.2 Types of Data—The technology used by a laboratory informatics solution varies with each vendor and platform. However, laboratory informatics databases are typically divided into two broad areas: (1) static or master data in which descriptive information is defined to prepare the system for use (for example, users, locations, profiles, tests, calculations,

specifications, and related information; commonly found in “look up/reference/dictionary” tables) and (2) dynamic data in which sample and result/determination information is stored as samples are logged and results are entered. The terms “static” and “dynamic” represent a general characterization of laboratory informatics data, reflecting the frequency of change. The laboratory informatics implementation team should assess the current full scope of laboratory information and workflow to gather the data necessary to set up the system to support their laboratory processes.

6.2.3 Statuses—Laboratory informatics solutions are generally capable of maintaining information on the status of various items or activities as they progress through their associated workflows. Status information is normally updated automatically as each laboratory transaction takes place. This status information provides a representation in the system of the progress of items and activities through their defined workflows or life cycles. This ability to track status of work items in a central solution is one of the essential benefits of managing a laboratory with informatics solutions. For example, experiments, orders, samples, and individual tests or determinations are all items that may have a status associated with them in the solution to indicate where they sit in the progression of their workflow. In addition to using statuses to track progress, laboratory informatics solutions may also allow certain system actions to be taken or not taken by end users based upon the status of an item. For example, some solutions may be configured such that samples or orders should not be approved until all tests and samples are reviewed. Individual functions/workflows may also be configured to have certain actions automatically triggered upon a change in status. For example, some solutions may be configured such that samples are automatically moved to an approved status once all tests on that sample are reviewed and have passed defined acceptance criteria. Statuses are also useful within these solutions to filter information or objects for actions. For example, some solutions may be configured such that samples can be made available for testing or placement on an electronic worklist only if they have been received in the laboratory as indicated by their status.

6.2.4 Data Load and Migration—A laboratory informatics solution is capable of maintaining a broad range of business and laboratory data required for the effective operation of the laboratory. Laboratory informatics solutions contain data that not only reflect the current operational state of the laboratory but also historical information on past performance and events. When implementing a laboratory informatics solution in a previously manual environment or replacing a legacy electronic system, it should be determined how much and what type (if any) of historical data should be carried forward (that is, loaded, “migrated,” or re-entered) into the new laboratory informatics solution. Static data are normally loaded into the laboratory informatics solution (either manually or through electronic means) as part of preparing the system for use. The decision on how to deal with historical dynamic data should be evaluated on the basis of risk, value, specific customer requirements, and record retention policies. Appropriate strategies for dealing with this data include migration, preservation, and archiving. In cases in which a new solution is replacing an

existing laboratory informatics solution, it may be possible to use technical means to migrate dynamic data from the legacy laboratory informatics solution to the new target solution. Migration of data needs to be carefully analyzed and planned, including considering the hardware and software differences (such as database type) of the originating and destined locations. The plan should include steps to verify that the data are successfully migrated to the new database without error.

6.3 Sample and Test Life Cycle:

6.3.1 Sample Registration—Sample registration may precede or follow physical sample collection and arrival in the laboratory. The initiation of a request for testing/sampling generally starts the sample workflow process. Sample requests may include manual forms, electronic forms, phone requests, web requests, process-driven requests, time- or calendar-based requests, ad-hoc requests, and system-generated requests. Information obtained from the sample request should include biographical, client, test, and safety information. Some laboratory informatics solutions allow both the laboratory to pre-log or post-log samples and the client to pre-log samples through a self-service request interface. In some business processes, the sample request needs to be approved by project leaders before the samples are retrieved for the study to ensure that appropriate samples are allocated to high-priority projects.

6.3.2 Sample Identification—The laboratory informatics solution should assign a unique number to each sample registered (that is, submitted for testing). The unique number can be a system-generated sequential integer or a user-defined sequence. Multiple samples submitted together for testing should be logically “linked” in the laboratory informatics solution (for example, all samples for a particular lot or research project). The system will normally provide functionality to capture descriptive information about the sample(s) such as who submitted the sample(s), cost, sample description, storage conditions, and what tests are to be performed on the sample. Other information may also be important, such as the priority of the tests, what level of accuracy and precision of testing is needed, what hazards the sample might present to laboratory personnel, what approximate levels of components are expected, and what should be done with the sample when analysis is complete.

6.3.2.1 A confirmation report is often issued (sometimes emailed) to assure requestors that the system accepted the sample request and may accompany the physical samples as they are delivered to the testing laboratory. Often, laboratory informatics solution statuses are updated for the sample/order and may be used to record the fact that an order was made (for keeping operational metrics) and when it was made so the system can track the time intervals for the remaining steps of the process. This will also allow laboratory management to determine turnaround time, sample status, and various overdue conditions.

6.3.3 Sample Collection—Sample collection may be a manual, automated, or robotic process. Sample collection logistics may become more efficient by having the laboratory informatics solution print collection lists and generate labels (for example, bar codes) for the sample containers. Sample collection can precede or follow sample registration as defined

by the laboratory's workflow. The laboratory informatics solution can provide information on how to collect samples, specific sample plans, container and preservation requirements, safety information [such as safety data sheets (SDSs)], sample storage requirements, and sample routing information. Chain of custody for samples is often tracked by the laboratory informatics solution, generally for location and status information. Chain-of-custody may be required to provide documented evidence of control and traceability of sample containers and their contents. Examples of situations in which chain-of-custody requirements may be required include handling of controlled substances, pieces of evidence (forensic) supporting legal court cases, or radioactive materials. Note that this functionality may not have all legal chain of custody requirements for specified sample types as defined by governmental or law enforcement agencies. The implementation team should review these requirements carefully during the planning/implementation phase.

6.3.4 Sample Receipt—The physical receipt of samples in the laboratory may be recorded in the system and may also include initial sample checking and labeling. Sample orders or groups of samples may be reviewed against customer or project sampling requirements. Additional information such as the number (or quantities/amount) of samples received and the arrival time may be recorded and the status of samples may be updated for the sample/order from “logged” to “received.” Where collection lists are used, a “missed sample” report should be used to indicate those samples that were not received as expected.

6.3.4.1 The laboratory informatics solution may be configured to specify the aliquot requirements for a sample based on the tests to be performed. Upon sample receipt, any issues, such as an unexpected color or physical state, may be noted and recorded within the sample record. The laboratory informatics solution should be flexible enough to allow preliminary sample treatment, such as addition of a preservative, to be performed and documented.

6.3.4.2 Store/Retrieve Sample—An often overlooked benefit of using a laboratory informatics solution is the ability to manage inventories for reference samples, laboratory reagents, standards, QC samples, time-based samples (shelf-life stability), and sample storage requirements in addition to normal samples. Inventory functions may provide critical business information with respect to resource and consumables management as well. This could include such information as expiration dates, vendor information, restock quantities, and so forth.

6.3.5 Sample Distribution—Distribution processes often include important laboratory informatics solution functions such as work lists, resource allocation, sample routing, and sample storage location tracking or chain of custody.

6.3.5.1 The laboratory informatics solution should provide a listing of all the tests that are to be performed, the amount of material required, and where samples are to be sent. The date and time of sample distribution is important since it designates when the sample becomes available to the various laboratory workstations for analysis. Sample status may be updated to indicate samples are available for analysis at this time as well.

6.3.6 Work Assignment—Once samples arrive in the laboratory, the work should be scheduled and allocated against available resources, people, or equipment, or combinations thereof. Resource availability and management may be handled through the laboratory informatics solution, if configured to capture this information. By utilizing the laboratory informatics solution appropriately, resources may be forecast, allocated, and tracked to improve the overall efficiency of the laboratory. The laboratory informatics solution may also be configured, in some instances, to group samples automatically into runs or sequences and schedule work (tests) for each sample/order, as well as be configured to allow authorized users to perform these functions manually.

6.4 Analysis:

6.4.1 Sample Preparation—Most samples require some preparation before analysis. The laboratory informatics solution may be configured to provide sample preparation directions for these preliminary processing and sample preparation steps; however this information may also be available in the form of standard operating procedures, technical documents, or work instructions stored externally to the laboratory informatics solution. In addition, it may be configured to capture who prepared the sample and when sample preparation was completed.

6.4.2 Sample Analysis—Analysis activities will vary from laboratory to laboratory. Depending upon the laboratory's requirements and data model, a common objective is to record as much of the information gathered during this phase as possible within the laboratory informatics solutions. In general, the analysis phase contains the following subparts:

6.4.2.1 Perform Test—Test results/determinations are the main output of the analysis process. Intermediate and final test results for the samples, standards, and their associated QC samples may be reported out in hard copy, electronic formats, or both. In addition, the measurement process may produce values for additional internal blanks, standards, and instrument self-checks. The definition of what is the laboratory's “raw data” and what needs to be retained for legal evidence may be defined differently for each client or agency involved and should be a fundamental part of the data model design process. The laboratory informatics solution should provide a means of contemporaneously capturing all activities undertaken during test performances, ensuring accuracy, traceability of actions, originality, and authenticity.

6.4.2.2 Retest Loop—Retests can be initiated at multiple points in the laboratory informatics solution workflow. A retest is defined as one or more additional determinations on the original sample/order container. These retests would normally be ordered if a given test was suspected to fail for reasons that may include failed quality control parameters, instrument malfunction, or technical judgment. The laboratory informatics solution should document each retest along with an appropriate justification.

6.4.2.3 Resample Loop—Resamples can also be initiated at multiple points in the laboratory informatics solution workflow. A resample is defined as one or more additional samples. The laboratory informatics solution needs to establish forward and backward links to samples that are added by way of the

resample loop. These resamples would normally be ordered if a given sample was compromised during initial testing (for example, sample tube dropped or contents spilled). They also would be reordered when a sample was suspected to have failed for reasons that include (1) insufficient amount of the sample was available for a retest, (2) technical judgment that the original sample was not appropriate for the test performed, or (3) test failure shall be confirmed/overcome.

6.4.3 Data Capture—The results of the analysis should be captured within the laboratory informatics solution. The amount and type of supporting data to include with the result data should be carefully evaluated and defined during the data model design, allowing for reconstruction of the critical activities that led to the test results. When a test result/determination is captured, the statuses of the sample/order and result determination should be updated. The associated date/time records should also be captured so that statistics of work accomplished and test progress can be tracked. The laboratory informatics solution should have validated electronic audit trails that record information about each transaction, both for initial entries as well as modifications to entries.

6.4.3.1 While data capture may be a manual process in which results and parameters are manually entered by an analyst, the true power, efficiency, and data integrity benefits of laboratory informatics lies in electronic data capture. This can involve automated capture of instrument data files and data from simple devices, automatic extraction of information from one part of the laboratory informatics implementation and transfer into another one, and automatic issuing of reports. In cases in which instruments are bi-directionally interfaced to laboratory informatics solutions, the sequence of unknown samples and control standards may be transferred to the instrument to streamline instrument setup before analysis.

6.5 Analysis Review:

6.5.1 Test Result Review and Interpretation—A laboratory should require that each test result undergo one or more levels of documented review and interpretation before release of results to the customer. To facilitate this process, the laboratory informatics solution may be configured to document multiple levels of review. The original sample result would typically be reviewed and interpreted by the primary analyst for any anomalies associated with the performance of the test method. This review can be documented in the laboratory informatics solution. Laboratories often require that results be reviewed by a second independent qualified person (this is industry specific and dependent on regulatory requirements) to ensure that the tests were properly executed, documented, entered, and interpreted.

6.5.1.1 To help in this process, the laboratory informatics solution may indicate the unusual (for example, out-of-trend) or out-of-specification results as flagged for further evaluation. If normal values are known for the substance being tested, they can be displayed. Results outside of normal can be highlighted or displayed separately for closer review. The laboratory informatics solution can be configured to enforce laboratory SOPs that require the reviewer to be a different person from the tester. Corrections or changes to laboratory informatics solution data made during the verification step should be audit

trailed and require authorization with a change comment. Audit trails should contain original data and all changes to the result record including date/time of change, who made the change, and the reason for the change. Electronic signatures may be used to confirm changes in status to the laboratory informatics solution records if the regulations or guidelines require. Not all laboratory informatics solution implementations require audit trails. The laboratory informatics solution implementation team needs to determine whether audit trails are important, what information should be retained and reviewed, and whether reasons for changes should be recorded during the data model design phase. Management may need to know when results are verified, another milestone in the progress of a test/sample/order.

6.5.2 Test Result Analysis—In R&D organizations, laboratory test results are the basis of decision making in designing the next round of experiments. To facilitate such decision making, advanced visualization capabilities may be provided by the laboratory informatics solution. Advanced data visualization allows the users to display large datasets that are often high-dimensional. Trends may be calculated or identified visually by manipulating the visualization. In cases in which the laboratory informatics solution does not support advanced visualization, it is desirable to integrate the laboratory informatics solution with a best-of-breed data visualization tool.

6.6 Sample Disposition:

6.6.1 A laboratory generally requires that samples undergo a documented review/approval process to disposition the samples, indicating that they have been evaluated against established criteria. The laboratory informatics solution can be configured to document the review/approval process. This process varies across industries but in many cases requires data reviews by multiple groups and levels (for example, peer or supervisor). The laboratory informatics solution should be capable of managing such scenarios, including multi-level reviews. Since the laboratory exists to generate information for the parent/client organization, the laboratory may organize and configure results to make interpretation and decision-making easier. Analysis is frequently done to confirm quality or properties of a material. In this case, material specifications may be entered into the laboratory informatics solution so that results can be checked against acceptable values to determine whether the sample meets/does not meet specifications. Electronic signatures may be used to document the sample review/approval process and update the sample status in the laboratory informatics solution records. In addition, certain industries/regulations prohibit final sample approval by the analyst who performed the test. Restrictions of this nature need to be identified during the implementation design phase so that the laboratory informatics solution configuration will support the constraint.

6.6.2 The output of the review/approval process is verified data and may be in the form of data reports, test verification reports, COAs, or direct process control actions. Often, the interpretation function coincides with the reporting process. In many laboratory informatics solutions, data are interpreted in a reported format either in electronic or paper forms. Sample disposition will also allow generation of comprehensive audit

trails and custody information for the sample through the laboratory informatics solutions.

6.6.3 Result data itself can undergo a separate evaluation and disposition process from the sample. In some industries and research organizations, the pass/fail (or approved/rejected) status of an individual result data point is captured, yet the overall sample is approved because the science generating the result value is sound. In such cases, sample disposition and analysis result evaluation are two distinct functions to be handled in the laboratory informatics solution. This is a key element that the implementation team needs to incorporate into the data model design.

6.6.4 Samples are frequently associated with a parent level entity such as a lot, batch, stability study, project or clinical study. Sample disposition may also trigger a review/approval process for these parent entities, and their status may get updated as a result. During implementation of the laboratory informatics solution, the team should ensure that the synchronization between these dependent entities is handled as per the business requirements.

6.7 Sample Disposal and Retention:

6.7.1 *Disposal of Samples*—Sample disposition is also the trigger for sample disposal or sample retainment processes. The proper documentation of sample disposal following analysis is an important responsibility of the laboratory, and the laboratory informatics solution may be used to record any special disposal instructions and track final sample disposal.

6.7.2 *Retention of Samples*—Depending on the industry, the sample may need to be physically retained for a predefined period of time. It may also be sent back to the client wherever sample testing is done in a collaborative, outsourced environment. The laboratory informatics solution can often be used to handle different scenarios for sample disposal, retainment, and shipment.

6.8 Reporting:

6.8.1 Following verification, data reported to the customer may include test results (including quality control data), auxiliary data such as sample demographics, accompanying pass-through data that is not “generated” by the laboratory, and other data necessary for data evaluation such as sample characteristics (for example, pH or temperature). Reports serve an important function by organizing the resultant data in a predefined or specified manner, or both, for both internal and external use. Modes of data transfer can include hardcopy reports, electronic data deliverables, and web-based systems. Report generation should be flexible enough to accommodate the different reporting needs of individual clients. Laboratory informatics solution vendors should provide basic formats for the most common hard copy and electronic reporting formats. Conventional COAs are commonly supplied as an example of a hard copy report. Many clients are now relying on the use of various electronic formats that support the transfer of data from the laboratory informatics solution database to the client’s database using electronic transfer formats such as XML. In some cases, the data is maintained in a data warehouse and accessed on an as-needed basis, rather than actively sending reports.

6.8.2 Reports can be developed for internal laboratory management to understand better laboratory requirements and performance. The volume of incoming samples provides visibility to the resource needs of the laboratory from both a personnel and instrumentation/materials perspective. Laboratory informatics statuses updated throughout the sample life cycle for the sample/order creates time stamps at various points in the process that can be used to highlight cycle times/turnaround times and laboratory throughput. Throughput data may include the number of samples processed at each workstation by hour of day, shift, or day of week and can help identify peak demands, roadblocks, and other potential process problems. Audit reports by sample can indicate the processing time for a test/method in an instrument and provide visibility to laboratory productivity. Turnaround times also show the laboratory’s responsiveness to customer needs, while work in progress (WIP) and overdue results help managers to determine how well the laboratory is responding to current demands. Reports on the number of samples retested and resampled can give an idea of the training needs of the personnel and other workflow problems.

6.8.3 Instrument utilization reports help in understanding instrumentation requirements for the laboratory. This includes instruments operated per day (in hours), the number of test methods used, and the types of samples received. Instrumentation usage information can aid in developing test costs for standard and nonstandard analyses. The total number of tests and test methods can be used to estimate the consumption of reagents and supplies for planning purposes.

7. Laboratory Informatics Infrastructure, Integration, and Interfaces

7.1 *Cloud Computing*—The innovation of cloud computing has helped facilitate a digital revolution grounded in high-performance computing, big data analytics, and AI, all with significant potential to impact laboratory productivity and capability. However, as cloud-computing technologies rapidly evolve, they create new demands and challenges for laboratories to maintain and control security, privacy, data integrity, system integrity, and regulatory compliance. Laboratory informatics solutions can be deployed on a computing infrastructure installed and supported entirely by a company’s internal resources. Alternatively, companies can now choose to use computing services purchased from external suppliers and delivered over the public internet or a private network as a service (in the cloud). The benefits of using a cloud-computing service model to handle laboratory informatics tools include lower start-up costs with ongoing expenses (rather than upfront capital purchases), flexibility in the number of licenses deployed, scalability in performance, and guaranteed service levels for the application. Ideally, organizations can also increase productivity and lower the total cost of ownership (TCO) through the reduction of capital equipment and supporting infrastructure (for example, cooling systems) expenses while only paying for services used. Some of the downsides of cloud-based laboratory informatics offerings include more complicated compliance issues, upgrade requirements (for

example, web browser versions, desktop applications, operating systems, database environments), user and data security considerations, and system integrations.

7.1.1 Cloud-Computing Deployment Models—The public cloud is the most well-known deployment model, essentially offering a cloud provider’s resources or services, or both, for a public audience over the internet. In some cases, users can access a public cloud via “direct connect” or dedicated network services, which promise to streamline networking and reduce costs. The private cloud is operated for a single entity, meaning personalized on-premises or offsite data centers owned by an entity provide scalable, self-service resources to meet its needs. A private cloud does come with added costs and considerations, however. A hybrid cloud is any combination of public/private cloud with a minimum of one public and one private cloud. Enterprises may have resources spread over several physical locations (private clouds) and use different public cloud resources to provide a broad spectrum of services to its customers, employees, and business partners. Consideration of cloud workloads and responsibilities is required, however, to determine which business unit or vendor will provide and manage identified computing services; when those services will operate; where data will be stored; and how different services will integrate with business applications, customers, and employees. Additionally, public multi-tenant cloud systems contain special considerations that can be problematic around governance and timing of upgrades or changes in regulated environments in which long lead times are generally required to support change control, training, and impact assessments.

7.1.2 Cloud-Computing Service Models—Many types of public cloud-computing service models are available to laboratories, though three broad categories stand out. Infrastructure as a service (IaaS) gives laboratories the option to use a service provider’s large pools of data center resources (that is, information technology infrastructure) to run virtually operating systems, applications, and, in some cases, networking components. The laboratory also has control over data storage. Platform as a service (PaaS) is similar to IaaS, though the service provider controls the operating system, networking, and storage. The laboratory is able to handle application deployment and, in some cases, environment configuration settings. Software as a service (SaaS) is even more controlled than PaaS in that the service provider controls most aspects of the installation, with laboratory personnel simply having access to application software and databases. In some cases, user-specific application settings may also be controlled by the user. Laboratories can also tap into other services over the cloud, including storage as a service (STaaS; cloud-based data storage), security as a service (SECaaS; cloud-based internet security), data as a service (DaaS; cloud-based data management and distribution), and test environment as a service (TEaaS; cloud-based, on-demand software testing environment).

7.1.3 Key Considerations of Cloud Computing—Cloud platform providers use a stacked layer architecture of services, allowing cloud architects the ability to select the level of responsibility for managing the cloud assets. For example,

deployment of virtual servers in virtual private networks affords complete control over the server configuration at the expense of increased support and management costs. Support and management may be offered as a mixture of in-house and externally managed services to provide operational support, but policies are still required to be written and demonstrably followed by all parties involved, in a coordinated manner. In the case of applications or services used in highly regulated industries or where sensitive data are processed and stored, it may be necessary to host these systems on premise or on an internal cloud to ensure their security. In other scenarios, applications or services may be used in a shared hosting environment that is managed by the cloud platform provider. When planning the laboratory cloud architecture, the following areas should be considered:

7.1.3.1 Architecture, design, and security are at the forefront of decisions regarding cloud deployments. Full realization of the benefits of the cloud will depend on the degree of integration between private and public infrastructures and the ability to apply safeguards and controls to protect intellectual property and communications. Particularly in hybrid deployment models, the location of users, devices, and private cloud data centers, as well as the location of public cloud data centers, shall be considered. Consider deploying applications to different cloud data centers. The identity and access management (IAM) framework for managing digital identities is another key architectural aspect. Laboratory employees, customers, business-to-business (B2B) systems, and devices will need access to cloud assets, and organizations will need to control and configure what and when cloud assets and data can be accessed and by which identities. Consider deploying applications in containers to isolate the resources with which a user will have access.

7.1.3.2 Governance represents another major consideration for laboratories. Regardless of where the computing assets reside, the laboratory is still responsible for the protection and responsible use of computing assets and data gathered and stored in those assets. Rules of governance described in this document still apply for cloud-based computing assets. If third-party vendors are providing SaaS services integral to B2B or laboratory operations, controls and checks shall be in place to ensure that third parties are meeting security, quality, and compliance requirements. Additionally, periodic audits should be performed to verify continued compliance. Using a risk-based approach, organizations can determine the best level of protections.

7.1.3.3 Management of cloud-based services requires the same level of rigor as in-house systems. In-house personnel may have to be educated in new technologies and toolsets to manage systems effectively in these new service delivery models; managing that increased complexity, particularly in hybrid environments, is a key success factor as the use of cloud services increases. Laboratory data and operations may depend on a set of disparate services managed by combinations of internal and external resources. As issues arise, having a clear understanding of which business unit owns the workload for a given step in the process shall be understood to have an efficient support response. Service level agreements shall be

used and match business needs to keep operations continuous. Factors such as when and where the workload runs should be considered.

7.1.3.4 Availability of cloud-based services should be assessed. A production system may require availability in different regions of the world on a daily, 24-h basis, while smaller laboratories may require a less stringent availability. Determine the level of availability required for the laboratory, understand how the provider defines that availability level, and discover what tools the service provider is using to improve the availability of its service (for example, checkpoints, redundancy model, and data replication).

7.1.3.5 Scalability of cloud services gives laboratories options they did not have before; rather than purchase more IT resources than required, companies can pay only for what they need, when they need it. Be aware of the options the provider has in helping to monitor service (response time), make service-level changes, and control costs.

7.1.3.6 Dynamic performance measurements and evaluations are important in laboratory informatics cloud implementations. Cloud infrastructures provide for increasing the number and power of virtual processors to address changes in demand (that is, number of users). Performance bottlenecks can be tied to an application and its design or deployment. Security and regulatory requirements may require significant encryption efforts, and if the application was not designed well to take that into account, performance may be impacted. Additionally, monitoring tools and even web browsers interacting with the cloud application may impact performance.

7.1.3.7 Encryption schemes are available to transform and encode data before it is uploaded to its intended storage point. Encryption should be available for securing data and cloud-based messaging between all endpoints, though the method may differ depending on the cloud provider. Be clear on how the provider handles encryption and if it meets necessary regulatory requirements.

7.1.3.8 Privacy concerns demand that laboratories consider what data are stored in the cloud, where the data will be stored, and how that data will be moved and kept secure. In particular, understand how, when, and where the provider is moving sensitive data. Data may be collected and stored in different geographical locations in an attempt to ensure that all users and systems have timely access to data. Yet special considerations shall be made of ePHI, as national and international laws may differ, causing problems with compliance; make sure ePHI is not excluded in the terms of service. Additionally, be sure the service provider makes it clear in its agreement what party is liable in a noncompliant scenario.

7.1.3.9 Compliance auditing is required to ensure data assets are not being compromised. This includes analyzing audit trails, security logs, and changes to cloud configurations. Additionally, any audit data, the data collector, and the log server should also be attestable. Determine the provider's capabilities, including what auditing capabilities are built into their infrastructure and what analytical tools they may be using. Tools need to be in place to support regulatory audits, for example, by boards of health for regulated health care applications.

7.1.3.10 Service level agreements (SLAs) are vital and should be considered carefully. A solid service level agreement will cover the important considerations of a cloud deployment and will delineate who is responsible for what. If moving from an in-house services model to a managed services model, extra time should be taken to understand the changes that will be introduced. Ask questions when in doubt. Many of the considerations already mentioned should be addressed in some manner in the SLA.

7.1.3.11 Mobile devices are playing a more significant role in the expansion of business and services. For example, as data sources they can be used to collect data in clinical trials, physical and chemical sampling, image and video data, and customer surveys. Modern mobile phones and tablets are capable of connecting to cloud-based services to send and receive data, participate in collaborative workspaces, and communicate electronically via email and instant messaging. Be mindful that devices connecting to cloud-based services should be authenticated and identified, and necessary checks and controls should be applied to ensure that mobile devices don't compromise data or compliance.

7.1.3.12 Adaptation, migration, and archival plans are pivotal for a successful cloud deployment. Business goals, workflows, workloads, software, infrastructure, legacy integration, data types, data structure, data relationships, and data destination all play a role and shall be assessed early on in any adaptation and migration plan. Additionally, determine any archiving strategies that may be associated with the migration, including whether the data will need to be archived warm (near-production access levels) or cold (long retention with lower response times).

7.1.3.13 High-performance computing (HPC) in the cloud is still in relative infancy, but cloud providers are beginning to support HPC environments, albeit with slightly different methods. A major consideration when working with the few service providers offering HPC support is how terabytes or even petabytes of data will be transferred to the service provider. Providers such as Amazon Web Services (AWS) offer physical data transfer services in the form of storage appliances of various sizes. Once the data has been moved and uploaded to the cloud by the provider, data updates can be managed incrementally by the laboratory.

7.2 *Internet of Things (IoT)*—Laboratory devices, acting as autonomous agents that connect to the cloud, provide additional ways to collect and process laboratory data and further compound the challenges in establishing and maintaining compliance requirements. The IoT, occasionally “cyber-physical system” for a more autonomous and networked sensor-based system, as a concept assumes that more than just desktop computers and mobile devices can connect to and perform functions on the internet. The idea has been touted in the consumer product realm since the late 1990s, and the underlying technology has in time made its way to a variety of laboratories, including clinical and pharmaceutical research laboratories. The IoT has the potential to improve laboratory automation and increase the likelihood of making valuable research discoveries. Yet how internet-connected devices move

and store data—including sensitive ePHI data—adds complication to the IoT’s implementation and evolution, particularly in relation to laboratory informatics.

7.2.1 Device connectivity is at the core of the IoT. Devices traditionally connect to a laboratory informatics solution either directly via a data connection or through middleware but with geophysical limitations such as cable length or wireless signal strength. IoT-enabled devices, however, can be located in laboratories in several campus buildings or even laboratories separated by thousands of miles and still surpass those geophysical limitations, linking together over the internet while all being monitored from a central online dashboard. This sort of connectivity gives laboratories more opportunities to ensure instruments and equipment are functioning as expected, even when nowhere near the laboratory. As the technology behind the IoT expands, an increasing number of instruments and equipment will be manufactured to have connectivity to a main server or cloud service. However, older devices may still be connected using middleware (a hub) to allow bidirectional communication. Messaging protocols and buffers should be considered in any setup. Determine if data message protocols are data-agnostic, if they limit message size, and what the buffer limit is should an application or device fall behind live data processing.

7.2.2 Security of IoT devices is, as with cloud computing, a vital consideration. The use of IoT devices—especially in the cloud environment—increases the attack surface of an enterprise, potentially exposing intellectual property or compromising system and data integrity. If not properly secured, IoT-connected devices acting as laboratory data sources may spread malicious payloads through inter-device messaging to the laboratory or enterprise network, even potentially siphoning off valuable data to the cloud. As such, proper device management is critical. To avoid such attacks and breaches, all people, devices, and systems shall be authenticated (using device IDs, unique access tokens, or some other method) and, once identified, access to computing and data resources shall be controlled and monitored. These security considerations are built in at various layers, from the device and communication network to the server where the data will be stored and managed. However, manufacturers are not solely responsible for ensuring these and other security features are addressed in medical equipment; the end user also has responsibility in managing any necessary software and firmware updates/patches to ensure compliancy of equipment’s original validation. Be sure an IoT device allows for such security patches, updates, reboots, and factory resets, preferably able to be performed remotely.

7.2.3 Sensor data collection, management, and use is a typical end goal for laboratories. Regulations and best practices dictate how environmental monitoring should be performed inside and outside a laboratory, and sensors play an increasing role in that monitoring. From vaccine manufacturers to medical device testing laboratories, processes shall be put in place to ensure environmental conditions do not threaten sample and instrument integrity. Sensors placed in sample freezers, incubators, and hypoxic chambers can evaluate temperature, humidity, carbon dioxide levels, or just about any other type of

measurement and transmit that data real time when connected via the IoT. However, intelligent policies shall be put in place on how this sensor data is monitored and audited, including setting up alerts properly for when instrument failure or unauthorized access occurs and ensuring error logging is active. The Open Web Application Security Project’s (OWASP) IoT Framework Assessment²⁸ (discussed further in Section 10) evaluation criteria provide a solid starting point for building such policies.

7.2.4 Quality of data generated from IoT devices shall be monitored to get the most out of an IoT-connected laboratory. Setting up alarms and alerts for when an unwanted test result or out-of-range measurement occurs will certainly aid in the quest for quality. However, such data at times may be missed by the machine. This requires data analysts to be vigilant as well. Should an analyst spot unusual data, they will want to contact quickly those responsible for managing the related device(s) so that further troubleshooting can occur. Additionally, IT personnel—or even vendors servicing IoT equipment—should communicate with laboratory testing personnel before and after they make upgrades and changes to equipment; odd data results may be a product of that process.

7.2.5 Predictive maintenance and condition monitoring give laboratories implementing IoT solutions the tools for improved maintenance scheduling and shorter periods of laboratory downtime. An IoT-connected device can be monitored for changes in acoustics, vibration, fluid levels, temperature, and more. Should these measurements begin to trend outside the set of failure parameters, a laboratory can take action early to replace or repair the instrument before its actual failure. However, this requires a thorough analysis of what that instrument’s failure parameters should be and how an abnormal state can accurately be determined. The tools to react shall also be in place, from personnel alerts to reaction engines that can shut the instrument down when an anomaly occurs. Also determine what sort, if any, of an uptime guarantee can be made by the vendor if predictive maintenance is applied.

7.2.6 Embedded programming was once the domain of skilled software engineers using programming languages such as Assembly and C or of electrical engineers designing specialized circuit boards. Embedded device programming and configuration is now accessible to school-age children using high-level languages such as Python and Java to develop games, plant-watering systems, and motion detectors with video capture. Laboratory researchers, including bioinformaticians, are also among those working with and developing applications for embedded systems, mainly out of necessity for the collection of scientific data. Consider creating a sandbox for embedded development using tool sets available through cloud-based IoT providers to facilitate innovation in the laboratory.

7.3 Hardware Infrastructure:

7.3.1 *General Requirements*—The hardware platform is an important factor in the deployment of all laboratory informatics solutions. This platform includes the computing requirements

²⁸ Available from OWASP Foundation, 1200-C Agora Drive, #232, Bel Air, MD 21014, <https://www.owasp.org/>.

(processor capability, memory, disk space) and the network requirements (bandwidth, security, instrument connectivity, LAN/WAN). The architecture of the hardware platform should be driven by the requirements provided by the supplier of the selected laboratory informatics applications(s) and of the business it supports. Acquisition and deployment of the actual hardware devices can be staged to match the software implementation schedule.

7.3.2 Key Considerations—Infrastructure design for laboratory informatics solutions is often driven by size and scope of the businesses and users supported. For example, small single-site implementations may require only modest infrastructure, while large multisite implementations may require much more thought and planning. In the case of these larger implementations, teams shall decide whether a centrally hosted single-instance design or a regionally distributed design makes more sense for their situation. When evaluating the infrastructure design requirements, the following capabilities and features should be considered:

7.3.2.1 Distribution of User and Sites—For large implementations involving multiple sites, the geographic distribution of the sites, their proximity to the data centers hosting the infrastructure, and requirements regarding data sharing and aggregation are just some of the factors to be considered.

7.3.2.2 Concurrent Users—The number of laboratory informatics solution users and their geographic distribution is important, especially when considering system performance at peak times of the day and when defining the appropriate infrastructure distribution model.

7.3.2.3 Number of Records Created per Year—The number of samples, average number of tests per specimen, and the amount of data generated during the testing is important in estimating the system resource requirements and, especially, planning for adequate online storage of records.

7.3.2.4 Online Storage—This includes the number of records to be maintained online as well as the data generated during testing and associated audit trails. Audit trail records can often consume large volumes of online storage.

7.3.2.5 Archival Storage—Both the amount of storage required and the length of time records are required to be stored are important factors in determining the requirements for archival storage of laboratory data. Data retention requirements are often driven by regulations or internal legal policies, or both, and may represent a significant technical or cost burden to the laboratory. If the cost of maintaining the infrastructure and expertise needed to store large volumes of laboratory data is prohibitive, archiving data to the cloud may be a more cost-effective option.

7.3.2.6 Number and Type of Reports and Labels Required—The number of reports to be generated during a work day, the geographical location of the printers, and even the types of printers may have an impact on system load and performance and should be considered in the infrastructure design.

7.3.2.7 System and Instrument Interfaces—The geographic location of interfaced applications and instruments, bandwidth requirements of the interfaces, operating system requirements,

network security, and separate or shared corporate network should be defined and considered when determining infrastructure requirements.

7.3.2.8 Network Bandwidth—Latency and network speed are often the limiting factors in overall system performance with centrally or regionally deployed infrastructure and may need to be evaluated on a site-by-site basis. For example, it is not uncommon in a centrally or regionally distributed model for smaller sites with low-capacity wide area network (WAN) connections to their corporate networks to experience degraded performance compared to other sites.

7.3.2.9 Application Load Balancing—The software architecture determines how well the laboratory informatics solution can be balanced over multiple processors. The ability to add hardware components (hardware scalability) to meet demands is important.

7.3.2.10 Network Security—Systems accessible from the public internet need additional measures to support the appropriate level of security to secure them from cyberattacks and data breaches. Instruments connected to the network shall also meet security requirements.

7.3.2.11 Distributed Computing—Global application deployment requires support for computing and connectivity across multiple regions and continents. Systems may be designed as single- or multiple-instance solutions and may use on-premise or cloud-based infrastructure services.

7.3.2.12 Data Backup Requirements—The criticality of the data and the difficulty in recovering lost data should be taken into consideration when determining data backup strategies for laboratory informatics solutions. For example, if recovery time is not a critical factor for a laboratory, then traditional but slow methods for recovery such as restoration from backups may be sufficient. In other cases such as with larger systems that are mission-critical to the core business, recovery time objectives are very short, and infrastructure options such as server mirroring or other data replication approaches are potentially costly but critical to ensure continuity of the business.

7.3.2.13 Data Aggregation and Analysis—The decision whether to implement a single- or multiple-instance infrastructure can be influenced by data sharing or aggregation needs of the business. For example, a single-instance system may make data sharing and aggregation much simpler, but WAN capacity and performance may need to be upgraded to provide adequate performance to end users.

7.4 Database Recommendations:

7.4.1 General Requirements—The database component of the laboratory informatics system requires the greatest level of resilience because of the ever-increasing demands of information exchange between the laboratory and the enterprise. The laboratory informatics system should be based on a commercial database management system that is reliable, effective, and supported by company standards. Commercial relational database management systems can be organized, configured, and tuned to meet a wide variety of usage and performance scenarios.

7.4.2 Key Considerations—The following features should be considered as part of the database platform evaluation:

7.4.2.1 Standardization—Many laboratory informatics applications support multiple database platforms. However, it is critical that the selected solution also conforms to the database platforms and standards currently the organization has in place.

7.4.2.2 Core Design Flexibility—The database schema design provided by the laboratory informatics vendor should be well documented and sufficiently flexible to accommodate the capture and storage of common laboratory data types, including a full range of numeric, date/time, and text data types. Other data types required may include images, multimedia, XML, and other proprietary or binary data files. Administration tools are normally provided through the application to facilitate administration and configuration tasks, such as the maintenance of users, modification of workflows, referencing of information in lookup tables, and the potential addition of user-defined fields.

7.4.2.3 Extended Design Flexibility—The database should support the ability to modify the database structure as needed, including adding/modifying fields, indexes, relationships, and tables. However, careful consideration should be given to the impact on laboratory informatics solution functionality before making these changes, especially where alignment across multiple locations to minimize maintenance costs is desired. The laboratory informatics vendor should provide guidance on how to achieve database modifications in a controlled manner.

7.4.2.4 Data Replication and Backup—The database platform should support industry best practices for data replication and backup in an efficient and expedient manner. These functions are critical to protecting and maintaining information in the laboratory informatics system in the event of catastrophic failure of the primary system. Laboratories that use electronic records should ensure the protection of information captured within the system. Typical snapshot or incremental backup processes run nightly but provide limited protection between backup intervals. Data replication tools within the database layer (and sometimes between the storage layers of, for example, a storage area network [SAN]) provide added protection against data loss by replicating all transactions between data centers or remote servers. Data replication timing (the time interval between the initial data capture and the time when the data is replicated to a different environment) can be adjusted based on the risks and costs of data lost. High-value laboratories typically replicate data within minutes.

7.4.2.5 Multiple Environments (development, quality, and production platforms)—The database platform should support the ability for multiple environments (copies) of the database and migration tools to move objects between environments. Typical implementations include a development environment for code/configuration development, a quality environment for formal testing and master data building, and a production environment for production information. The database environments include application development components for data administration, application customization, and integration. These tools often include the capability to develop stored procedures, views (stored queries), and functions for access by scheduled processes and other applications or application

modules. Consideration of additional database application licenses should be made for systems that reside on separate hardware.

7.4.2.6 Maintenance—The database platform should allow the database administrator to fine-tune the performance and security of the database with functions such as indexing, table space management, and process schedulers.

7.4.2.7 Personnel—In situations in which some or all of the laboratory informatics system will be managed and maintained in-house or by parties other than the application vendor, consideration should be given to the availability of technical personnel with skillsets in the technologies used by the system.

7.5 Laboratory Informatics Application Platform:

7.5.1 General Requirements—Laboratory informatics systems are developed on a wide variety of application platforms, and many standards and programming languages are used that provide adequate features and functions. It is, therefore, important that the vendor provide detailed documentation on the technical capabilities and design of the application architecture. The documentation should allow for the clear evaluation of application flexibility relative to the capabilities needed by the laboratory. An overriding requirement for many organizations is to evaluate the application architecture in the context of its ability to be configured, customized, and integrated with other systems.

7.5.2 Key Considerations—When evaluating the application platform, the following capabilities and features should be considered:

7.5.2.1 Modularity—The application's functionality at the architecture level should be clearly separated into logical modules with clearly defined standard interfaces between modules. These modules can be defined based upon feature groupings or layering of the application architecture (for example, presentation, business logical, data interfaces), or both. Often this is accomplished through the use of object-oriented design techniques in conjunction with other industry standard best practice approaches to application design. Design modularity can minimize unexpected consequences of application changes through configuration, customization, or integration by isolating the areas of the application affected and facilitating application testing. A modular design also allows the user to deploy additional functionality as the system matures.

7.5.2.2 Configuration Tools—The laboratory informatics solution should provide an extensive administrative interface so that end users can configure the application without programming or direct database intervention. These configuration tools should be evaluated for the ability to add, remove, and change design and form elements on the screen to create productive forms and workflows with minimal programming. Additional configuration functionality is often provided through the use of a scripting language to tailor system behavior or build calculations within the application. Ideally, the laboratory informatics solution should allow for customization that incorporates content from external systems such as embedded multimedia (chromatograms, gel plates, short training videos, operating procedures, and so forth).

7.5.2.3 Laboratory Informatics Solution Software Development Kits (SDK)—The laboratory informatics solution should provide a programming tool such as an SDK to address situations where the user requirements cannot be met by the application. The programming tools allow staff to extend the functionality of the application to meet business requirements. The use of industry standard programming tools by laboratory informatics solutions increases the availability of qualified resources to implement and support the system. Caution is recommended when customizing a vendor-supplied laboratory informatics solution to ensure that the system is compatible with future vendor software upgrades.

7.5.2.4 Laboratory Informatics Application Data Structure—The laboratory informatics solution and its underlying technology should closely match an organization's laboratory workflow requirements and information structure. The application and database architecture of a system should be assessed on its flexibility to configure, customize, and integrate the system to fit the organization's needs.

7.5.2.5 Performance Design—The application architecture should be designed to use the operating system and hardware platform specified as efficiently as possible. This includes evaluation of concurrent usage, peak usage, and the ability for individual end users to multitask (that is, open multiple screens or application functions in the same user session) without losing work. The use of test automation tools and building a performance-testing model of the system early in the process provides significant benefits and can be used to qualify the final hardware used for deployment. The automation tools can be used to monitor changes in performance, perform regression testing when changes in software are applied, and tune the system as it is developed and even during the operational and maintenance phases of the system life cycle.

7.5.2.6 Personnel—In situations in which part or all of the system will be managed and maintained in-house or by parties other than the application vendor, consideration should be given to the availability of technical personnel with skill sets in the technologies used by the application. Knowledge of supported operating systems, programming languages, and design techniques used to customize and integrate with the application is important.

7.5.2.7 Application Programming Interfaces (API)—If customization or integration of the system is anticipated, the vendor should be able to provide a well-documented API for interfacing with the application, with a clear model for interfacing with the application's functions at a granular level.

7.5.2.8 Integration Standards—The laboratory informatics system should provide a means to integrate and exchange data based on common methods, such as the XML-based SOAP protocol or based on the integration methods supported by applications that commonly integrate with the laboratory informatics solution, or both, such as document management systems, manufacturing execution systems, and enterprise resource planning (ERP) systems.

7.6 Integration of Laboratory Informatics Solutions:

7.6.1 Integration—The ability of the laboratory informatics product to exchange information with other software systems is an important consideration for most laboratories. In Fig. 1, the

categories of systems that are often integrated are illustrated. This may involve either the exchange of data with other applications or the exchange of application logic and functionality with other applications, or both. Integration allows for laboratory informatics system to leverage the features of other applications without adding custom features to laboratory informatics system itself. Integrating laboratory informatics systems may require design changes to the database or application architecture of the laboratory informatics system. These modifications should, ideally, be minimal if the product is flexible and configurable for integration with other systems.

7.6.2 Common Laboratory Informatics Integration Activities—Examples of external systems that can be integrated with laboratory informatics solutions include:

7.6.2.1 Document management systems (standard operating procedures, chain of custody management, reports, and so forth);

7.6.2.2 Training and e-learning systems;

7.6.2.3 Enterprise resource planning systems (ERP);

7.6.2.4 Laboratory equipment, inventory, and calibration systems;

7.6.2.5 Chemical inventory systems;

7.6.2.6 Manufacturing execution systems (MES);

7.6.2.7 Business support systems;

7.6.2.8 Laboratory support systems;

7.6.2.9 Web portals;

7.6.2.10 Data warehouses; and

7.6.2.11 Field data capture systems.

7.6.3 Business Considerations for Integrating Laboratory Informatics Systems with Other Applications—One of the most critical evaluation criteria in the selection and implementation of any system is the organization's need for the customization and integration of the laboratory informatics system and the capabilities of the candidate system to perform these functions flexibly. Integrating the laboratory informatics system with other systems can profoundly impact product selection, implementation, ongoing management, and total cost of ownership of the system. The decision to customize the laboratory informatics solution for a particular business purpose or integrate the application with another information system shall follow a formal process that forms a part of the system life cycle. For a more detailed explanation of the business and management considerations of system integration, see Section 8.

7.7 Enterprise Computing Architecture—Fig. 1 provided a conceptual model of related laboratory informatics solutions. Fig. 7 illustrates an example of an integrated enterprise computing architecture that spans from enterprise systems down to the bench-level laboratory analytical systems (instruments). Other architectures are also used such as configuring a laboratory information management system (LIMS), scientific data management system (SDMS), laboratory execution system (LES), or electronic laboratory notebook (ELN) to serve as the enterprise integration platform without the extra service layer depicted in Fig. 7. A chromatography data system (CDS), on the other hand, as well as other instrument control and data analysis software could form an additional layer between LIMS/SDMS/LES/ELN and the actual instruments.

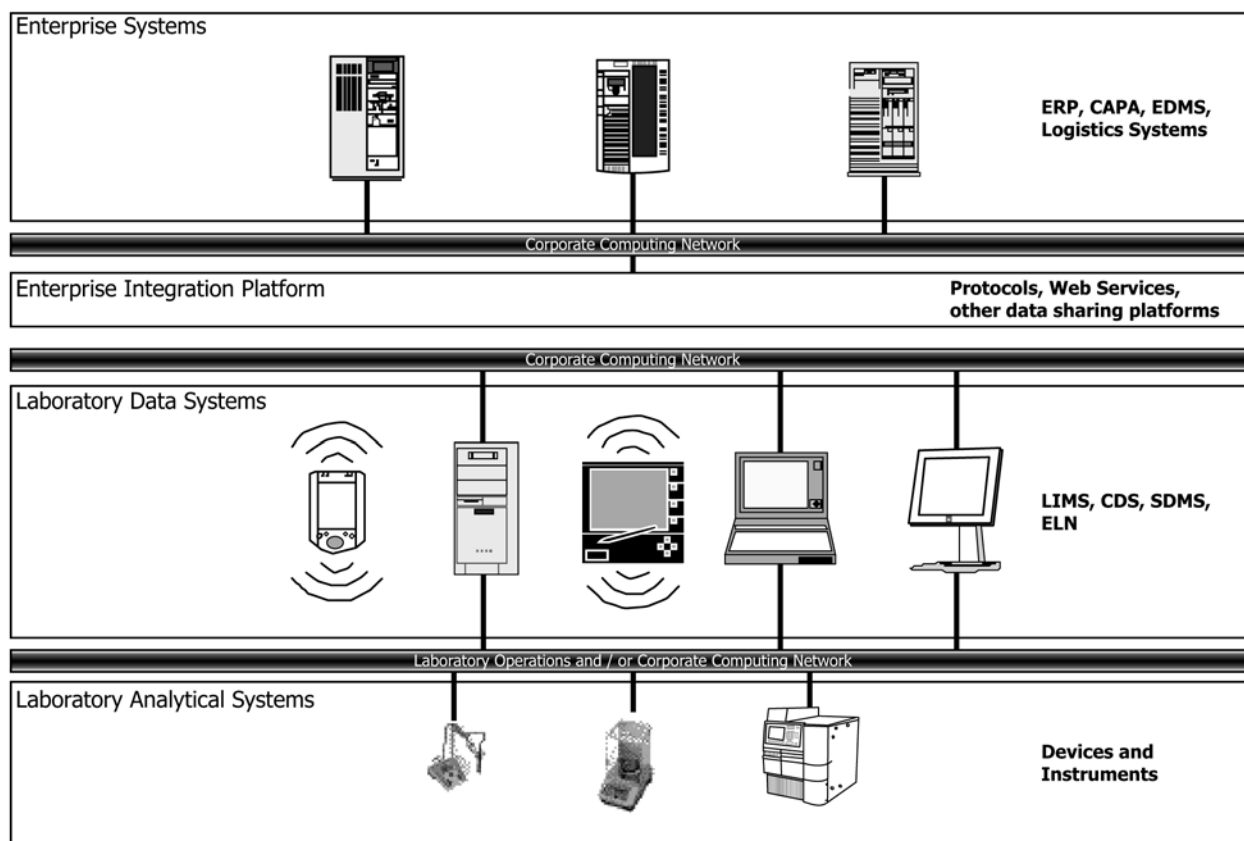


FIG. 7 Conceptual Laboratory Informatics and Corporate Computing Architecture

7.7.1 Multisite Deployments (Globalization or Corporate Multisite Deployment)—Globalization is the process and environment by which companies conduct business (internal activities and commerce with others) in many countries across the globe. The advent of advanced communications technology such as the internet and the rapid expansion of trade (among other factors) have greatly distributed scientific activities in general and laboratory environments, in particular, across the world. In **Table 1**, the benefits and challenges of different deployment strategies to be considered during multisite laboratory informatics solution implementations are described.

7.7.2 Content Localization—Content localization provides a user interface that reflects the specific geographic needs of each user and generally involves three elements (language, character sets, and time zones), with each value usually set in the user's application profile or in the local installation.

7.7.2.1 Language localization is a translation of the language in the user interface for the local user. It is generally a configurable item in global deployments, usually driven by database language content, configuration files, or an XML-based user interface framework, or combinations thereof.

7.7.2.2 Character sets refer to the set of characters used to express the selected language in the user interface. The choice of character sets is particularly important when considering user interface customizations. Unicode character sets are often used to specifically address local language issues without impact to the rendering of the user interface.

7.7.2.3 Time zone refers to the specific local time zone used by the system for time stamps and audit trail purposes. It is especially important to be mindful of time zone configurability in a system when storing, comparing, or aggregating data across multiple time zones, such that the audit trail can be preserved. Expressing the time as an offset of Coordinated Universal Time (UTC) for each time zone unifies this information. For example, U.S. Eastern Standard Time is UTC–5 h and is commonly used within laboratory informatics implementations.

7.7.3 Regulatory and Functional Issues—Globalization issues include specific functional needs across regulatory jurisdictions and other functional issues derived from the local laboratory environment. These issues can either be handled through configuration (for example, FDA 21 CFR Part 11 support) or customization. It is important in these scenarios that customizations are performed with upgrade path, support, and impact to users in other locations in mind.

7.7.4 User Security—User security varies greatly by technical implementation across laboratory informatics systems, but in general, systems should be evaluated according to the following basic user security issues: (1) Does the security framework provided by the system provide audit trail and permission control in a comprehensive manner as compared to the needs of the organization? (2) Is the security framework flexible and granular enough to allow control of security at a functional or task level? (3) Can the security framework be

TABLE 1 Multisite Laboratory Informatics Solution Deployment Strategies

Description	Benefits	Challenges
Central deployments house the entire laboratory informatics solution in one or more data centers with all application functions centrally managed. The user interface is delivered via a web browser or thin client terminal (including remote desktop).	• Consolidates support resources. Changes to the system are available to users in all locations quickly. • Minimum local IT support required. • Deployment to new locations can be quicker and easier than with distributed systems. • Data are centralized, making it easier to gather and analyze business intelligence across organization. • Regulatory and management standards can be easier to harmonize and enforce across sites.	• Application architecture is critical to the flexibility of the laboratory informatics solution for local laboratory needs. • The system shall be highly configurable to smoothly handle local language and regulatory issues. Customized features shall be analyzed to ensure that the entire user community benefits (and is not otherwise harmed) by the changes. • Infrastructure shall be carefully planned for network availability and redundancy.
Regional deployments combine the economies of scale of the central model with the flexibility of the distributed model. Clusters of laboratory informatics solution users are defined with common geographic, regulatory, or usage communities so that laboratory informatics solution implementations can be centralized into a few installations while still meeting the core application needs of each group.	• Delivers a happy medium between cost savings from economies of scale and delivering the right return on investment to meet user needs. • Local IT support is still minimized since this method generally uses a “thin client” approach. • Deployment to new locations is still relatively quick and easy compared to a fully distributed model. • Changes and upgrades can be delivered quickly while maintaining flexibility. • Data are still more easily aggregated across common business needs. • Regulatory and management standards can be easier to harmonize and enforce across sites.	• User clusters shall be defined carefully, or the result is simply a higher-cost centralized model. • User requirements shall be clearly aligned across language, regulatory, and functional needs so that the system can be properly configured. • Customizations and upgrades shall be carefully synchronized, both among the region’s users and across the organization. • Careful infrastructure planning shall be done to ensure system availability is not affected by regional events.
Distributed deployments consist of either stand-alone fat client desktop applications, or a client-server model, with a server installation at each location. Distributed models are generally not used when globalization is a significant issue.	• Content localization is out-of-the-box, since there is an installation for each location, which is configured for local needs. • Configuration and customization can be performed largely without regard to needs beyond the local deployment. • Ideal where each location’s needs are clearly separate from one another or functional differences by location are clearly delineated, or both. • Network availability is generally confined to more manageable local network (LAN) issues versus more complex wide area network issues.	• Deployment, maintenance, upgrades, infrastructure, and support can be very costly since each specific installation shall have these resources available. • Determining a global strategic direction for aligning laboratory informatics solution with the organization is difficult due to a high degree of local variance. • Common regulatory and management standards are difficult to automate across the organization, and benefits from one deployment are difficult to translate to another location. • Data aggregation to analyze and report on global operations is nearly impossible with highly distributed data sources. • Disaster recovery plans are difficult or costly to implement, or both, since each plan and implementation will vary by location.

conveniently administered, with a single security framework for all modules of the system, and can it be integrated with other enterprise frameworks? Laboratory informatics system security frameworks generally support the following capabilities at the user, group, and enterprise level:

7.7.4.1 User Security:

(1) *Audit Trail*—The system should, at a minimum, be able to provide a user ID and timestamp on all data activities a user performs (insert, update, and delete) and allow that data to be accessible for audit trail reporting.

(2) *Single Sign-on*—Most systems have a unified authentication scheme whereby a user can log in once and access all of the functions for which they have permission without requiring an additional log in.

(3) *Session Timeout*—The authentication scheme should also allow a configurable setting for requiring the user to enter user name and password again when the user has been idle on the system for more than a configured period of time (this is sometimes managed outside of application by the enterprise network settings).

(4) *Password Policy*—The system should meet company standards for requirements on password renewal, password

combinations (that is, minimum character lengths and combinations of characters and numbers), and encryption strength of database password storage.

(5) *Regulatory Compliance and Electronic Signatures*—The system should comply with applicable regulatory standards and company standards where electronic signatures, user acknowledgement of electronically recorded actions, or both, are used.

7.7.4.2 Group Security:

(1) *User Assignment*—The system should support the ability to assign individual users to system groups or roles.

(2) *Query Assignment*—The system may optionally allow assignment of users in batch by querying other user information, such as department.

(3) *Functional Assignment*—The system should allow for the assignment of permissions to groups for specific operations in the laboratory informatics system, such that a single group can have a standard set of permissions for a configurable set of granular activities (for example, performing specific data entry tasks, running specific reports or categories of reports, and so forth).

(4) *Audit Trail*—The system should provide an audit trail (user ID, timestamp, and reason) of all changes to group membership or group permissions.

7.7.4.3 Enterprise Security:

(1) *Enterprise Directory or Network Security*—A laboratory informatics system may optionally integrate with an organization’s enterprise security directory (for example, LDAP, Active Directory, Windows²⁹ domain, and so forth). This feature provides the benefit of a single set of credentials for multiple applications, seamless integration for laboratory informatics system into corporate security policies, and more convenient and robust security administration of users (for example, removing a user in the directory removes the user’s access to all applications, rather than requiring the user to be removed from user databases in each application).

(2) *Integration with Physical Security Systems and Public Key Infrastructures*—A system’s security framework may also optionally integrate with an organization’s physical security systems, such as security badges or biometric devices. These security systems supplement user identification/password credentials with encrypted certificates, smart card systems, and so forth. These features provide an additional validation and audit trail for authenticating users, and (with the appropriate device integrated with system access) may provide the additional convenience of passive authentication to the laboratory informatics system rather than manually typing in a user ID and password.

7.7.5 *Infrastructure*—The network and hardware infrastructure required by a laboratory informatics system varies widely by the particular software and feature package chosen, but it is especially important to understand what network and equipment resources will be needed depending upon the chosen deployment method (central, regional, distributed). In Table 2, a summary of the common infrastructure requirements and choices based on deployment strategy is listed.

7.8 *Exchange of Data between Laboratory Informatics Systems and Enterprise Resource Planning (ERP Systems)*—An

interface is a bridge that connects two separate software platforms using standard technology to enable smooth transmission of data between the systems. When multiple software systems are connected through various interfaces, the ecosystem of connected software systems along with respective interfaces are called “integrated systems.” Interfaces are classified into two separate groups based on the nature of software to which they are connected. When an analytical/medical device is connected to an information management system, it is termed as an “instrument or device interface,” whereas when two software systems are connected through an interface, it is called a “system interface.” A high-performance chromatography instrument controlled by a CDS platform or pH meter that is connected to a LIMS platform would be an example of the first, whereas an ERP system connected to a LIMS platform would exemplify the latter. ERP systems (the next level above manufacturing resource planning [MRP] systems that automate production planning systems; see Fig. 8) enable cross-functional integration so that an organization can evolve to a networked manufacturing company that uses real-time monitoring of business functions. The incorporation of product quality information from the laboratory within ERP systems is a clear priority for addressing product release or quarantine/reprocessing in a timely manner, as required by the FDA through process analytical technology (PAT) requirements.³⁰ Between the production plant and the laboratory that is analyzing data from production, there is a need for regular exchange of information about quality and analysis values. In order to leverage the full benefits of modern ERP solutions, organizations require automated access to all aspects of their business, including the process laboratory, as detailed in Fig. 9. The inspection requirements arise from the passage of materials along the supply chain managed by the ERP logistics modules. Data sharing between systems can facilitate a more rapid and efficient quality control/production process through the life cycle of each production batch (Fig. 9).

³⁰ See the Food and Drug Administration’s “OPS Process Analytical Technology (PAT) Initiative,” available via the FDA at <https://www.fda.gov/>.

TABLE 2 Infrastructure Requirements by Deployment Strategy

	Central	Regional	Distributed
Network Configuration	Wide-area network or virtual private network	Wide-area network or virtual private network	Local-area network
Security Infrastructure	Enterprise directory or internal laboratory informatics solution database	Enterprise directory or internal laboratory informatics solution database	Internal laboratory informatics solution database or local domain
Backups and Redundancy	Daily or continuous incremental data backups automatically to offsite location Redundant data centers Redundant network access points	Daily or continuous incremental data backups automatically to offsite location Redundant data centers, or shared failover between regional centers Redundant network access points	Daily backups with manual transport to offsite storage Usually no data center or network redundancy
Hardware Configuration	Mainframe or multiple-CPU servers with fully redundant components (server, motherboard, memory, disk, power supply, and so forth); thin-client configuration (terminal, remote desktop, web browser)	Multiple-CPU servers with fully redundant components (server, motherboard, memory, disk, power supply, and so forth) Multi-tier (desktop application with database and application server on back-end) or thin-client configuration (terminal, remote desktop, web browser)	Departmental server with minimal redundancy or no server (for deployments with a low number of clients) Client-server (desktop application with database and application server on back-end) or standalone desktop configuration with local data store

²⁹ Windows is a registered trademark of Microsoft Corporation.

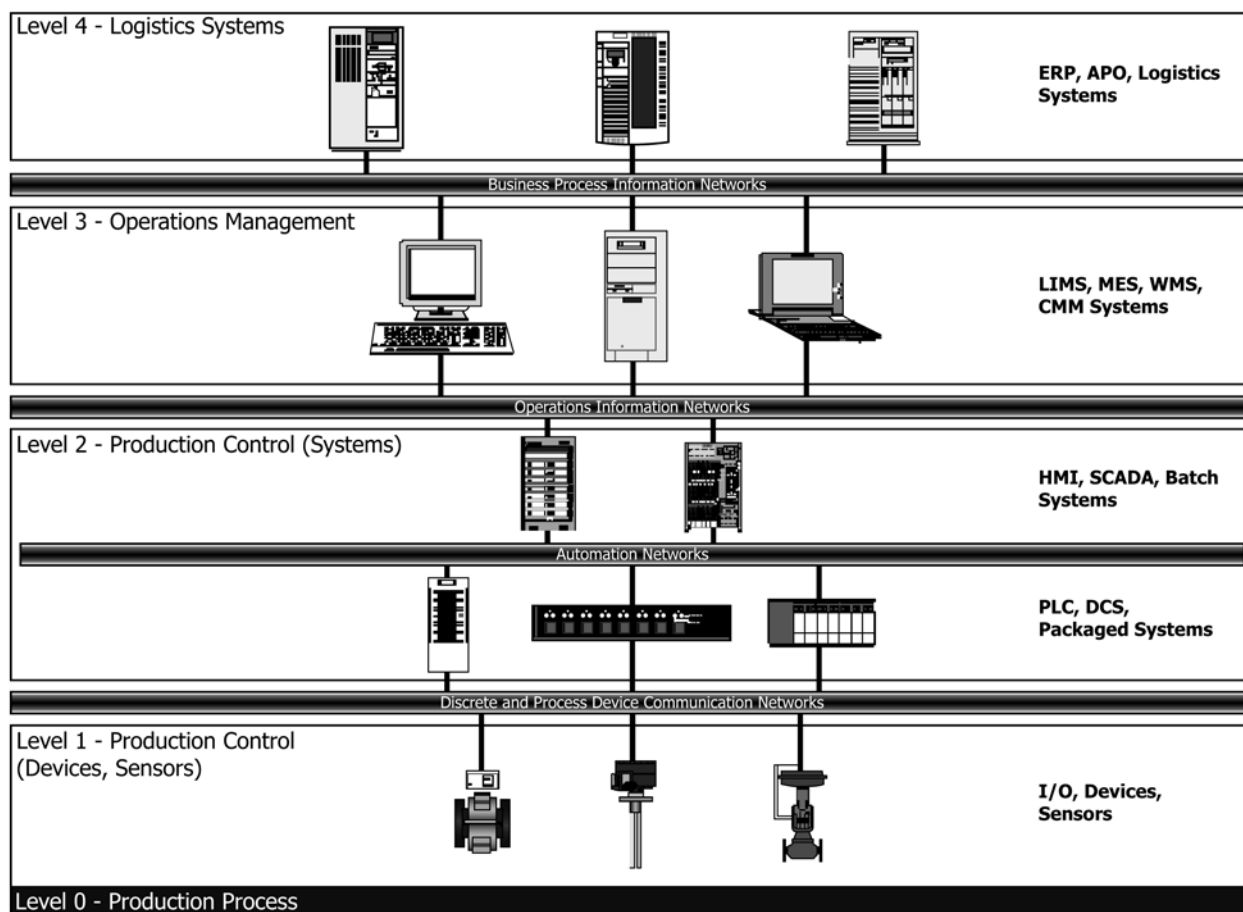


FIG. 8 ANSI/ISA-95.00.06 Enterprise-Control System Integration—Part 6: Messaging Service Model

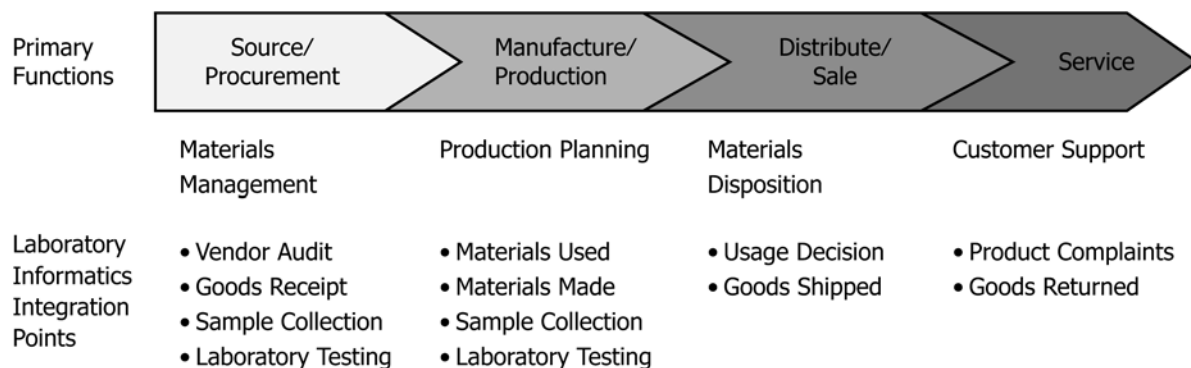


FIG. 9 Laboratory Informatics Solution—ERP Integration Points

7.8.1 Integration Points between Laboratory Informatics Systems and ERP—Depending on the nature of the laboratory and its ERP, lab informatics applications can connect through various modules. In Fig. 10 (a) and Fig. 10 (b), the basic integration points between laboratory informatics systems and ERP systems are outlined.

7.8.2 Integration Options—Approaches to facilitate laboratory informatics systems integration with other systems range from the creation of customized interfaces to vendor provided configurable solutions. An array of technologies and techniques exist to help facilitate integration. System vendors may provide specific certified interfaces for ERP integration, which

remove much of the burden of custom development and costly maintenance from the implementer. The ISA 95 Enterprise-Control System Integration standards are also highly relevant and widely supported by many ERP and MES vendors. The standards describe integration between the business logistic management layer, which includes ERP systems, and the manufacturing operations management layer, which includes MES and laboratory informatics systems (Fig. 8).

7.8.2.1 Integration strategies between laboratory informatics applications and ERP may take many forms, but two common ones stand out. First is a “heavy” ERP-centric interface where ERP drives the process and controls the sample

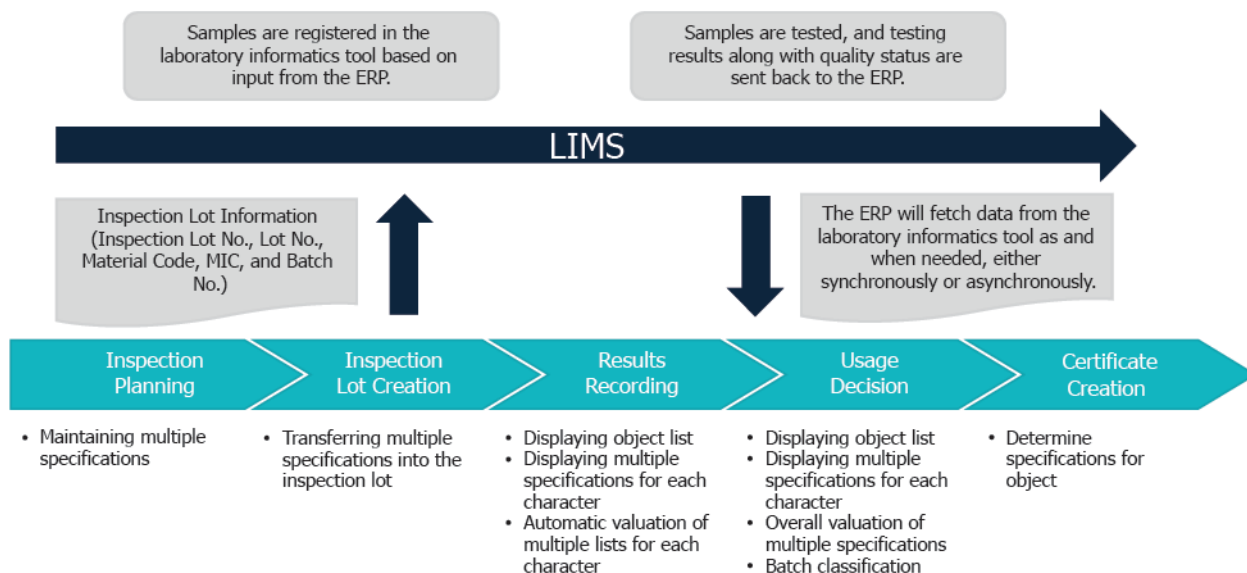
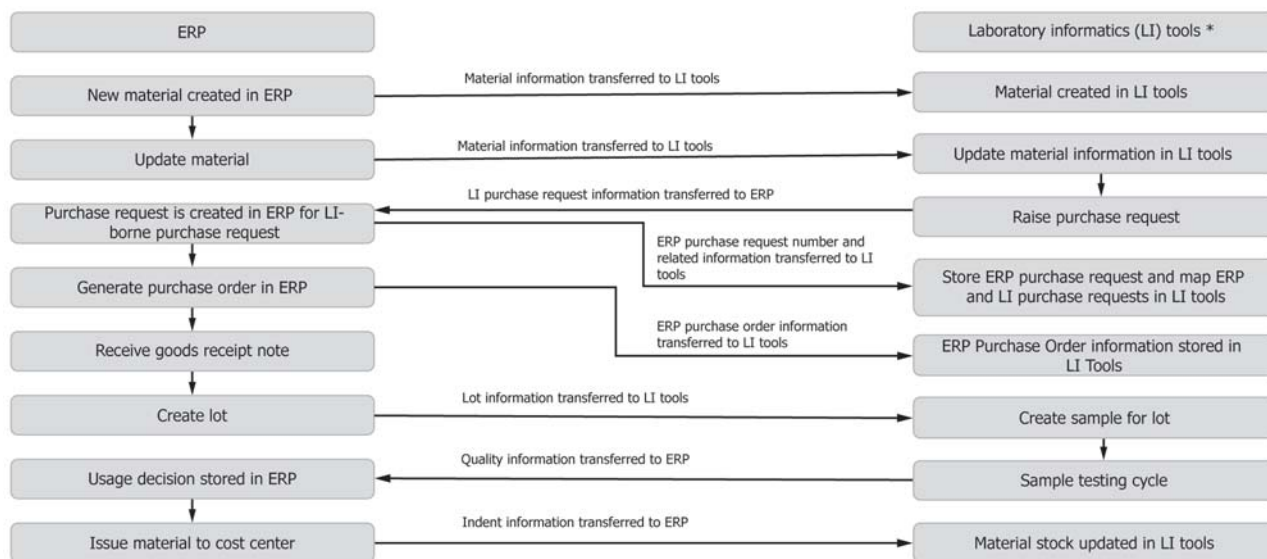


FIG. 10 (a) Possible Points of Data Exchange between ERP and Laboratory Informatics Solution



* Laboratory informatics (LI) tools: LIMS / ELN / ELS/ other informatics tools

FIG. 10 (b) Possible Points of Data Exchange between ERP MM and Laboratory Informatics Solution (continued)

test and acceptance criteria information provided to the laboratory informatics application, which uses it to guide the testing activities. When testing is complete, the laboratory informatics application provides back to the ERP all of the test data needed to evaluate material quality and generate a certificate of analysis (COA). Second is a “light” laboratory informatics-centric interface in which the ERP sends only the lot information and the laboratory informatics application controls all of the sampling, test, and acceptance criteria information used to guide the testing activities. When testing is completed, the laboratory informatics application sends only summary information and an indication of whether testing has passed or failed acceptance criteria back to the ERP.

7.8.2.2 Technical interfacing with an ERP may be implemented using many different approaches, but three are mentioned here: (1) a direct point-to-point interface between ERP and lab informatics applications where the ERP controls both outbound and inbound data movement with minimal assistance from the lab informatics applications, (2) a direct point-to-point interface between the ERP and laboratory informatics applications in which the applications trigger information flow with minimal assistance from the ERP to pull inbound information from and push outbound information to an ERP staging area, and (3) an indirect interface using a middleware application to control or broker data movement between the laboratory informatics application and the ERP. The third approach is

more flexible and adoptable since it does not have a dependency on the ERP or LIMS vendor to manage the interface scenario.

7.9 *Instrument Integration:*

7.9.1 Integration of laboratory informatics systems with instruments requires interfacing two moderately complex or highly complex systems. Interfaces between the laboratory informatics system and laboratory instruments (such as balances, pH meters, spectrophotometers, and CDSs) typically involve interfacing the system directly to the instrument or another data system controlling the instrument hardware, such as a standalone instrument computer system or CDS. The CDS is typically addressed as a specific category because of its inherent complexity.

7.9.2 Instrument integration with the laboratory informatics system can be a key factor in delivering a strong return on investment for the implementation through improvements in laboratory productivity and data quality. Some of the typical benefits of integration include improvements in data quality through the elimination of transcription errors and improved laboratory efficiency through the elimination or reduction of time-consuming manual data transcriptions, second person verifications, and corrections to transcription errors.

7.9.3 Because of the inherent complexity of managing instrument sample test sequences (in some cases, an auto sampler) in both the laboratory informatics system and the laboratory instrument data system, it is highly desirable that the laboratory informatics system send a test sequence, including sample identifiers from the laboratory informatics system to the laboratory instrument data systems via an interface to ensure accurate exchange of information.

7.9.4 In a typical scenario, the laboratory test is performed on the instrument, and test data and metadata is transferred to the laboratory informatics system via the interface for further processing. In many cases, the result transfer from the data system to the laboratory informatics system is required to be done in real time such that further result processing (cross-technique calculations, approvals, reporting, and so forth) be done as soon as the measurements have been performed.

7.9.5 Technical interfacing of instruments to the laboratory informatics system may be achieved through functionality available within the laboratory informatics system itself or use of an intermediary application specifically designed and implemented to connect lab instruments or their control systems, or both, to the lab's core application. These intermediary applications typically perform bi-directional communications between the instrument and laboratory informatics systems via data stream, file exchange, or web services. These integration applications are typically housed as an on-premise application within the business infrastructure; however, with the advent of widespread cloud services for many purposes, instrument integration services housed in the cloud are becoming available.

7.9.6 Instrument interface data exchange processes and formats have evolved over the years from simple RS-232 cables connected directly to a computer port or flat (text) files manually transferred over the network to more secure and robust communication methods and data exchange formats.

Some of the earlier standards or formats developed such as ANDI, JCAMP, and others were often developed for specific analytical techniques and so were not broadly applicable for integration of all instrument types with laboratory informatics systems. Perhaps because of that, there has not generally been widespread adoption of the standards by instrument and laboratory informatics suppliers for instrument data exchange. Although still not widely adopted for instrument integrations, newer, more general XML-based data formats for the storage and exchange of laboratory and instrument data standards such as AnIML²⁷ and ASTM's chromatographic (Specification E1947 and Guide E1948) and spectrometric (Specification E2077 and Guide E2078) standards now provide greater opportunity for standardized connections to instruments, requiring little to no configuration or programming. The XML data language format is a broadly supported industry standard for exchange of structured data between two systems. Although XML-based formats can be permanently stored as files, they can also be the basis for data exchange as a stream between two connected applications, typically via web services. Elimination of the file exchange process greatly increases the security and decreases the complexity of the instrument interface process. Because of its broad acceptance, there are many industry standard tools which can be used to easily transform the data representation from one supplier format to another, increasing the value of an XML-based interface.

7.9.7 Choosing which system will perform automated calculations should be guided by the functionality each system (laboratory informatics system versus instrument data or instrument integration system) provides. For example, a CDS typically includes calculations for system suitability, peak integration, and calibration curves, while a LIMS performs cross-technique calculations, correcting for moisture and other factors, as well as results trending across multiple techniques and time periods. Typical approaches for linked LIMS-CDS calculations include the use of ratios in which the CDS reports a sample response over a standard response, and the LIMS is used to calculate the final result and compare it to specifications and further reporting.

7.10 *Industry Data Exchange Technology:*

7.10.1 With an increase of electronic communication, it is common for data users to request and report laboratory data in a standardized electronic format, also known as an electronic data deliverable (EDD). There are many types of EDD formats and potentially multiple versions within a format. Using EDDs saves time by sending data directly from a laboratory informatics system such as a LIMS, minimizing and possibly eliminating manual data entry through automation, reducing transcription errors, allowing data delivery in a secure manner, and decreasing the need to harmonize and cleanse data. The Center for Disease Control and Prevention's (CDC) Electronic Laboratory Reporting (ELR) initiative³¹ is an example of a national effort to share clinical laboratory health data with public health agencies and has been a driving force for data

³¹ Additional information available from Centers for Disease Control and Prevention (CDC), 1600 Clifton Rd., Atlanta, GA 30333, <https://www.cdc.gov/>.

standardization along with reporting clinical and environmental data for national emergency response activities and regulatory compliance across many state and federal agencies.

7.10.2 Both public and private laboratories together can be viewed as nodes to a large network of laboratories that share data and can work in a coordinated way to integrate their collective data into a single data feed. This data originates at a laboratory informatics system. The ability to integrate this data is complicated by the many variations in laboratory informatics systems and the many different functional requirements and “standards” used by recipient agencies and clients. As the client/agency needs differ both in content and in formatting of the data message, the laboratory results communicated from their laboratory informatics system reflect different program needs, vocabulary, data elements, message structure, and secure transport.

7.10.3 Utilization of standards for laboratory informatics implementations is an effective way to reduce implementation costs, improve the networking capability of both public and private laboratories, and achieve efficiency of laboratory informatics interoperability. Laboratory informatics solutions should take into account the many system architecture options that a laboratory may use to submit electronic data on reportable laboratory results to clients/agencies as required by state or local laws and practice.

7.10.4 In cases in which data is exchanged between a large, diverse set of public and private laboratories where patient information or other metadata and demographic data is also exchanged, data standards for message and content structure such as Health Level Seven (HL7) have been developed. Data content standards such as Systematized Nomenclature of Medicine-Clinical Terms (SNOMED-CT) and Logical Observation Identifiers Names and Codes (LOINC) provide consistent ways to store, retrieve, and aggregate clinical data across specialties and sites of care. There are many other data exchange standards, too numerous to list here, used in other industries. Data exchange standards applicable to the laboratory’s business should be researched and, where needed, integrated into the laboratory informatics system requirements for implementation. Examples of standards from other industries include Specification E2369; ISO/IEEE 11073-10101; National Council for Prescription Drug Programs (NCPDP); Standard for Exchange of Nonclinical Data (SEND); Environmental Sampling, Analysis and Results Data Standard (ESAR); and Chemistry Industry Data eXchange (CIDX).

NOTE 1—

Health Level Seven (HL7)—Health Level Seven International is an ANSI-accredited standards developing organization dedicated to providing a comprehensive framework and related standards for the exchange, integration, sharing, and retrieval of electronic health information that supports clinical practice and the management, delivery, and evaluation of health services.

Specification E2369—Messaging standard for digital images. DICOM is produced and managed by the DICOM standards committee, which consists of vendors, user organizations, government agencies, and trade associations.

ISO/IEEE 11073-10101—A family of medical device communications standards that allows hospitals and other healthcare providers to achieve plug-and-play interoperability between medical instrumentation and computerized healthcare information systems, especially in a manner that is compatible with the acute care environment.

NCPDP—The NCPDP Batch Transaction Format provides practical guidelines and ensures consistent implementation throughout the industry of a file submission standard to be used between pharmacies and processors, or pharmacies, switches, and processors.

SEND—SEND is a Clinical Data Interchange Standards Consortium standard for nonclinical studies and is one of the required standards for data submission to the FDA.

CIDX—Formerly a CIDX standard, transferred to the Open Application Group, Inc. in 2008.

7.10.5 The CDC’s ELR initiative attempts to standardize the electronic transmission from laboratories to public health of laboratory reports which identify reportable conditions. ELR has many benefits, including improved timeliness, reduction of manual data entry errors, and reports that are more complete. Electronic laboratory reporting has been promoted as a public health priority for the past several years, and its inclusion as a meaningful use objective for public health³¹ serves as a catalyst to accelerate its adoption.

NOTE 2—The EHR “meaningful use” regulations that address automated electronic laboratory reporting to public health. Data is transmitted using HL7 messaging via secure transport such as PHIN-MS or NHEN Direct.

7.11 *Enterprise Application Integration and Middleware*—Organizations with a defined enterprise application integration strategy, or a standard enterprise middleware platform should evaluate the laboratory informatics system for its ability to integrate easily with the middleware’s supported standards. Integration with a central middleware platform can substantially reduce integration costs and complexity. A single integration implementation of the system to the middleware platform can then potentially support information exchange with multiple enterprise applications connected to the same middleware hub.

7.11.1 The middleware platform can consist of standards-based integration broker software. An integration broker, built primarily on messaging middleware, provides an end-to-end integration platform to handle components of data exchange between laboratories and data consumers. Vocabulary, messaging, and transport are automated across the extended enterprise, which includes the data exchange partners. It provides wide-ranging, prebuilt application adapters and bidirectional connectivity to multiple applications, including packaged and mainframe applications. In this configuration, the integration broker component of the system filters and maps the data, converting local codes to standard codes, and generates valid message structure and content before securely transmitting it using the agreed-to transport mechanism. The message broker/integration engine can be a separate standalone capability or integrated with the laboratory informatics solution. See the bibliography for clinical and health data standards and more information on networks which are used to exchange different types of information, including public health, food alerts, drug safety, and environmental data at a national level.

7.12 *Digital Content*—A wide array of digital media (images, site and corporate SOPs, reports from miscellaneous instruments, user training records, supplier reports, and so forth) are stored in a secure way in the laboratory. Storage and management of these supporting documents is typically handled by an external SDMS or data archive system with links

to the laboratory informatics system. Access to these digital content sources can be accomplished through a wide range of technical solutions (enterprise data share, content management system, and so forth). Access to this data storage from the laboratory informatics system is highly desirable, especially if integrated into the system workflow through configuration.

7.13 Electronic Laboratory Notebook (ELN) Integration—The boundaries between laboratory informatics applications such as an ELN, LES, and LIMS have been blurred as suppliers expand the functionality of their solutions. Consequently, the integration of two laboratory informatics solutions such as an ELN or LES with a LIMS can take many forms depending on the functional capabilities of each system. One strategy might be full implementation that covers the laboratory bench and instrument integration up to final results and material (lot) disposition (with LIMS performing the function of an ELN). More limited implementations may see the LIMS manage the final result but not fully cover the laboratory bench or instrument integration functions, with the ELN handling those tasks while passing the resultant information to the LIMS. ELN integration is not limited to LIMS, however. An SDMS or other laboratory informatics solution is a common destination as well.

7.14 Reporting and Business Intelligence—Many laboratory informatics solutions include the ability to generate simple reports and charts that may be printed or exported in a variety of formats for distribution or to support formal reporting processes ranging from a COA to research reports. When

capabilities of included reporting tools are not sufficient, integration of the laboratory informatics application with other reporting applications can provide additional features such as more sophisticated quality control charts or data visualization, data mining, business performance management, statistical calculations, and analysis or other complex data analytics. Reporting and business intelligence may also include configuration of data exports or links to external reporting systems, data warehouses/marts, information dashboards, and portals to provide business intelligence or visibility on laboratory data and performance.

8. System Life Cycle—Laboratory Informatics

8.1 Introduction—The laboratory informatics (LI) system life cycle refers to the activities that are taken to implement, operate, and eventually retire a laboratory informatics system. A laboratory's typical overall system life cycle is illustrated in Fig. 11. While many of a laboratory's activities may occur simultaneously, the system life cycle may be better visualized by organizing the activities into sequential phases for the purpose of facilitating planning and providing checkpoints for managing the project. The system life cycle begins with implementation. This may be the introduction of an informatics system to replace manual procedures in the laboratory or the replacement of a system that is reaching obsolescence. The implementation life-cycle phase is followed by an operational and maintenance phase when the system is in use. The system will evolve during this phase as new requirements are

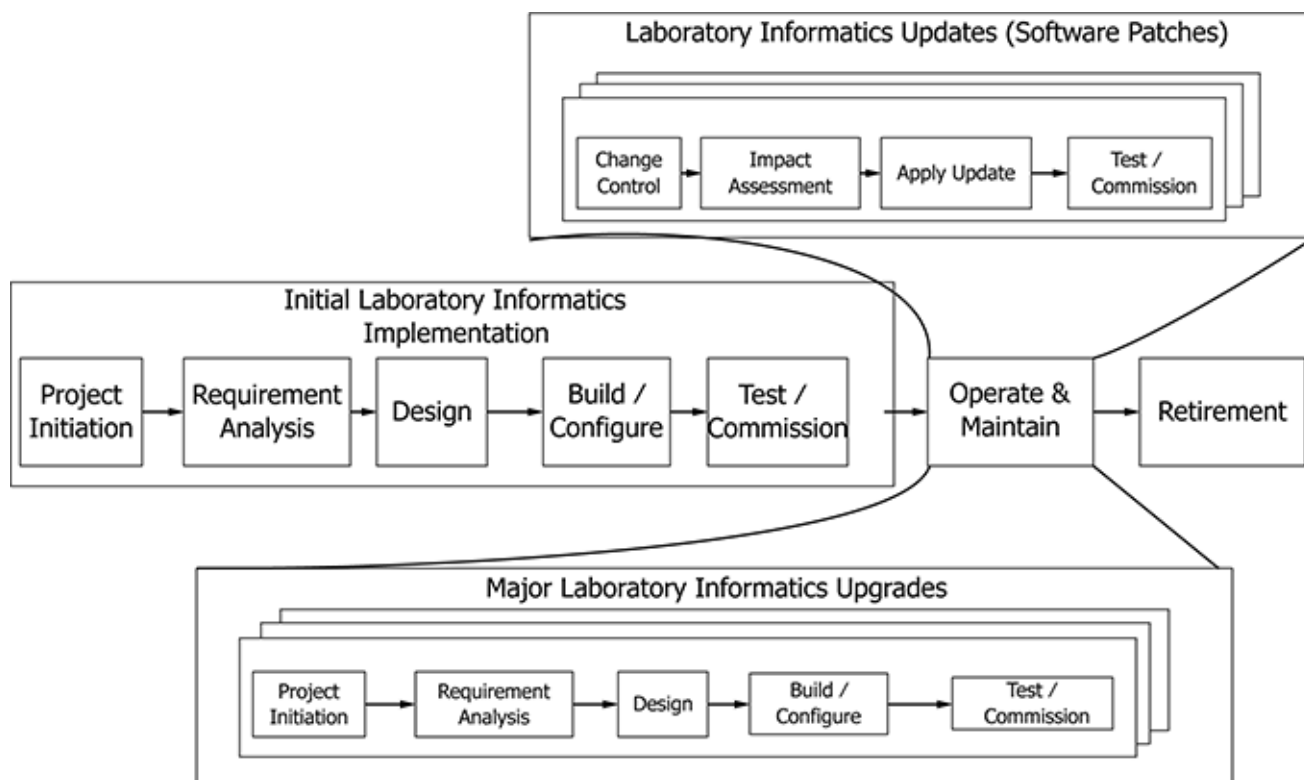


FIG. 11 Laboratory Informatics System Life Cycle

implemented, new releases are made available by the supplier, or underlying infrastructure (software and hardware) is updated. Retirement occurs if the system becomes obsolete (or if business pressures force a change) and a new system is implemented.

8.1.1 Laboratory Informatics Life-Cycle Phases—Overview:

8.1.1.1 Project initiation occurs first during the initial implementation of laboratory informatics solutions. This is also when the potential for future system upgrades and developments is considered. During this phase, the project or upgrade is conceived, the business case is developed, and the initial scope and boundaries of the project are determined. The ultimate outcome of project initiation is a “go” or “no go” decision for the project as a whole.

8.1.1.2 Requirements analysis occurs next and is important for determining the specific needs of laboratory informatics systems. Requirements analysis occurs during the initial implementation. It is also required for system upgrades and developments that occur during the operational and maintenance phase. Requirements provide a basis for the selection of a laboratory informatics system and may be functional or non-functional. Functional requirements cover the day-to-day use of the system and the workflow it shall support. Nonfunctional requirements define other considerations that may include the IT infrastructure in which the system shall work and regulatory considerations. Typical regulatory considerations define the need for the system to support whatever regulations and auditing requirements to which the organization may be subject, including any potential need for the system to undergo formal validation.

8.1.1.3 The design phase translates functional requirements into detailed logical and physical design specifications. Design occurs during initial implementation, though system upgrades and developments occur during the operational and maintenance phase. For commercial-off-the-shelf (COTS) systems, this phase may consist of specifying how the laboratory

informatics product features will be used, configured, or customized, or combinations thereof, to meet the requirements.

8.1.1.4 The build/configure phase acts as the construction phase of the laboratory informatics system. Activities include customization, configuration, and development testing of the solution being implemented. Build/configure activities occur during initial implementation, though system upgrades and developments occur during the operational and maintenance phase.

8.1.1.5 Testing/commissioning occurs as the solution is being put into operation. Activities include final data loading, user acceptance testing, training, and deployment. Formal validation may occur, if required; reference GAMP 5 for detailed guidance on validating the informatics solution in regulated environments.

8.1.1.6 The operational and maintenance phase is when the system is used to support the laboratory’s operations in production. Further work may be required during this phase to ensure the solution stays aligned with changing business and technology needs. The vendor or laboratory IT personnel may provide updates or new releases that resolve identified issues, provide enhanced functionality to the base system, or provide support for changing technology. These updates can be applied to the system after their impact is analyzed. Applying upgrades can be a major project and should follow a process similar to the initial implementation activities.

8.1.1.7 Retirement is the final phase in the system life cycle. Usually driven by technology obsolescence or major organizational changes, retirement involves planning for the next generation solution, including providing access to relevant historical laboratory data.

8.1.1.8 *Laboratory Informatics Life-Cycle Implementation Project Flow*—The flow for a typical system implementation and life cycle is illustrated in Fig. 12. Fig. 12 illustrates relative resource utilization during various phases of the informatics system life cycle. Three types of resources are illustrated,

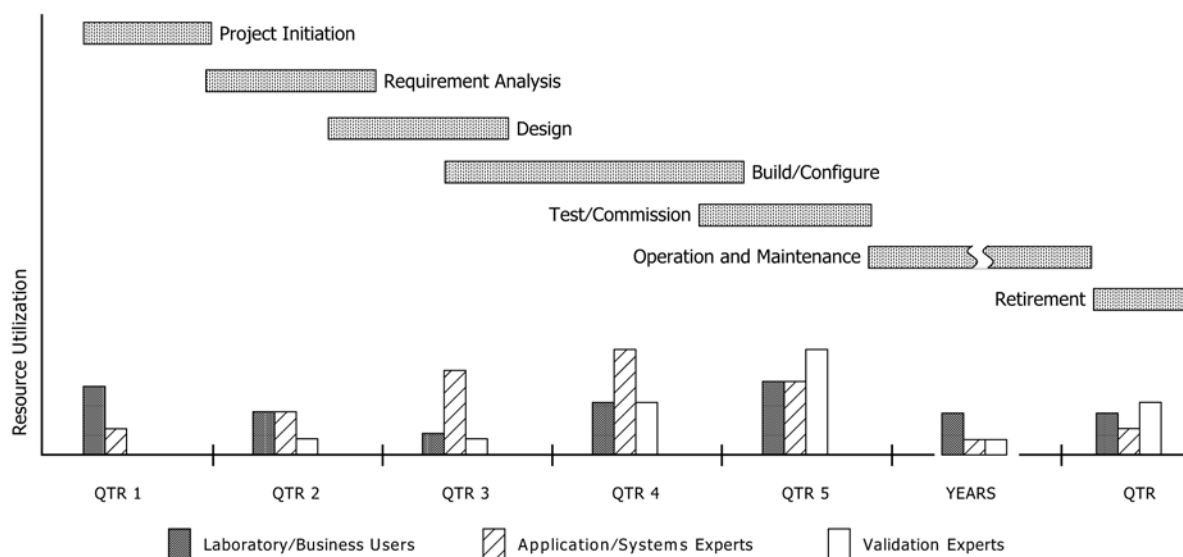


FIG. 12 Laboratory Informatics System Life Cycle—Resources and Timeline

including laboratory/business users, application/systems experts and developers, and validation experts.

8.1.1.9 The time frames depicted in Fig. 12 are illustrative only, and many factors affect the time line. These include complexity and scope of the required solution, the fit of the selected solution to the identified requirements (which influences the development, customization, or configuration required), the willingness of the organization to change working practices to match a COTS solution, the impact of nonfunctional requirements such as validation for regulatory purposes, resource availability, time constraints, and budget. Those responsible for managing the implementation of an LI solution shall ensure they work with their key stakeholders to ensure that timelines are achievable and realistic

8.1.1.10 While the phases occur sequentially in this guide, some activities may occur in parallel. For example, design could start before completion of requirements and build/configure could start before completion of design. The use of agile, iterative, or prototyping development techniques may further blur the distinction between the phases. However, the phases provide a logical approach to the implementation of a laboratory informatics project, and an agile approach can be seen as a way in which the phases are compressed into shorter but more numerous iterations. More comprehensive descriptions of the activities in each of the system life-cycle phases are provided in 8.2.

8.1.1.11 The resource utilization bars shown in Fig. 12 depict the continued involvement of the laboratory/business users, the application/systems experts, and the validation experts throughout the life cycle, as well as the times of higher levels of engagement. As the end users of the solution, laboratory/business users are more heavily involved in the initiation, requirements, and test/commission phases. Systems experts are most heavily involved between the design and test/commission phases. Validation experts are primarily engaged in the later part of the implementation during the build/configure and test/commission phases.

8.1.2 *Keys to Successful Laboratory Informatics Project*—Laboratory informatics projects shall be approached in the same way as any other complex or business-critical IT project. This section outlines some of the key factors affecting the success of any laboratory informatics project.

8.1.2.1 Standard project management techniques shall be used for all laboratory informatics projects; however, it is outside the scope of this guide to try to cover all aspects of project management in detail. There are a number of generally accepted project management methodologies available, and individuals can become certified project managers (that is, project management professional certification from PMI or Prince2 certification). A good project management methodology is scalable to reflect the size and complexity of the project. People undertaking a laboratory informatics project can benefit from involving experienced project management resources from their own organization if they do not have the experience themselves. Third-party project managers can be employed as well. The suppliers of the laboratory informatics system should also have project management skills, and these can be used and actively involved in the project as well. Use of a formal

management control structure and project management tools can help in understanding and communicating project complexity, priorities, and variables, leading to better planning and decision making from software acquisition through deployment.

8.1.2.2 Constraints bound all laboratory informatics projects. These constraints are time, cost, and scope. However, quality or “fitness for purpose” may be better suited to describe the third constraint, as it encompasses potential nonfunctional requirements such as system validation. When running a laboratory informatics project, these constraints shall be fully understood; what does the system need to do, how much time is needed to do it, and what are the budget constraints? It is also vital to ensure that the expectations set for these constraints are compatible. In other words, does the organization have enough time or money, or both, to implement the system to the required quality? Any change to one constraint will almost certainly have an impact on one, or both, of the others; increasing the scope of the project will result in an increase in costs and probably time. When working with a vendor, it is important that they are aware of all the constraints. In all circumstances, change control mechanisms shall be used to manage the impact of change on the project.

8.1.2.3 Long-term stakeholder involvement and commitment further ensures the success of a laboratory informatics system’s implementation. Stakeholders include the user community and management and, potentially, the organization’s quality and regulatory functions and customers; other functions may be identified as well depending on the scope of the project. In a commercial testing organization, the finance department may be a key stakeholder, as they will require information from the system to ensure accurate and complete billing. If the organization has an IT group that will be involved, they shall be identified as stakeholders. Stakeholders may be involved in the project to a greater or lesser extent depending on how much they or their function will be impacted by the project. In all cases, continued communication with stakeholders is vital throughout the project. This communication should include the reasons for the project, the impact it will have on the stakeholder, progress reports (including financial reporting where appropriate), and, after the project has been completed, communication of benefits realized. It shall be remembered that any supplier that is involved with the project is also a key stakeholder. Stakeholders shall be committed to the success of the system throughout the life cycle. Longer, more complex projects require continued communication/updates to stakeholders—particularly if/when changes occur—to ensure their continued commitment.

8.1.2.4 A business case is fundamental to a sound project plan and should be created as part of project initiation. Guidance on how to create a business case is prevalent; however, two of the key reasons for developing a business case are to define the reasons for carrying out the project and how success will be measured once the project has been completed. Additionally, define how the business value of the project will be quantified. For example, before project startup, compare laboratory performance metrics or key performance indicators (for example, samples processed per unit of time, employee,

and instrument; cost per test; quality defects per batch) against the results post-implementation, that is, every three to twelve months after commissioning the new system.

8.1.2.5 Avoid customization, configuration, and change orders not supported by the business case. All of these add time and complexity to development and testing activities and may introduce additional sources of error. Scope creep is subtle and insidious; use formal change control processes to approve any additions or changes to scope, no matter how small. Carefully evaluate any potential warranty, technical support, and software maintenance (product upgrade) implications before making changes to the system.

8.1.2.6 Establish sound procurement processes and selection criteria based on the organization's guiding directives, the business case, and identified requirements. Requirements should focus on the problems to be solved rather than dictating or architecting the solution. Nonfunctional IT requirements should be validated against the organization's technology standards and technology roadmap rather than individual preferences or existing computing infrastructure. In estimating total cost of ownership, factor in costs for internal customization and support, savings through automation and operational improvements, projected product life cycle, add-on licensing costs, and software maintenance contracts.

8.1.2.7 Identify the implementation methodology to be used for the project. There are a number of methodologies that may be used, ranging from a waterfall methodology through spiral, iterative, and agile methodologies. It is outside the scope of this guide to describe or recommend an implementation approach. Organizations may have already developed their own implementation methodologies, in which case it may be worthwhile using this internal experience as part of the project team. A good approach can be to adopt the methodology used by the laboratory informatics system vendor if they are involved with the project. Additionally, third-party consultants could be used.

8.1.2.8 Implementing a new system presents an opportunity to re-examine business processes and implement best practices supported by the new system. Avoid replicating outdated paper-based systems, retaining disconnected and disparate data sources (for example, spreadsheets/macros), or automating workflows that no longer make sense. However, recognize that these additional goals will inevitably impact the resource and time requirements, and decide in advance, and with stakeholder involvement, whether the tradeoffs are justified.

8.1.2.9 Manual (paper-based) or standby systems should be established (and periodically tested) to maintain operational continuity in the event automated systems become temporarily unavailable.

8.2 Key Phases in the Laboratory Informatics System Life Cycle:

8.2.1 *Phase 1: Project Initiation*—The purpose of this section is to outline the activities that occur during the project initiation phase of the system life cycle. The section contains three parts, including project initiation and planning, developing a business case, and developing a quality plan. It will also emphasize the importance of change control mechanisms. Using this section will provide the reader with guidelines to follow as they initiate a new informatics project.

8.2.1.1 The project initiation phase is the process in which the idea of implementing a LI project or program either becomes a live project, a program, or is rejected. A key concept for any project or program is that it should start, achieve something, and stop. Project initiation lays the ground work for these three outcomes. For example, a business case should provide evidence of the value of the project as well as projected costs, a project brief may provide outline requirements that provide an overview of functional and nonfunctional requirements, a quality plan will outline the quality approach required for the project, and a communication plan can outline how project progress is to be communicated to all key stakeholders. The final output, or deliverable, of the project initiation phase will be a decision based on the information gathered during the phase as to whether the project or program will proceed or not. Accepted project management techniques explain the project initiation phase in detail and the possible deliverables of the phase. What is included in project initiation and how long it takes will depend on the nature and scope of the project. Project management techniques are scalable to allow different size projects to be managed effectively. As the project initiation phase provides and supports the rationale for the project, it should be included in some form or another for all LI projects, from a full implementation to the creation of a new report. Obviously, the size and scope of project initiation will vary enormously in these examples, but the outcomes are the same; at the minimum, an understanding of the reasons for the project (business case), and agreement as to whether to continue with the project. A decision to go ahead can only be made if funding is available for, at least, the next phase. Next are described three key deliverables of the project initiation phase: the project brief, business case, and quality plan, while also emphasizing the importance of change control mechanisms.

8.2.1.2 A project brief is designed to give an overview of the project, to place it in the context of the organization as a whole, and to position it with other projects or programs that may be running. The project brief describes the issue or issues the project is designed to address; this could be a problem that needs to be solved, an opportunity that can be exploited, or a requirement that needs to be addressed. These may be very specific, for example, the introduction of a new regulatory requirement that needs to be incorporated into an existing LI solution or more general or larger in scope, such as the replacement of an existing paper-based management system. The project brief should document the project objectives and the currently identified project deliverables, which may include the outline requirements. These requirements may be very high level at this time, such as the implementation of a biobank sample tracking system to replace the existing paper-based system. Items that are specifically excluded should be documented as well. This applies especially when it could be assumed that the item would be included. For example, if a specific department or group will not be part of an LI implementation, it can be stated here. In the case of a biobank system, automated freezer management hardware and software might be explicitly excluded from the project. The project brief may define known constraints. These can include, among

others, time constraints (for example, the deadline for implementing new regulatory requirements), financial constraints, or technical constraints (for example, the corporate standard requires a cloud-based virtual system or the corporate standard specifically prohibits a cloud-based virtual system). The project brief may also identify projects, products, and program interdependencies. For example, the implementation of the biobanking software may be dependent on the purchase and commissioning of new freezers, or the implementation of a laboratory information management system (LIMS) may be dependent on the building of a new facility. Project briefs may also contain any known risks to the project, and an outlined project plan, if required, by the organization's project management methodology.

8.2.1.3 A business case is used to justify the proposed project by defining the business benefits and costs. The business case developed during the project initiation phase may be a high-level document, especially as far as costs are concerned. For example, the requirements may not be fully defined at project initiation, a product or platform to base the project on may not have been selected, an approach to implementing the system may not have been decided on, and negotiations will not have started with the chosen supplier (internal or external). However, at project initiation the business benefits should be well understood, as these form the basis of why the project should be considered at all. The business benefits may have been outlined in the project brief, but the business case provides an opportunity to expand on them. The business case is a document that is subject to review during the course of the project to ensure that the rationale for it is still valid. As an extreme example, if the aim of the project is to install a LI solution at a site or in a specific laboratory, the business case becomes invalid if the organization closes or sells that site or laboratory and the project should stop. The benefits defined in the business case should, wherever possible, be measurable as they provide the best way of proving the success of a project once it has been completed. Developing a business case may be easier for some projects than for others. For example, regulatory reporting or compliance requirements may be easier to justify than the anticipated benefits of automation or quality improvements. Good cost-benefit analysis requires time, solid understanding of the laboratory environment, and careful analysis of the expected benefits. The cost-benefit components of inaction—not implementing the system—shall also be addressed. This is sometimes called the do-nothing option. An analysis of costs should consider the total cost of ownership (TCO), including the initial purchase costs, implementation (including anticipated customization) costs, and ongoing maintenance costs. Benefits are normally characterized as tangible (hard benefits that are easily calculated) or intangible (soft benefits that are recognized but more difficult to quantify). As stated above, these may not be fully known at project initiation, and therefore, the business case should be regularly reviewed and updated as required during the project. If possible, a financial value should be associated with each benefit. If a financial value cannot be associated with a benefit, some other way of measuring or proving its value should be identified. For example, for an internal service

laboratory that does not generate revenue, increased sample throughput is only a valid business benefit if there is, or will be, a demand for more samples to be processed. It is the case that “soft” benefits will often be challenged when used to justify a project. Protecting brand reputation is often cited as a benefit; if using this as a justification, it shall be backed up with how the system will help protect the brand and, if possible, the cost associated with failures. In the food industry, a LIMS may protect the brand reputation by preventing the shipment of products that later need to be recalled. Such recalls affect the brand reputation, but there is also a direct cost associated with the recall that can be measured and used in the justification. **Table 3** provides a list of the types of benefits a LI project could provide, in this case specifically a LIMS; note that the list does not attempt to define the measurable value associated with each benefit. Any assumptions associated with each benefit calculation should be documented so that its validity can be confirmed each time the business case is reviewed. Avoid creating unrealistic expectations as disillusionment over the expected benefits by management or end users often leads to project failure. When creating a business case, it is not uncommon to overestimate the potential benefits. Sources of error may include underestimating the cost of implementing the system, especially the cost of staff involvement and the cost of reworking ill-defined requirements; overestimating the potential productivity gains; overestimating the demand for the improved laboratory services; failing to account for the initial impact of the introduction of the new system on productivity because of lack of familiarity with the system (this should be short term); and assigning unrealistic values to “soft” benefits. The business case is a living document that should be reviewed and updated as required throughout the lifetime of the project. As the project progresses, items such as cost estimates will become more accurate and actual cost incurred can be included. Assumptions used in the business case can be monitored and updated as required and the validity of the project confirmed. However, it is possible that reviewing the business case may show that the project is no longer viable or necessary, in which case it should be stopped. A project that is halted for the right reasons is not a failure but rather should be looked at as a success for good project management.

8.2.1.4 A sound, well-considered, quality plan helps ensure a smooth, successful implementation and, if required, validation of a laboratory informatics system. Quality expectations for the LI project shall be identified. These will vary from organization to organization based on factors such as the level of regulation to which the organization is subject. Key stakeholders in quality planning will be the organization's quality or regulatory functions, or both. These functions shall be involved in identifying the quality requirements of the project. They may also need to be involved in monitoring adherence to the quality plan and final sign-off of the system. If they are not involved, they may prevent the system going live if it does not meet the defined quality standards. A quality plan should include a quality policy and objectives, identified expectations, and any assumptions made. A quality plan needs to assess the risk to the project, users, and ultimate customers impacted by the system. These risks will help guide the level of effort

TABLE 3 Example Benefits Attributable to Laboratory Informatics

Tangible Benefit	Description/Comment
Laboratory throughput and capacity	Increase in the number of samples processed in the laboratory per unit time, analyst, department, or instrument, or combinations thereof.
Laboratory turnaround	Decrease in sample processing time from start to finish.
Labor savings	Reduction in the number of analyst hours required to process samples and data.
Reduction in error rates	Automation systems reduce data entry, transcription, and calculation errors. Lower error rates improve processes by minimizing process waste/rework, including time spent re-analyzing samples and reprocessing data.
Reduction in time spent on laboratory investigations and quality inspection activities	Quality improvements allow more time to be spent in value-adding activities, and may allow for a reduction in the frequency of process inspection and data review steps.
Reduction in cycle time to release manufactured products	Integration with MRP/ERP systems enables release of product as soon as laboratory results are approved.
Reduction in customer service costs and time	Providing self-service, read-only access to sample status, test results, and other laboratory information eliminates end-user dependence on customer service personnel.
Earlier detection and correction of errors	Informatics technologies can alert users of out-of-control conditions or other possible problems that will trigger corrective actions as early as possible in the process.
Reduction in time spent supporting audits	Ready access to electronic data, SOPs, audit trail documentation, and other quality records can significantly reduce the amount of time dedicated to audits.
Improved sample management	Sample tracking functionality can provide detailed knowledge on the status and location of all samples, reduce the likelihood of mislabeling or misplacing samples, and document sample traceability from collection through disposal.
Intangible/Other Benefits	Description/Comment
Better service management	Benefits will accrue from the ability to integrate information more closely with suppliers and clients.
Access to information/tools	End-user access to more data and better tools will improve self-service reporting and analytics. Data access will also facilitate routine and nonroutine problem solving.
Easier to achieve and demonstrate regulatory compliance	Improvements in work processes, standard operating procedures, and electronic record keeping will make it easier to enforce and demonstrate compliance with quality and regulatory requirements.
Improved laboratory management	Better tools to enable more consistent workflow and management of laboratory operations.
Enforcement of Lean/Six Sigma practices, earlier intercepts of drifts in product quality	Closer coupling between the laboratory and manufacturing will provide quicker indication of quality trends and allow for near-real-time adjustments in manufacturing processes.
Improved customer satisfaction and competitiveness in the marketplace	Better quality data and services may improve customer satisfaction/confidence, the laboratory's reputation, corporate brand value, and overall competitiveness in the marketplace.

required for validation and support during implementation and throughout the system life cycle. There are several well-documented reference materials associated with risk assessment. (See, for example, ICH Quality Guideline Q9). Users should evaluate projects against their business objectives and the risk associated with the informatics project. Risk parameters include, but are not limited to, overall project risk, product risk based upon the impact that the new system will have on the product quality, people risk in the area of change management, and business risk. Business risk may include risk introduced by software customization. The quality plan should also identify any internal or external quality standards that need to be met, as well as any identified procedures to be followed. These procedures may define how the system will be implemented, validated, and maintained in a validated state during routine use and retirement of the system (system life cycle). In addition, the quality plan may define the required documents for a project such as a user requirements document, functional specifications document, system design document, test scripts, and validation plan. Remember that the implementation of the quality requirements will have an impact on both the time required to implement the project and the cost. Therefore, the level of quality applied to the project shall be appropriate to the size, potential risks, and regulatory requirements associated with the project. If working with a supplier, either internal or external, they should have their own quality system in place. It is important, therefore, to ensure there is no conflict between the organization's quality plans and the supplier's. The best option is to agree to a single unified quality approach. For organizations or individuals not used to working in a quality environment, it may be appropriate to adopt the

supplier's quality approach. In all cases, however, the supplier shall know, understand, and be given access to the relevant quality information.

8.2.1.5 Nothing puts a project at risk more than uncontrolled change. Changing functional requirements, commonly referred to as scope creep, represents a type of change that can affect project success. However, other changes that affect project constraints will have the same effect. Change is almost inevitable, necessary, and often beneficial, even in the best planned projects. Potential change, therefore, needs to be managed through some form of change control mechanism, which shall be in place from the start of the project. During the project, and as changes are identified, they should be reviewed to ensure they are appropriate and required, and their impact should be assessed. Based on this assessment, they may be accepted for inclusion. Changes will impact the project constraints and can affect one or more of them, that is, the functionality (quality), the cost of the project, and the time to implement it. This cannot be forgotten when requesting changes; does the change affect existing agreed functionality, does the project have the budget to pay for them, and can the time line for the project be changed? When working with either internal or external suppliers, they shall be involved in managing and controlling change control as, in many cases, they will be best placed to assess the impact of the proposed changes and will be directly affected by them.

8.2.1.6 The initiation phase forms the groundwork for the entire project and is important to get it right. To help ensure this, the correct individuals and functions shall be involved. Who and what these are will vary depending on the nature and scope of the project. This guide provides some guidance but is

not a complete list of all possible stakeholders. When developing a project brief, the key stakeholders will be the person or group whose idea the project is. Creating the project brief will involve users and others who may be affected by the project and may require the involvement of the organization's project management function. If the project progresses, the chosen supplier should have access to the project brief so that they understand the background of the project. If the project brief contains sensitive material, this may need to be removed. The key stakeholders for the business case will include the potential funding body for the project, the people who will sign off for the project, and the people or functions that will benefit or be affected by the project (for example, laboratory users, laboratory managers, quality and regulatory functions, and, potentially, customers). In developing the business case, the involvement of the finance function may be required to help calculate potential financial benefits, as well as the organization's project management function. If the project progresses, the chosen supplier should have access to the business case so that they understand the background to the project. The supplier may also be given the opportunity to contribute to the business case based on their experience of similar projects. The quality plan key stakeholders will include the project manager and the quality/regulatory function of the organization. People working on the project should be aware of the implication of the quality plan, for example, any implication for people creating user requirements or carrying out testing of the system. The supplier will be a key stakeholder as they will need to adhere to the quality plan. If undertaking a selection process for a system and supplier, details of the quality plan and the expectations placed on the suppliers shall be included so that they can confirm adherence to the quality plan and account for any costs associated with it in the overall system costs. If the organization does not have experience of defining quality plans, it may choose to adopt the supplier's own. When considering a change control mechanism, the key stakeholders will be the project manager and whoever is responsible for the project budget, as they will be most affected by changes to the project. However, it is important to make sure that other stakeholders, especially users, understand why change control is important and why all proposed changes, no matter how small, should be run through the change control mechanism.

8.2.1.7 Deliverables of the project initiation phase include the project brief, business case, quality plan, change control mechanisms, as outlined above. The deliverables may also include any other documents that the organization's project management function requires. However, the key deliverable of project initiation will be a "go" or "no-go" decision for the project. Such a decision indicates that the fundamentals of the project have been prepared and reviewed, and the project is at least worthy of moving to the next stage. This decision should be accompanied by a guarantee of funding for at least the next phase of the project.

8.3 Phase 2—Requirements Analysis—This section outlines the activities that occur during the requirements analysis phase of a system life cycle.

8.3.1 Workflow Analysis—Current and Future State:

8.3.1.1 Modeling the current state of laboratory practices helps the organization fully understand its existing working practices and how they will be impacted by the LI solution. This will provide insight into the needs and requirements of the system. It may also identify anomalies in current working practices that need to be resolved before the LI solution can be implemented. There are a number of ways in which the current state of laboratory practices can be modeled. However, the important element of this exercise is to involve the correct people from within the organization. These may include end users, laboratory managers, and external users (customers) of the system, together with representatives of other areas that may be affected by the system such as manufacturing and production management, finance, and customer service. The aim of modeling the current state is to identify anomalies, problems, and bottlenecks in the current working practices.

8.3.1.2 Working from the modeled current state, the future state of laboratory practices should also be defined before system implementation/selection. This is also the opportunity to harmonize working practices across the organization. This may be processes within a single laboratory, across multiple laboratories, or across a global organization. Harmonizing or standardizing processes where possible will ultimately simplify system implementation and leverage the effort across the enterprise.

8.3.2 Business Requirements Analysis:

8.3.2.1 High-level business objectives and strategies shall be understood and considered as a prerequisite for developing detailed business requirements for a laboratory informatics solution. These should have already been defined in the business case. The business requirements should be defined and documented as to what is needed and should be approved by key stakeholders of the business process. The business requirements should be realistic in terms of what can be expected of commercial systems or what can be expected for custom-developed systems based on known resources and their capabilities.

8.3.2.2 Business requirements provide a base for the laboratory informatics solution to perform properly and meet defined business needs, including defined inputs, behavior (function), and outputs. Business requirements may be calculations, technical details, data manipulation and processing, and other specific functionality that define what a system is supposed to accomplish. Behavioral requirements describing all the cases in which the system uses the business requirements can be captured in use cases. Business requirements are supported by nonfunctional requirements (also known as quality requirements), which impose constraints on the design or implementation (such as performance requirements, security, or reliability). Generally, business requirements are expressed in the form "system shall meet the specific requirement," while nonfunctional requirements are expressed in the form "system should meet this requirement."

8.3.2.3 The user requirement(s) document (URD) or user requirement(s) specification is a document that specifies what the user expects the software to be able to do. User requirements and functional requirements are often combined into a single functional requirements document. The URD should be

developed with the input of end users, subject matter experts, and various business stakeholders. Once the required information is completely gathered, it is documented in a URD, which is meant to spell out exactly what the software shall do and becomes part of the contractual agreement. A customer cannot demand features not in the URD, while the developer cannot claim the product is ready if it does not meet an item of the URD. The URD can be used as a guide to planning cost, timetables, milestones, testing, and so forth. The explicit nature of the URD allows customers to show it to various stakeholders to make sure all necessary features are described. Formulating a URD requires negotiation to determine what is technically and economically feasible. Preparing a URD is both a science and an art requiring both software technical skills and interpersonal skills. The URD often includes priority ranking for each requirement. A typical URD might include mandatory requirements (M) that shall be built into the final system, desirable requirements (D) that should be included unless the cost is too high, optional requirements (O) that can be built into the system as needed, and potential future enhancements (E) that may be included in the final system but are largely features desired in and of themselves. It may also be useful to have a “won’t have” ranking which covers requirements that have been identified but, for whatever reason, have been rejected. Each functional and user requirement should be unique and precise, not contradict each other, and be measurable. Requirements should not be assumed; they should be specifically itemized and addressed in the various requirements documents and subsequently shared with the prospective vendors during the request for proposal phase. Requirements should support the business need; if it is not possible to link an identified requirement to the business case, its validity should be questioned. The functional and user requirements documents should be managed through a change control process.

8.4 Phase 3—System Design:

8.4.1 *Functional Requirements Analysis*—Depending on the system development life cycle or project methodology being followed, functional requirements may be defined in a separate document from the previously discussed user requirements. Where user requirements describe in business language what the system is required to do to support the business process, functional requirements on the other hand describe, in the language of the functionality of the system, how the system will satisfy the business requirements. In other words, functional requirements are the translation of business requirements into application-specific language. Functional requirements should be specific, measurable, and realistic to ensure testing can easily confirm that requirements have been satisfied.

8.4.1.1 When business and functional requirements are defined in separate documents, industry regulations may dictate that they be linked using a traceability matrix to provide complete traceability between business and functional requirements. Even if industry regulations do not dictate this, it is good practice to have this traceability in place. It can help ensure that all agreed user requirements will be included in the final system and ensure complete coverage of the system when it comes to testing.

8.4.1.2 Functional requirements may be documented in the form of a checklist of major features and functions of the system that will be configured and used. The laboratory informatics system functions map (Fig. 3) can be used as a starting point in developing a list of laboratory informatics system functions. Also see ISO/IEC/IEEE 29148 for a general guide on developing software requirements. Specific hardware and software standards in place at a particular laboratory or company may also need to be adhered to and therefore specified in functional requirements.

8.4.1.3 Functional requirements analysis is normally performed after the selection of the laboratory informatics system is complete. The selection will have been made based upon the supplier’s ability to meet the business requirements through analyzing how the selected system will meet the user’s needs. Clearly the supplier has a key role in developing the functional requirements document; however, the customer shall also be involved. At a minimum, the customer shall be available to clarify and answer questions the supplier may have and to review and approve the final document. Even if a process does not dictate the creation of a functional design, the supplier’s own quality system or process might. Alternatively, because of the size or nature of the project, the supplier may recommend the creation of such a document. In these cases, the supplier’s recommendations should be adhered to and time and funding made available for the creation of the document.

8.4.1.4 The importance or weighting of functional requirements should reflect the importance or weighting of the user requirements to which they relate. If user requirements have been identified but flagged as not to be included in the system, there is little point in developing the functional requirements for them. However, for clarity, they may be referred to in the functional requirement but flagged as not required.

8.4.1.5 Rapid prototyping development techniques may be used to assist users and the development team in elaborating high-level requirements into a detailed set of functional specifications. When working with COTS products, this process may be beneficial in helping constrain user expectations and system scope if the worst excesses of configuration and customization are to be avoided.

8.4.2 *System Design Document*—Following the completion of the functional requirement document, a system design document (SDD) that details the system design specifications may be produced. The system design document is a technical document that may be used by the suppliers or developers of the system to document the technicalities of the system, including items such as database table structures, specific algorithms, and low-level communication protocols. It is used as the reference document if a team of developers are to be employed in developing a system. Especially with a commercial LI system, a system design document may not be required. The user requirements and functional requirements provide the information required to implement the system based on the existing functionality and configuration or customization tools available. The supplier should be consulted as to whether a system design document is required. If a system design document is required, it should again be possible to trace each

element of the SDD back to the functional requirements and therefore back to the user requirements.

8.4.3 Test Plans: Once the functional design document and, if required, the system design document have been developed, formal testing documents can be developed as required. The development of testing strategies for software or software systems is outside of the scope of this guide. However, testing may include unit testing, system testing, integration testing, and some form of user acceptance testing. Organizations may want input from their IT or quality departments on the testing approach to take. Alternatively, the supplier will be able to provide advice on successful testing strategies that have been employed previously, or a third-party consultant could be used. When testing a system, it should again be possible to trace the testing carried out back through the functional requirements and to the user requirements.

8.4.4 Key Stakeholders in the System Design Phase—The design phase is where the users' identified and agreed requirements are transformed into how they will be included or built into the system. Interaction between the users of the system, the project team, and the suppliers will be important during this phase, and the key stakeholders will come from these areas. The supplier will be involved with developing the functional specification, but the project team members will also be involved, especially those responsible for developing the user requirements. Key users are also involved, especially if there are any questions or issues raised about the user requirements. On completion of the functional design, it shall be signed off by the supplier and authorized project team members. The quality or regulatory team may need to review and approve the document and the way it was developed to ensure it adheres to quality or regulatory requirements, or both. The system design document is generally a document used internally by the supplier in the development of the system. However, it may still be a good idea for the organization's project team to review and approve the document. It could also contain items that need to be reviewed and approved by the IT department and quality and regulatory departments. Key stakeholders for the test plans will be project team members and the suppliers as well as the people who will be developing and running any test scripts that are used. Quality and regulatory teams may also need to review and approve the test plans.

8.4.5 Deliverables of the System Design Phase—The functional requirements document agreed upon and signed by at least the responsible member of the project team and the supplier is likely to be the most important deliverable of this phase. If required, an agreed upon system design document may also be a deliverable. Formal test plans and test cases may also come out of this phase, as well as any other documents that the organization's project management function require including, for example, change control requests and logs.

8.5 Phase 4—Build/Configure:

8.5.1 System Configuration and Development—In this phase, the system software is configured, interfaces are configured or developed, any required customizations coded, and the components, including the IT infrastructure, are integrated into a complete system.

8.5.1.1 Configuration is often just seen as using settings and flags in commercial software to change the behavior of that software. The ability to successfully configure a full system in this way is dependent on the supplier anticipating all the possible variations in laboratory workflow, something that is difficult, if not impossible, to do. However, configuration can be seen in a broader sense. The configuration of a car is based on bringing together a set of standard options (engine size, transmission type, color, and so forth) from the supplier that has been designed to work together. By changing the configuration, two buyers can end up with very different versions of the same car. However, they are based on the same platform and standard components, are covered by a warranty, and can be maintained and serviced by the supplier. Similarly, a LIMS configuration brings together a set of functional objects to create a specific configuration of the system. It should be possible to manipulate these functional objects using sophisticated tools to create a system that meets the needs of the users (both graphically and functionally) but which continues to be supported by the supplier and can be upgraded as new versions of the software become available. The functional objects are not changed, but the interface and context in which they work do.

8.5.1.2 Customization is most commonly defined as software development activities that involve writing procedural programming code in the laboratory informatics system vendor's proprietary development language or other third-party languages. Furthermore, customization may alter the way base system functions were intended to be used or add new functions not present in the vendor's base product. Customizations may be built to make the system meet the exact needs of a specific organization if the configuration capabilities of the product do not provide the flexibility to do this. When customizing a system, take time to understand the impact of what is being done. Specifically, be clear on the ability of the supplier to support the system as well as address issues surrounding the upgradability of the system. For example, ask whether customization will make it difficult to upgrade the system as new releases become available; if it does, the system will be at risk of becoming outdated, unsupported, or obsolete. It is important to understand the system's fundamental design with respect to configuration and customization extensibility. If the software can be configured without compromising ongoing product warranty or support, the impact on project scope, testing/validation, and maintenance can be minimized. Modifying core system code may have more far-reaching consequences. There are many methodologies for system configuration, development, and testing. ISO/IEC/IEEE 12207 can adequately explain the different approaches to software development. The approaches range from phased implementation to rapid prototyping. Rapid development characterized by iterations of the design, build, and test activities, supported by a cross-functional team (representation from business, IT, and quality), can be an effective method for implementing LI systems; however, it does require the involvement of staff experienced in this way of running a project.

8.5.1.3 Custom code modules should be developed according to the defined quality plan if one exists. If using the supplier or third parties for the development, they should adhere to the quality plan; if no quality plan has been developed, the supplier or third party should at least be working to their own quality plan or procedures and shall be able to show they are adhering to these.

8.5.1.4 During the build/configure phase, especially where custom code is being developed, the quality plan or internal procedures may require elements of unit testing, integration testing, and system testing. In regulated environments, proof that these activities have taken place (validation) may be required. In unit testing, each developmental unit or module is tested against defined test parameters and scripts based on the previously developed requirement and specifications. Integration testing brings the modules together into functional elements and again tests them against defined test parameters. Integration testing is designed to test the entire system, including any interfaces that may have been developed for third-party systems or instruments. This testing is done as part of the build/configure phase by the parties responsible for building or configuring the system and is distinct from the more user-based testing of testing/commissioning.

8.6 Phase 5—Test/Commission:

8.6.1 *Validation*—Validation of the laboratory informatics system may be required for regulated industries. If validation is required, this will have been identified in the project initiation phase. Specific validation requirements exist for industries regulated by agencies such as the Food and Drug Administration (FDA), the U.S. Nuclear Regulatory Commission (NRC), and European Medicines Agency (EMA). Properly managed execution of validation principles that represent the use of best practices can save time, resources, materials, and expenses in the long run. Traditional validation methodologies such as those defined in GAMP can add significantly to the implementation time of a laboratory informatics system. The application of intended use methodologies or a risk-based approach can reduce time and cost within the regulated framework. Documentation plays a key role in the validation process for a laboratory informatics system. If validating a system in a regulated environment, the organization's quality or regulatory function should be involved from the start of the project, and the whole project should be run with reference to the validation requirements. A number of standards, documents, guides, and requirements are available that cover software validation and testing, including GAMP 5, IEEE 1012, IEEE 1028, and NRC 10 CFR Part 50.

8.6.2 *Testing*—The testing requirements should have been identified as part of the quality plan defined during project initiation. Testing will typically be based on following test scripts that define the steps to be undertaken when testing the system and its functionality. The test steps should be traceable back to individual requirements that have been defined and agreed upon to use. It is useful to identify the requirement or requirements that a test step relates to in the test script, allowing a full traceability matrix to be completed that links user requirements to the system design documentation and the delivered system. Even where agile or rapid development

techniques are deployed, testing is important and shall be traced to identified requirements. A pass/fail result should be recorded for each step, and it should be possible to enter comments as well. Comments for failed steps should be as comprehensive as possible and describe, in detail, the nature of the failure. Providing this detailed information will make it simpler to identify the root cause of the problem and to resolve it. Comments such as “It doesn't work” are not helpful when it comes to resolving identified issues. Test scripts may need to be run more than once to ensure identified issues are resolved. Part of the testing strategy will identify the data that needs to be in the system to adequately test the system. ISO/IEC/IEEE 29119-4 and ISO/IEC/IEEE 29119-5 provide further information on software testing.

8.6.3 *User Acceptance Testing*—Before a laboratory informatics system is released for production, all or some of the intended users are given an opportunity to evaluate whether the system does indeed support their daily activities and whether none of the users' daily tasks is impeded by the system. Scripted testing can help verify proper operation; unscripted testing may help uncover missed requirements.

8.6.4 *Training*—End user and system manager training require appropriate planning and resources. Training should be completed as close to the general operation date of the system as possible. Training covers all aspects of laboratory informatics system operation, from training on the day-to-day system functionality to system manager training on how to maintain the system. Training can be tailored to the roles defined in the system that control the functions to which the user has access. Consideration should be given to whether **formal training is mandatory for the use of the system, especially in regulated environments**, and, therefore, how training records will be maintained. If users move from one role to another, they may require training for that new role. The organizations own procedures and policies may determine if retraining is required at specific intervals; however, retraining may also be required if major upgrades or changes are made to the system during its lifetime.

8.6.5 *Documentation*—The required documentation should be in place. Documentation includes manuals supplied by the vendor and user-developed documents. Examples of vendor-supplied documentation include user operational manuals, technical reference manuals, and validation documents, including installation qualification (IQ) manuals if the system is to undergo formal validation and quality control documentation. User-developed documents include all SOPs and training documents. Ensure that all other required project and quality documents are available at this time, including change control forms, acceptance-testing records and test scripts, problem report logs, backup and recovery logs, audit reports, and security records. ISO/IEC/IEEE 26511 and ISO/IEC/IEEE 26512 provide further information.

8.6.6 Data Loading:

8.6.6.1 Master data (sometimes referred to as static data) can be seen as the management data within the system. In a LIMS, the master data would include elements such as users, products, specifications, test definitions, and calculations. A large laboratory with hundreds of tests, calculations, and

specifications can spend significant amounts of time entering and verifying master data, and this time shall be included in the project timelines. The ability to migrate data using automation (if it is possible) can greatly reduce costs in terms of time, transcription errors, and verification.

8.6.6.2 If moving to a new laboratory informatics system from an existing one, it may seem to be a good idea to migrate the active data to the new system. The active data is the data the system generates during its day-to-day operation. In a LIMS, this may include batch information, the samples that are created, the test assigned to the samples, the results entered, and the outcome of any specification checking. However, if data migration is to be carried out, it requires a carefully considered plan to assess the transfer of data and metadata from one system to another. The plan should account for synchronization of field types, field length, field mapping, and manual verification. (For further information on migrating information see 8.8.)

8.7 *Phase 6: Operation and Maintenance*—Operation and maintenance tasks include data backup, data recovery, various database management and optimization tasks, user account maintenance, resolution of user issues with the system via a support process, and general administration such as service contracts administration and software/hardware maintenance and upgrade as well as continuous improvement/system enhancement activities.

8.7.1 *Laboratory Informatics Staff and Organizational Placement*—Organizations may choose to have staff dedicated to the support of the LI solution or solutions on a full-time basis; this is likely to be the case for larger organizations. Other organizations may not be able to justify these resources and rely more on the supplier to support the system. However, in all cases someone shall have overall responsibility for the LI solutions, that is, be identified as the system manager. They will be responsible for ensuring day-to-day running of the system, liaising with suppliers, and managing changes to the system, including the implementation of new or additional functionality and the management of system upgrades. The computer and system skills required of the laboratory informatics system staff vary with the technology used. The ideal candidates for laboratory informatics system staff include personnel with both laboratory and computer experience. Laboratories have been successful in retraining existing laboratory personnel to acquire the appropriate computer skills.

8.7.2 *Security*—Policies and procedures need to be established to document users' access to data as well as privileges to update, insert, and delete data. Laboratory informatics system privileges should be assigned in terms of business needs, assuring data integrity, quality considerations, business ethics, segregation of duties, and customer confidence in privacy and safety of data. A procedure to document changes in users' privileges should be established covering the full life cycle of the user. Changes to data that have previously been identified as requiring auditing, for example, because of regulatory requirements, should be subject to audit trail recording within the system at the time the changes are made. Certain transactions within the laboratory informatics system may be subject

to electronic signature approval dependent upon the regulatory requirements the system is operating within.

8.7.3 *Backup and Archiving*—A backup policy needs to be established. The policy needs to detail the type and frequency of backups. Careful consideration should be given to the tolerance for data loss. Backup and recovery procedures shall be tested at regular intervals to ensure they work correctly. Additionally, data may need to be archived periodically to manage system storage space and performance. Similarly, considerations, tolerances, and tests should be considered for such archives. Note that backups and archives should not be assumed to be the same thing. The role of a backup policy is to provide assurance against data loss and should be based on the organization's tolerance for data loss. Archiving is used to manage storage space and performance.

8.7.4 *Disaster Recovery*—A disaster recovery procedure should be established. Disaster recovery exercises should be conducted at a specified frequency, including evaluation of business continuity plans. The disaster recovery procedures shall be tested at regular intervals to ensure the system works correctly.

8.7.5 *System Administration*—Special system software, audit trails, and LI system reports are used to monitor the fidelity of system data and information. New instruments may be connected to the LI system for transferring information. Links to external systems are maintained and serviced. Preventive maintenance tasks are performed per a predefined schedule. Repairs are conducted on failed hardware units. Software support is conducted with the laboratory informatics system vendor.

8.7.6 *Regulatory Requirements*—Laboratory informatics systems used in laboratories falling under a country's regulations should be compliant with that country's regulations and guidelines. Regulations and guidelines should be reviewed regularly to ensure the system remains compliant.

8.7.7 *Maintenance and Support*—Commercial laboratory informatics systems generally have maintenance agreements and services that cover technical support with varying degrees of service. The service agreements can include written or implied provisions for software upgrades and training as well as clear definitions of both user and vendor support expectations for the life of the arrangement. The service agreement should spell out how disagreements over service will be mediated and should be made a part of the contract with the laboratory informatics system vendor. Arrangements for escrow control of the source code may be made as part of the overall support and maintenance for the laboratory informatics system.

8.7.8 *Change Control*—Change control/configuration management plays an important role during laboratory informatics system operations. Changes in hardware, software, laboratory staff, and laboratory environment and requirements need to be carefully monitored and controlled according to the agreed change control mechanism. Examples of activities that trigger change control include the installation of product updates provided by the software vendor and customizations made by the system support staff.

8.7.9 Key Stakeholders in the Operation and Maintenance Phase—The operation and maintenance phase sees the LI system being used on a routine basis. During this phase, the system should not only operate reliably but also be capable of changing to meet evolving business needs and to accommodate potential updates to the underlying IT infrastructure. This includes updates to the LI software as they become available and if they are deemed necessary or beneficial. To help ensure correct operation, the correct individuals and functions shall be involved. This guide provides some guidance but is not a complete list of all possible stakeholders. The key stakeholders will include the person identified as the system manager, who should have overall responsibility for the system. Other key stakeholders will include the users of the system, and it can be a good idea to ensure at regular intervals that the system is still meeting their needs. The organization's IT department may be a key stakeholder if the system is running within their infrastructure and if they have responsibility for other IT infrastructure such as networking. They may need to be involved if upgrades to the system are planned. Other key stakeholders may include the quality/regulatory department, who should be involved with ensuring the system meets any new or changing regulatory requirements. It may be necessary to include the project management team if major updates or changes to the system are planned. The supplier (internal or external) is also a key stakeholder and it is worthwhile consulting them if major changes are required.

8.7.10 Deliverables of the Operation and Maintenance Phase—The deliverable of the operation and maintenance phase is a system that meets the identified and agreed requirements of the organization. During this phase, changes will be required. The implementation of these changes should be treated as projects in themselves and, therefore, be managed as described previously. Therefore, they will generate their own set of deliverables such as a business case, quality plan, and requirements.

8.8 Phase 7: System Retirement (Replacement, Archive):

8.8.1 Planning—An LI system may need to be retired for a number of reasons, including obsolescence of the system, closure of the facility, the ending of the program using the system, or a corporate directive for standardization on a different system. The retirement of an application needs to be carefully planned to avoid unintentional disruption to the business now or in the future. A comprehensive plan addresses the following elements: verification that business functions are no longer needed or will be available in a different application; identification of all users and dependent applications; inventory of hardware and software; a timeline for discontinuing or disabling functionality; archival of data in electronic or human-readable form; and a strategy for migrating data to the new system if necessary. The plan needs to be communicated to users and other parties that may be impacted by the retirement.

8.8.2 Verification of Obsolescence—This step is relevant in cases in which an application is not being replaced or is being replaced by an application with a different functional footprint. It shall be confirmed that the LI system can be taken out of service without negatively impacting the organization. In the

case of replacement, it should be verified that the new system will include any required functionality that exists in the current system.

8.8.3 Users and Dependent Applications—All users of the application and all owners of other applications that depend on it need to be identified so that they can be informed of the planned retirement. They should be informed of the possible impact of the retirement or replacement of the existing system. Users should also have the opportunity to comment on the planned replacement or retirement and any previously unforeseen impact it may have.

8.8.4 Inventory of Hardware and Software—One objective of a retirement may be cost savings. It may be possible to redeploy hardware and third-party software used by an application, or they may be retired altogether. This includes hardware and software that may be dedicated to nonproduction environments. An inventory of hardware and software associated with the LI system to be retired should be carried out. Maintenance and support contracts for retired systems should be cancelled.

8.8.5 Retirement Timeline—Retirement often encompasses many steps that need to be performed in a proper sequence to avoid disruption to the business. A detailed project plan and timeline need to be created, communicated, and managed. All parties need to be kept informed of ongoing progress and reminded of upcoming events. The retirement plans should be supported by a business case.

8.8.6 Data Archival—Laboratory data will almost certainly need to be maintained minimally for business and likely for regulatory reasons. Retention schedules will need to be adhered to, and these can vary greatly dependent upon the industry, company, and any regulatory requirements. With advances in technology and virtualization, companies may explore converting to flat tables for future retrieval. Data can be maintained in online read-only state with minimal infrastructure cost.

8.8.6.1 Other options include exporting data to a third-party provider to archive it and then retrieve when needed again. There are costs associated with this option, and there is risk of the vendor going out of business. Additionally, there may also be limitations with retrieving the data in a timely manner.

8.8.6.2 If data storage in human-readable form is required, one option is to print the necessary data to paper and archive it in a secure storage repository. Alternatively, the information can be converted into an electronic human-readable format, though the format should be chosen carefully to ensure it will be supportable in the medium and long-term; the PDF format is commonly chosen.

8.8.6.3 Careful considerations shall be given to how data are archived according to regulatory demands. For example, do the regulations require that the raw instrument data (such as a chromatograph) can be reprocessed using the original software (such as the chromatography data system)? If so, then the raw data may need to be preserved in its original form, and the original software may need to be preserved in a state where it can be run. This in turn may require the preservation of the hardware environment on which the original software ran.

8.8.7 Migration of Data into a New Application—When replacing an application with a new one, an opportunity may exist to leverage new capabilities within the solution and to move the existing laboratory data into the new application as part of the implementation project. When migrating data, business rules need to be established and applied in the requirements-gathering phase relating to how much historical data and which data are to be migrated to the new application. Data migration can be complex, time-consuming, and expensive. The feasibility of data migration depends on the complexity of the data and the data structures—including database types—that exist within the old and new system. It may be a relatively simple operation to migrate user information from one system to another. However, when it comes to migrating the linked batch, sample, test, results, product, and specification data from one system to another, or migrating stability study data (especially if there are currently running studies to be migrated), it becomes a much more difficult exercise and may require the development of complex data transformation routines and a complex and time-consuming validation exercise. Any organization contemplating data migration should carefully weigh the costs and benefits to make sure it makes financial and operational sense. Part of the analysis should include the level of automation associated with the migration. Some data may easily be migrated manually, whereas other data, because of volume/complexity, may be better served using a fully automated migration scheme. Additionally, the compatibility of the existing data with the type of database it is to be migrated to should be considered. For example, migrating data to a database from a different supplier may be more difficult than migrating to the same type of database. If the decision is to migrate data, then rules or policies shall be established by the organization and should take into consideration any compliance, risk management, legal, IT, and business requirements regarding how much and which data need to be migrated to the new LI system. The definition of these rules will be influenced by how the historical data will be used. If historical data are to be used in reporting, the organization may have additional challenges with any necessary data conversion or report enhancements, or both. Organizations should also investigate whether gaps will be created with other business systems as a result of migrating historical data. If an organization makes the decision not to migrate any—or migrate all—of the data and start with an empty, or at least not fully populated database, then a strategy should be formed regarding how to process out the operational data (especially any data from long-term studies that may extend beyond the introduction of a new system) and archive the historical data.

9. Lean Laboratory and Continuous Improvement Concepts within Laboratory Informatics

9.1 Lean Laboratory—Laboratory informatics can be used to support and facilitate lean concept implementation, resulting in significant benefits when they are integrated successfully. Lean concepts can be applied in laboratories to improve productivity, quality, and efficiency while reducing costs. Lean concepts most likely to be facilitated with laboratory informatics include workload leveling and flow, visual management,

continuous process improvement, and waste reduction. Each of these concepts is briefly described below.

9.2 Workload Leveling and Flow—Leveling strategies are used to balance the incoming workload and maintain a consistent workflow to make the best use of the resources available in the laboratory. Leveling is the smoothing of the variability of the incoming demand for work by ensuring that each work day has a consistent workload. Continuous flow concepts keep the work moving through the laboratory processes while minimizing queuing or backlog between steps. (Examples of areas in which workload leveling and flow can be applied include functions C-1, C-2, C-3, C-4, C-5, and E-14 of the laboratory informatics functions map in Fig. 3.)

9.2.1 Laboratory informatics systems contain the data needed to develop workload leveling and flow strategies. This includes expected average incoming workload demand, expected turn-around times for sample testing, actual testing times, optimal testing batch sizes, required sample result due dates by customer, current amount of work in the laboratory, and available staff and equipment.

9.2.2 Laboratory informatics can be used to automate the release of work into the laboratory based on the workload leveling strategy for each laboratory, thereby minimizing the scheduling and planning effort required to level the daily workload.

9.3 Visual Management—Visual management implementation allows quick assessment of workflow processes at strategic points and is meant to provide the opportunity to indicate whether a process is working optimally. (Examples of areas where visual management can be applied include functions C-3, C-4, and many of the E functions of the laboratory informatics functions map in Fig. 3.) Laboratory informatics can support this lean concept by visually displaying summarized data and compiling all needed information into one location to allow all users to identify quickly workload requirements as well as where review and action is required. Laboratory workflows can also be visually displayed by laboratory informatics systems, showing sample queues, sample locations, test status, samples/tests ready for review, and areas that need attention (that is, laboratory investigations).

9.3.1 Color coding, symbols, and icons that are easily understood and recognizable can be used to allow users to understand statuses and identify issues quickly. For example, dashboards can have colored gauges to represent the percent of work completed on-time, the current turnaround time against a six-month average, or the amount of scheduled work as a percentage of capacity.

9.3.2 Visual management dashboards can be used to provide real time electronic updates on sample status for customers of the laboratory.

9.3.3 Other examples include real time-control charts showing key performance indicators versus their warning and control limits, graphs of error rates pinpointing areas of opportunity for improvement, pop-up alerts indicating imminent deadlines, and so forth.

9.4 Continuous Process Improvement—Continuous process improvement tools are used to map actual workflow and can

help identify potential failure points or places where consolidation or separation of steps would be beneficial. (Examples of areas where continuous process improvement can be applied include many functions but in particular C-3, E-5, and E-13 of the laboratory informatics functions map in Fig. 3.) A key to success with continuous process improvement is to understand the workflow and identify waste at the ground level of laboratory processes, with subsequent implementation of small changes continuously rather than major changes all at once. Laboratory informatics can support this by rendering data onto dashboards and reports and into control charts and production graphs, including data such as error rates, turnaround times, inventory control, and so forth. These can be used to identify bottlenecks and vulnerable steps in the processes and also to monitor the effectiveness of improvements. The data in the laboratory informatics systems can be used to measure and monitor key performance indicators before and after implementation of changes, as well as inform future decisions.

9.5 Waste Reduction—Waste reduction as a concept covers many areas, all related by the goal to decrease the amount of effort or time that does not add value to the product from the customer's point of view. Continuous process improvement can be used to reduce waste in laboratory processes. Some key opportunities for waste reduction within laboratories are planning and scheduling work, reviewing and approving data, filing paperwork, documenting, and entering or transcribing data. The following are examples of waste reduction strategies that can result in significant benefits for a laboratory:

9.5.1 Review by exception is a waste reduction strategy that uses laboratory informatics systems to monitor key process parameters of mature, highly repeatable batch processes, and to evaluate them against specifications that have been configured and validated within the laboratory informatics system. Visual management tools such as color coding or symbols allow out-of-specification results identified by the laboratory informatics system to be flagged for laboratory analyst/supervisor review, while in-specification results are confirmed by the system and do not proceed to a manual review. (Examples of areas where review by exception can be applied include functions C-4 and E-9 of the laboratory informatics functions map in Fig. 3.) Examples of laboratory transactions that can use review by exception concepts include fit for use of equipment, raw materials, and consumables; training records; deviations from standard operating procedures; and so forth. Evaluation of only the failing parameters reduces the time spent reviewing and approving, resulting in a faster time to release, lower cost, and higher throughput.

9.5.2 Automation is another waste reduction approach to reduce time spent on processes in which there are set formulae, rules, or steps by using the laboratory informatics system to perform these types of transactions instead of a laboratory analyst. (Examples of areas where automation can be applied include functions E-6 and C-6 of the laboratory informatics functions map in Fig. 3.) Examples of these processes within laboratory informatics include calculations; batching of samples; parsing of data from instruments, spreadsheets, and tracking systems; passing or sharing of information from one

system to another; and data compilation. Automation of these processes removes the need for the secondary manual review of accuracy, allowing for more productive work to occur.

9.5.3 Paperless laboratory processes are waste reduction tactics to reduce the amount of activities executed on paper. Paper-based transactions can be error prone and require manual reviews to confirm accuracy, are difficult to search for data and information when there are investigations, and require physical handoffs that can increase wait times in laboratory processes. In addition, the paper itself creates wasteful nonvalue-added tasks, as paper shall be purchased, handled, filed, stored, and destroyed. (Examples of areas where paperless processes can be applied include functions E-1, E-6, E-8, and E-13 of the laboratory informatics functions map in Fig. 3.) Paperless laboratory processes have a high potential of reducing nonvalue-added steps, a key factor in implementing lean. Laboratory informatics is a critical component of paperless laboratory processes, as its associated systems are able to display, store, and transmit information electronically. Laboratory informatics systems are also highly searchable electronic storage systems that allow for rapid retrieval of stored items, or links to files stored in the informatics system. Going paperless with laboratory informatics includes activities like capturing data directly from a balance, pH meter, or other instrument; linking between systems to allow the sharing of one document without having it stored a second time; and covering the implementation of capturing logbook or notebook entries when a touchscreen or keyboard is used instead of a pen.

9.5.4 Mobile devices such as smartphones and tablet PCs that are able to receive notifications from an informatics system regarding imminent or actual issues or that are able to access inventory applications, dashboards, reports, and so forth, also support the lean concept of waste reduction. (Examples of areas where mobile devices can be applied include functions E-5, E-7, E-8, and E-10 of the laboratory informatics functions map in Fig. 3.) With such easily accessed information available, decisions regarding remedial or corrective action can be made in a timely fashion, resulting in a quicker resolution of issues, faster turnaround, and greater productivity.

9.5.5 Streamlining laboratory informatics support functions is important to both the initial implementation as well as keeping support costs low. Examples include use of leveling, flow and standard work, and visual management concepts for administrative and support functions like master data maintenance, help desk support calls, change control monitoring, and user account maintenance.

9.5.6 Laboratory informatics can support the implementation of lean concepts in many ways. Informatics systems contain the data needed to summarize and evaluate performance markers and processes. They are responsible for handling the import and export of data, and for the controlled access to those data. Laboratory informatics systems are key elements in the improvement of productivity and efficiency, as well as the reduction of time and effort spent processing laboratory work, decision-making, and improving laboratory performance.

10. Data Integrity, Cyber Security, and Protection of Personally Identifiable Information

10.1 *Data Integrity*—Laboratory data shall be accurate, reliable, and protected. Therefore, data within the laboratory informatics system shall be protected against data's loss, improper or inconsistent use, unauthorized manipulation, obscuration, and distribution that violates privacy.

10.1.1 *General Requirements*—A laboratory informatics system's ability to ensure data integrity is fundamental. Therefore, a complete data governance program shall start with a thorough understanding of data integrity concerns, whose rigor should be driven by the requirements of the business. These requirements shall take into consideration the criticality of data as it contributes to the system's intended use, the impact on the safety of the subject or environment, how it may affect the quality of the product or service, the regulations of governing and certifying agencies, and the validity of decisions derived from it. Although much attention is given to computer system data integrity, all systems (people, machines, and methods) shall fully meet the data integrity standards set by the laboratory's business. Because many organizations already publish standards and guidance regarding data integrity, refer to the following for more detailed information:

10.1.1.1 FDA Data Integrity and Compliance with CGMP: Guidance for Industry.

10.1.1.2 MHRA GxP Data Integrity Definitions and Guidance for Industry.

10.1.1.3 GAMP Guide: Records & Data Integrity.

10.1.1.4 PIC/S Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments.

10.1.1.5 WHO Technical Report Series, No. 996, Annex 5 Guidance on good data and record management practices.

10.1.2 *Key Considerations*—When evaluating the data integrity compliance of a system, the following should be considered:

10.1.2.1 Data are facts and statistics that have been collected for reference or analysis. Data should be attributable to the person or system that generated it, legible and permanent, contemporaneously recorded, original or true-copy in nature, and accurate. These principles—referred to as ALCOA—ensure the completeness, consistency, accuracy, content, and meaning of data endures through its life cycle, and that it is controlled against intentional and unintentional changes. Laboratory informatics solutions are designed to emphasize the importance of data by supporting the principles of data integrity: data should be attributable, legible, contemporaneous, original, accurate, complete, consistent, enduring and available (ALCOA+5; from the PIC/S DI standard, which incorporates a bit more than the standard ALCOA principles).

10.1.2.2 Raw data (source data) represent original records (or true copies) of data, metadata, transformations, and reports of data that have been generated or recorded at the time of the activity. Raw data is stored as a permanent record on durable storage in any format (textual, audio, visual, and so forth) and may be used to reconstruct fully and evaluate the activity when—and to the extent that it is—required. While a true copy of manually captured data may be created later (for example,

an accurate reproduction such as scanning to PDF), the reverse does not constitute raw data (raw data captured and stored electronically then printed to paper).

10.1.2.3 Without metadata, raw data has no meaning. Therefore, the laboratory informatics system shall protect metadata against loss and misuse using the same data integrity standards as applied to raw data. Further, the relationships between the data and the metadata shall be preserved in a secure and traceable manner so that they are perpetually available during data review and reporting activities.

10.1.2.4 The need to control the integrity of data, such as GxP records, is often driven by government, industry, or market regulations. When referring to GxP records, all data generated to satisfy the regulation—both manual (paper) and electronic—shall be maintained in a manner that complies with this section. Data may only be excluded when there is valid, documented, and scientific justification that the data is anomalous or nonrepresentative. The justification for the exclusion shall be documented and remain available for consideration during data review.

10.1.2.5 As a type of metadata, preservation of audit trails and logs is vital. Laboratory informatics systems shall provide details of those items required to reconstruct the manufacturing process or analytical activity. These details attribute the creation, addition, deletion, or modification of data to an individual (or when automatically generated, an original data source), and they include the date and time of the action as well as the reason or context within which the action occurred. When automated audit trail functions are not available, an equivalent control may be implemented in the form of an audit log or book, which describes the nature and justification for the change. As with all metadata, the audit trail shall be available during data review activities. Complex informatics systems—including LIMS, ELN, SDMS, and CDS—should include robust audit trail reporting, which is validated and readily available during review and inspection activities.

10.1.2.6 Electronic signatures should be supported and controlled by complex laboratory informatics systems. These may be comprised of any symbol—or series of symbols—that have been executed and adopted by an individual and declared to be legally equivalent to the full handwritten signature, initials, or other signing. For specific requirements on electronic signatures, how they are controlled as electronic records, executed and maintained by the individual, and declared for use in record submission, refer to the laws and guidelines of the regulating agency.

10.1.2.7 Regulatory agencies are placing more emphasis on the protection of electronic data that is placed in temporary memory in which amendments or deletions can occur before creating the permanent record and alterations are outside the audit trail function. Laboratory informatics systems shall minimize the length of time that data is stored in temporary memory, which minimizes the opportunity to manipulate data into achieving desired results.

10.1.2.8 Routine data review should be a supported feature of the laboratory informatics system. Data review should be able to be performed in a manner that is clear and accurate and

include raw data, metadata, audit logs, and electronic signatures. Routine data review is applied to an individual data set and should reflect the organization's risk-based strategies. This review is not intended to be exhaustive or forensic, and methods to streamline the review should be considered, for example, conducting reviews by exception.

10.1.2.9 A periodic data review is used to verify that data governance controls are adequate to maintain the data integrity of the system and consider the possibility of unauthorized activity. This review should include the integrity of data throughout its life cycle, changes to system configuration, security and user roles, operational permissions, the ability to archive and restore data, and the training on and procedures of the system. The review should include an exhaustive analysis of certain critical datasets to verify the effectiveness of existing data integrity control methods.

10.1.2.10 The design of a laboratory informatics system should include a fundamental consideration for the data integrity topics discussed in this section. Complex informatics systems should specifically be designed and delivered out of the box to have detailed audit trails, configurable and compliant electronic signatures, robust data and audit review options, and automatic handling for data manipulations that may occur in temporary memory.

10.1.2.11 Good risk-based strategies shall be used when designing data integrity control methods within the laboratory informatics system. These strategies shall be fully documented and include all supporting rationale. Control measures should reflect a rigor that is equal to the data's criticality; the risk associated with quality, safety, and efficacy of decisions; and the complexity of the system.

10.1.2.12 Documentation of and training on data integrity control measures in the laboratory informatics system should be a part of an overall data governance program. Staff operating and supporting the system shall be trained on these control measures and shall be able to detect data integrity issues when they occur. Training should cover how data integrity controls reduce errors and omissions and how they are used to improve the safety and quality of the product or service.

10.2 *Cybersecurity*—Much like quality management, cybersecurity is a topic that affects many aspects of an organization.

10.2.1 *General Requirements*—Organizations shall develop a comprehensive security program that analyzes, assesses, and mitigates vulnerabilities of the laboratory informatics system, as well as other connected systems and peers, both upstream and downstream. For further information, refer to the following standards and resources:

10.2.1.1 National Institute of Standards and Technology (NIST) Cybersecurity Framework,

10.2.1.2 ISO/IEC 27000 family of standards, and

10.2.1.3 Open Web Application Security Project (OWASP).²⁸

10.2.2 *Key Considerations*—These are the key considerations used to evaluate and verify the security of a laboratory informatics system. These can be applied to any system, whether it is hosted internally or provided by a third party such

as in the cloud. These are minimum requirements and may be expanded for systems and data that have less risk tolerance.

10.2.2.1 The organization's information security management system (ISMS) should include cybersecurity as it relates to the laboratory informatics system. The ISMS should include defining the characteristics of the organization; risk assessment approaches; risks unique to the organization; and a suitable methodology to identify, control, and manage these risks.

10.2.2.2 An organization's ISMS should include a comprehensive set of policies and procedures and the appropriate training programs for IT personnel, developers, administrators, and end users. Training records should be kept and maintained as being accurate and updated to include new cybersecurity threats.

10.2.2.3 The laboratory informatics system—and its connected systems and clients—shall be protected against damage to hardware, software, data, and services. This includes controlling access to physical resources and property, and protecting them against harm coming across electronic connections such as Ethernet, Wi-Fi, serial ports (RS-232), energy infrastructure, electromagnetic radiation, and more. The organization's ISMS should likewise warn against cyber vulnerabilities that come through human interaction such as trickery, social engineering attacks, and intentional or inadvertent misuse. While policies and training are sufficient for many physical and human vulnerabilities, electronic measures shall be taken to secure servers, including a network firewall, malware protection, and active scanning to detect network penetration attempts.

10.2.2.4 The realm of web application security is constantly changing. Organizations shall receive regular updates on cybersecurity vulnerabilities that may apply to the laboratory informatics system. The OWASP maintains *The Ten Most Critical Web Application Security Risks*,²⁸ which is a good indicator for risks that may affect any application with computer network access, particularly those operating on the internet. Those ten risks include SQL, OS, and LDAP injection; broken authentication and session management; sensitive data exposure (for example, healthcare, PII); XML external entities (XXE); broken access control; security misconfiguration; cross-site scripting (XSS); insecure deserialization (hostile serialized objects); use of components with known vulnerabilities; and insufficient logging and monitoring. The organization's ISMS should require regularly scheduled scans of all network-connected applications and assess and mitigate the findings according to the risk-based strategy.

10.2.2.5 The security of mobile devices connecting to the laboratory informatics system is an often overlooked area of concern. Key cybersecurity considerations include the use of connection encryption (for example, VPN), the protection of stored data (for example, encryption), the use of security software and settings, and the implementation of malware protection (against viruses, worms, Trojans, and so forth).

10.2.2.6 Certain laboratory resources—for example, equipment and instruments—may connect to the laboratory informatics system as internet of things (IoT) devices. Some of these might be rudimentary, while others might have complex embedded software. These IoT devices can act as inputs to, and

can sometimes cause actions within, the laboratory system. Therefore, IoT devices shall be controlled according to the organization's security policy as if they were a computing device. Such security policy should take into consideration controlling for authorized access (for example, uniquely authorized accounts, secure APIs), using a firewall or proxy, updating device firmware and embedded software, limiting access to only required functions, and implementing risk-based validation testing.

10.3 Protection of Personally Identifiable Information—Increasing globalization and utilization of electronic communication platforms are driving the need to redefine personal data privacy and security requirements. There are many national and international privacy-related laws that regulate the collection and use of personal data. Some apply to a certain type of data such as financial or health information. Others apply to certain data-related activities such as telemarketing and commercial e-mail. In addition, there are broad consumer protection laws that are not privacy laws per se but have been used to prohibit unfair or deceptive practices involving the disclosure of, and security procedures for, protecting personal information. Whereas legal statutes and requirements may vary between countries, some fundamental concepts apply to the protection and exchange of personally identifiable information. While it is not the intention of this guide to offer a definitive definition for personal data privacy, laboratory informatics implementations should consider certain fundamental elements of data privacy.

10.3.1 Developing a Data Privacy Program—The data privacy program should be based upon clear quality and legal directives with a focus on customer permission and protection and define an escalation process when data breaches are discovered. It shall be readable and understandable by everyone, yet be detailed enough to give specific guidance for handling data securely and with privacy in mind. Training, the assignment of roles and responsibilities, and the certification of those who might be exposed to data are important. In addition, a company's program should consider the protection of data being exposed to individuals outside of the company that may include vendors, consultants, and third-party data hosts, including data administrators as well as cloud and backup service providers. The data privacy program should include these eight guiding principles:

10.3.1.1 Collection and usage of personally identifiable data shall be based upon legitimate grounds with transparency as to how the data is collected, used, and distributed. An individual shall be given an appropriate privacy notice when collecting personally identifiable information.

10.3.1.2 Personally identifiable information shall be obtained only for specified and lawful purposes and not further processed such that it would no longer align with those purposes.

10.3.1.3 The personally identifiable information to be retained shall only consist of what is required for the reason that has been fully disclosed with no more data collection than is necessary.

10.3.1.4 Personally identifiable information shall be accurate and kept current as needed. Reasonable steps should be

taken to ensure that the data collected is clear and accurate with a consideration as to when and if it is necessary to update the data.

10.3.1.5 Personally identifiable information should not be kept for longer than necessary. Practices should be in place to establish length of retention and archival needs with policies that ensure secure deletion of data when it goes out-of-date.

10.3.1.6 Individuals have the right to a copy of their personally identifiable information. Once agreeing to supply their data, individuals have rights to object to use of that data that may cause harm or distress. In addition, the individual has the right to prohibit direct marketing of their data as well as use of automated decision making involving their data. It is the right of the individual to have any inaccuracies associated with their data corrected, blocked, erased, or destroyed, or combinations thereof. Such inaccuracies may lead to individual compensation for any damage resulting from the misuse, lack of security, or breach of security of their personal data.

10.3.1.7 Organizations that collect or store personally identifiable information are required to have appropriate security to prevent data from being compromised. Therefore, it is important that policies, procedures, and electronic systems are designed and implemented to accommodate the type of personal data being collected, processed, and stored. Policies and procedures should clearly indicate who is responsible for the security of data and require that audit trails, security logs, and intrusions are reviewed on a periodic basis. Policies, procedures, and systems should have the appropriate security and backup. This requires reliable and well-trained personnel to mitigate and minimize appropriately the impact of any type of security breach of personal data.

10.3.1.8 Given the different statutes and regulations across countries, transfer of personally identifiable information is complicated. For international transfer of data, equal measures shall be in place and maintained to ensure individual rights to personal data protections. Depending on the context, it may be necessary to notify individuals of the purpose of the transfer. Specific agreements may be required to define how the transferred data will be used, processed, and stored.

10.3.2 Summary—In general, the protection of personally identifiable information within the laboratory informatics system shall be a key consideration that starts during early system planning, continues through implementation and regular operation, and ends only when the data has been destroyed. Developing clear and focused policies and procedures is critical to complying with a complex regulatory environment both domestically and internationally.

11. Laboratory Informatics and Artificial Intelligence

11.1 Introduction—Artificial Intelligence (AI)—AI represents a new area of informatics, and the applications of these are evolving rapidly. AI was inspired by the structure of the human brain. At the core of AI is the ability to “learn” from experience without being explicitly programmed to do so and improve upon that learning in the future. This ability is referred to as machine learning. This happens, in part, by looking at a dataset, finding relationships, expanding on existing keywords to find more relevant ones, and expanding search algorithms

and variables to make discoveries from the results. AI has the potential to take real world—and real time—actions when operated in conjunction with laboratory IoT devices. As such, judicious use of AI is required until the technology is mature, including taking special precautions when analyzing its security vulnerabilities. If an AI function operates a laboratory informatics system as would a human user, then validation of the AI-enabled function should verify that it has no more access or operational rights than its human counterpart. Organizations using AI or machine learning in the laboratory should continue to monitor academic literature and the laboratory informatics industry for current risks and mitigating practices.

11.2 Machine Learning and Training—It is important to understand that training is not “learning” in the human sense, and it is not AI either. Training in the AI world refers to the numeric optimization for a set of model parameters to minimize a cost function. Without explicit instructions, the algorithm can infer from past learning, with continual iterations until the correct target is reached, leveraging the cost function. What is very exciting in machine learning is that once the training examples have been identified, the remainder of the “learning” process is purely a computational problem that does not directly involve humans. The ability to develop artificial intelligence by applying machine learning to the task at hand could allow laboratories to automate a number of processes. This could range from rudimentary tasks such as the development of an on-demand how-to-guide, report generation, and trend analysis to more advanced functions such as predictive analysis.

11.3 AI in Laboratories—There are numerous uses for AI-based approaches within the domain of laboratory informatics, chiefly among them the creation and definition of unique ontologies, that is, a set of concepts and categories in a subject area or domain that shows their properties and the relations between them. Using the above paradigm rules for data mining, aggregation, transformation, and reporting of data can be facilitated and used to support interrelationship analysis and begin to understand sublime relationships between different domains of data, which may be nonobvious and help drive decision rationale. Largely, AI has been relegated to interpreting text, voice, movement, and tactile expressions to improve the human-machine interface for various systems and applications of visualization of complex data sets. Additionally, so-called intelligent data mining and pattern recognition are vital in the analysis of raw and post-compute analysis sets such as genomic data. AI systems are also used to monitor infrastructure and interpret events that are too miniscule or transient to be interpreted in real time by human senses. Monitoring, interpreting, collating, and acting in an automated and systematic self-directed manner can provide a host of meaningful applications that free human operators from menial or, conversely, dramatically complex tasks to focus on transforming data into information and ultimately organizational knowledge.

11.3.1 Benefits of AI in the Laboratory—The benefits of AI-enabled tools are many. Chiefly among them is the ability to

process previously unmanageably large volumes of data and provide analysis of multivariable problems simultaneously while determining trends, outliers, and counterintuitive complex patterns not noticeable to humans and without emotional or intellectual bias. AI-based tools can capture, organize, and automate the addition of organizational knowledge from disparate sources while providing an ability to diagnose root causes of observed events to enable organizational enterprise decision support. It is possible for AI to apply existing knowledge to parse, analyze, and classify text- and voice-based data sources to provide visualizations to facilitate human analysis of these complex information forms and create data collections. Various AI applications are used to provide additional layers of security by using new ways of storing and managing data such as decentralized, deconstructed, disparate data stores. Another benefit of AI-based tools is the ability to create constantly increasing data stores from multiple sources to provide the simultaneous ability to enhance sharing of knowledge and information as timely as the information is collected and potentially transformed. AI concepts like knowledge mapping and ontology generation can help to identify and increase the likelihood of organizing data to create information and ultimately organizational domain-based wisdom. Data integrity checking of large data sets in near real time is a possible fit for AI in the laboratory.

11.3.2 Challenges of AI in the Laboratory—While the benefits are numerous, there are challenges to implementing AI technologies as part of core laboratory informatics landscape. The technological expertise of AI is typically lacking in laboratory environments, as well as end user buy-in for the need. This implies a shift in mentality of entire organizations, the requirement for significant upfront investment, and natural difficulties associated with implementation and support of AI systems. It is also a given that AI produces the most benefit from analysis of large data sets. However, depending on the industry, such data might only exist in paper and not digital format. This will inherently limit AI’s ability to connect the dots and will require additional organizational effort to convert data to digital format. Another factor to consider is the risk versus reward of this solution. The initial payoff associated with automation of trivial tasks is not significant, and true payoff is not realized until maturity of AI is achieved. However, AI automation can result in over-reliance of the organization on it for its success. As such, a flaw in AI learning could result in significant impact to the organization. Lastly, as AI learns more and is able to find desired relationships across desperate data points, the ability to validate this result and trace the path to conclusion becomes virtually impossible. The human nature of distrust towards AI indicates that persons will need to undergo a similar level of learning in order to arrive at the same conclusion. This process will be more challenging for more desperate data sets and more complex findings. Given these challenges, AI in the laboratory will evolve slowly and will be most likely incorporated into existing laboratory informatics solutions as enhancements to existing functions.

12. Keywords

12.1 chromatography data systems, CDS; electronic laboratory notebooks, ELN; laboratory execution systems, LES;

laboratory informatics; laboratory information management systems, LIMS; laboratory information systems, LIS; scientific data management systems, SDMS

APPENDIX

(Nonmandatory Information)

X1. LABORATORY INFORMATICS FUNCTIONAL REQUIREMENTS

X1.1 This functional requirements checklist covers functionality common to the various laboratory informatics systems discussed throughout this guide as well as requirements recommended as part of this guide.

X1.2 The laboratory informatics functional requirements checklist is an example of typical requirements that can be used to guide the purchase, upgrade, or development of a laboratory informatics system. While comprehensive in scope, note that this checklist is not meant to be exhaustive.

X1.3 *Use of the Laboratory Informatics Functions/Requirements:*

X1.3.1 The checklist is set up as a spreadsheet with ten columns, as described in the following:

X1.3.1.1 *Left-most Columns*—Separate columns exist for the major categories of laboratory informatics systems. Laboratory informatics functions can be satisfied by one or more of the major categories. Users of the checklist can include notations here to indicate whether the requirement is important to the proposed implementation. Recommended symbols to use here are “S” for suggested and “R” for required.

- (1) SDMS—Scientific data management system.
- (2) ELN—Electronic laboratory notebook.
- (3) IDS—Instrument data system (that is, chromatography data systems).
- (4) LIMS/LIS—Laboratory information management systems/laboratory information system.
- (5) OTHER—Users can let this refer to any other systems not otherwise listed.

X1.3.1.2 *#*—The requirement number, with reference to the laboratory informatics functions map in Fig. 3.

X1.3.1.3 *Laboratory Informatics Functions/Requirements*—The individual requirement.

X1.3.1.4 *C—Criticality*—Users can specify the overall importance of the requirement. Two examples for scoring criticality are as follows:

- (1) Simple: On a scale of one to ten.
- (2) Complex:
 - (a) *Out-of-the-box/user-configurable*: 5—The functionality is available without modification to the application or through configuration settings available to the end user.
 - (b) *Administrative configuration*: 4—The functionality is available through configuration settings or tools intended for use by trained application experts.

(c) *Minor administrative customization or dependence on third-party product included*: 3—Functionality is available through a third-party product included or delivered with the system and intended to be performed by system experts.

(d) *Major administrative customization or dependence on third-party product not included*: 2—Functionality is available through a third-party product not included or via advanced programming by client.

(e) *Partially met requirement or supplier-supported customization required*: 1.

(f) *Not available or supplier customization is not supported*: 0.

X1.3.1.5 *S—Scope*—Users can specify whether the requirement should be considered in- or out-of-scope.

X1.3.1.6 *P—Phase*—For projects that are to be implemented in phases, this column can be used to denote the implementation phase.

X1.3.2 *Guidance on Use of the Laboratory Informatics Functional Requirements:*

X1.3.2.1 Laboratories can complete the checklist as part of the requirements definition process.

X1.3.2.2 Laboratories can submit the checklist to prospective suppliers / vendors (with or without input from their own requirements process) as part of the vendor selection process in which each vendor self-scores.

X1.3.2.3 Laboratories can use the checklist as part of a formal scoring process during a vendor selection/demo evaluation process.

X1.3.2.4 Weighting/criticality can be visible or hidden from prospective vendors.

X1.3.2.5 Where multiple vendor solutions or third-party tools, or both, are used to deliver the laboratory informatics solution to the laboratory, each function should be tied to the vendor solution or third-party tool that is delivering the function.

X1.3.2.6 Each laboratory needs to consider unique requirements to their specific industry or type of laboratory.

NOTE X1.1—Regarding criticality scoring, it is important to identify clearly functionality that has been customized solely for the purposes of a vendor demonstration. This functionality will likely not exist in the core product and should not be considered a “5” on the complex scale.

X1.3.3 The requirements section headings (Table X1.1) map to the laboratory informatics functions map of Fig. 3 where applicable.

TABLE X1.1 Requirements Checklist

SDMS	ELN	IDS	LIMS/LIS	OTHER	# (Fig. 3)	Laboratory Informatics Functions/Requirements	C	S	P
Sample/Experiment Registration (Section C-1)									
					C-1-1	The system should allow for the registration of samples prior to, or after, physical sample collection, for example, via preregistering (before) and registering (after collection).			
					C-1-2	The system should allow for the generation of user-definable sample labels, including the use of bar codes.			
					C-1-3	The system should allow for automated sample registration via different triggering mechanisms, including, but not limited to, calendars, web requests, and system-to-system methods (through API, web services, database table, and so forth).			
					C-1-4	The system should allow for inclusion of metadata (for example, lot, demography, source, sampling point, and client information), depending on the industry (laboratory type) where it is implemented.			
					C-1-5	The system shall allow for the inclusion of the requested tests, the ability to modify predefined tests, and the ability to add/remove tests.			
					C-1-6	The system should include safety information regarding the submitted material, depending on the industry or domain where the system is implemented.			
					C-1-7	The system should include the ability to use predefined metadata templates, as well as the ability to create ad-hoc samples, single samples, or multiple samples, or combinations thereof.			
					C-1-8	The system shall have the ability to define sample collection details such as container types and sizes, collected by name, collected date/time, container size, number of containers, preservation method, retention period, holding time (prescribed by QAPP in environmental, CAP for clinical), and so forth. It should also handle storage location, preservation method, and associated safety hazards.			
					C-1-9	The system should provide for additional free text or listed descriptive information about a sample to be captured, such as conditions as the time of sampling or any observation made during collection..			
					C-1-10	The system should allow for creation of user-definable sample registration preferences/methods or input screens, or both.			
					C-1-11	The system sample receipt functionality should include the ability to record sample delivery information (location, time stamp, resource) for a preregistered sample.			
					C-1-12	The system shall assign a unique identifier to each sample using an incrementing integer or user-defined naming format.			
					C-1-13	The system shall provide facilities to acknowledge the physical receipt of sample material and the time of receipt in the laboratory (known as receipt or receiving).			
					C-1-14	The system shall provide a mechanism to compare received samples against customer or project sampling requirements to identify variances. It shall also be able to share such requirements with submitters.			
					C-1-15	The system should allow for documentation of undesired or unexpected characteristics of submitted samples.			
					C-1-16	The system should allow for recording of sample preparation activities. This is normally done in an ELN.			
					C-1-17	The system shall allow for the recording of every sample's distribution to personnel and location at all times while the sample is in the laboratory's possession (that is, aliquoting and chain of custody).			
Sample Management (Section C-2)									
					C-2-1	The system shall record chain of custody for standards and reagents.			
					C-2-2	The system shall allow a user to record the storage location for any standard or reagent.			
					C-2-3	The system shall allow for the restandardization (assigning a new standard value to a standard that had been previously standardized) of existing standards and reagents.			

TABLE X1.1 *Continued*

SDMS	ELN	IDS	LIMS/LIS	OTHER	# (Fig. 3)	Laboratory Informatics Functions/Requirements	C	S	P
					C-2-4	The system shall require the date to be recorded when a standard and reagent is first opened.			
					C-2-5	The system shall permit only active standards and reagents to be available for use.			
					C-2-6	The system shall allow authorized users to designate a standard or reagent as no longer available for use in sample testing.			
					C-2-7	The system shall provide for a mechanism to group logically associated samples together. The logical grouping shall be based on identified sample or test metadata (for example, sample type or product, test method, user workload assignment, or sample/test status).			
					C-2-8	The system should provide a facility for identifying a physical sample in the system (for example, barcoded label).			
					C-2-9	The system should provide the ability to track manually the inventory amounts of samples.			
					C-2-10	The system shall be able to assign tests and specifications for standards and reagents.			
					C-2-11	The system shall provide an option to create a sample and test to capture data for calibration, preventative maintenance, or service-event-related testing.			
Core Laboratory Testing/Experiment (Section C-3)									
					C-3-1	The system shall be able to record the standards and reagents used in laboratory testing and record them for each sample/test during results entry.			
					C-3-2	The system shall, by default, only allow users to select standards and reagents that are approved for use and not expired.			
					C-3-3	The system shall prevent the use of a media that will expire during testing.			
					C-3-4	The system will have the ability to check limits (physical, control, specifications) for an instrument sample as well.			
					C-3-5	The system shall be able to enforce the rule that only certified laboratory personnel may record or perform testing in the system for the test methods or procedures that they are certified to perform.			
					C-3-6	The system should allow result entry for all tests on a single sample and for multiple samples for a single test.			
					C-3-7	The system should allow for a user-definable result entry method, such as upload from a spreadsheet.			
					C-3-8	The system should allow tracking of outsourced samples for aliquoting, chain-of-custody, results entry, and approval.			
					C-3-9	The system shall be able to perform or assign a retest workflow. When assigning the retest, the system or authorized users shall be able to determine a retesting due date and whether a new sample is required.			
					C-3-10	The system shall be able to define the data type for test results. The following test result data types minimally shall be supported: floating point/real number, integer number, text, data, list, file, calculated, Boolean (yes/no or true/false), and interval.			
					C-3-11	The system shall include the ability to enter operators such as <, >, +, – with numeric test results.			
					C-3-12	The system should allow for intra- and inter-sample/ test calculations to be performed, including the use of advanced math functions.			
					C-3-13	The system should provide a warning to users for entry of out-of-specification test results (audible, color code, icon, screen message.)			
					C-3-14	The system shall be able to record all instruments used during testing.			
Results Review/Verification (Section C-4)									
					C-4-1	The system shall be able to verify specification limits for accuracy or tolerances and indicate to the user when the values are exceeded.			
					C-4-2	The system shall auto-authorize verification check tests and samples if they are within specification limits for interfaced instruments.			
					C-4-3	The system shall provide a full audit trail for modified results.			

TABLE X1.1 *Continued*

SDMS	ELN	IDS	LIMS/LIS	OTHER	# (Fig. 3)	Laboratory Informatics Functions/Requirements	C	S	P
					C-4-4	The system should provide one or more levels of review and interpretation of results.			
					C-4-5	The system should flag out-of-range results for further evaluation.			
					C-4-6	Entered results should be compared with predefined specification limits from master data and indicate to the user when the results exceed specification limits.			
					C-4-7	The system should provide an audit trail for all changes made to records in the system and should include personnel's identities, time/date of activity, and reason for such activity, wherever needed.			
Sample/Experiment/Study Approval/Verification (Section C-5)									
					C-5-1	The system should allow recording of the final disposition of samples.			
					C-5-2	The system should allow reviewers and approvers to have access to relevant metadata for each sample/test result, the instruments and reagents used, raw data from instruments, and reports during the review and approval process.			
					C-5-3	The system should provide appropriate functions to handle retest, resampling, recalculation, OOS initiation, and disposition of sample statuses like approved and rejected.			
Reporting (Section C-6)									
					C-6-1	The system shall have the ability to store multiple files (such as a PDF of the vendor certificate of analysis) in electronic format and associate it with a standard or reagent and samples tested.			
					C-6-2	The system shall alert users in advance for samples that might be out-of-spec or not meet expected turnaround time requirements.			
					C-6-3	The system should substantiate the status of verified results, for example using a certificate of analysis.			
					C-6-4	Generation of a certificate of analysis should be allowed when the results and associated metadata are available, verified, and approved by authorized persons.			
					C-6-5	The system shall provide the ability to generate reports for pending samples for approval, out-of-spec samples/ tests, instruments pending calibration, expired reagents to be disposed, certificates of analysis, trend charts of laboratory performance, and so forth.			
					C-6-6	The system should provide the means to interface with a third-party reporting tool.			
					C-6-7	The system should allow for custom reporting configurations in conjunction with standard reports. Examples include, but are not limited to, sampling route lists, sample registration reports, work and backlog lists, laboratory performance reports, instrument reports, graphical reports, statistical analysis reports, regulatory reports, audit trail reports, incident reports, chain of custody reports, quality assurance reports, and inventory analysis reports.			
Document Management (Section D-1)									
					D-1-1	The system should have document management capability that can help in creating various types of documents, as well as allow for the review and approval of documents with version and release controls.			
					D-1-2	The system shall provide readily accessible electronic documents, such as standard operating procedures, instrument manuals, and other applicable documents to the designated staff.			
					D-1-3	The system shall have the ability to designate the most current version of the document and archive prior versions.			
					D-1-4	The system shall be capable of associating the applicable standard operating procedure revision with the test performed using that revision.			
					D-1-5	The system shall maintain test methods applicable to the testing performed with sufficient information for the test method, method reference, specifications, assigned limits, holding times, and other information pertinent to the testing performed as required by the reference method or laboratory.			

TABLE X1.1 *Continued*

SDMS	ELN	IDS	LIMS/LIS	OTHER	# (Fig. 3)	Laboratory Informatics Functions/Requirements	C	S	P
					D-1-6	The system should be able to maintain training records and allow personnel to be associated with approved methods.			
					D-1-7	The system shall be capable of linking applicable documents to individual sections/tables/forms/modules within the system.			
Resource Management (Analyst Training Records, Analyst Training Verification, Qualification Status, Job Description, Time Tracking/Performance Monitoring) (Section E-1)									
					E-1-1	The system should have the ability to schedule training sessions for personnel.			
					E-1-2	Training materials should be made available to individuals who need to attend training.			
					E-1-3	Training attendance, course completion, and certification on successful completion of the course need to be recorded for each trainee.			
					E-1-4	Tests taken by individual trainees need to be uploaded and made available for reference.			
					E-1-5	A training matrix of individuals should be available.			
					E-1-6	The system should map tests/methods/techniques to available training paths, such that only trained and certified personnel are able to perform tests/methods/techniques.			
Compliance Management (Audit Trails, Electronic Records, Electronic Signatures) (Section E-2)									
					E-2-1	The system shall accurately and reliably be able to capture and stamp the date and time on all electronic records and any modifications to them.			
					E-2-2	The system should audit trail any event that creates, modifies, and deletes data.			
					E-2-3	The audit trail should record the user who did the action along with the date and time stamp.			
					E-2-4	The audit trail should record the previous value and current value of a field that was modified.			
					E-2-5	Electronic signature support shall be available within the application wherever a user needs to authorize actions such as review, approval, rejection, or disposition of a record.			
Instrument & Equipment Management (Section E-3)									
					E-3-1	The system should have the ability to track laboratory equipment/instrument usage.			
					E-3-2	The system shall have the ability to reserve or plan the usage of instruments for work planning purposes.			
					E-3-3	The system shall have the capability to put an instrument online and take an instrument offline both manually and via a configurable scheduler.			
					E-3-4	The system should be capable of recording instrument usage time for use in capacity planning/scheduling.			
					E-3-5	The system shall support user configuration and recording of multiple instrument events for the same instrument (different event types, different frequencies for same event type, and so forth).			
					E-3-6	The system will have the capability to group instruments together (that is, balance the group to include like-types, or to group by instrument type and by laboratory).			
					E-3-7	The system shall have the ability to set the frequency, with window of time, for calibration/maintenance activity. Frequency intervals can be in days, weeks, months, or years.			
					E-3-8	The system shall have the ability to provide an alert or email notification when an instrument is about to reach its calibration due date. Providing a window of time for such activities will help in planning and execution.			
					E-3-9	The system shall indicate out-of-calibration instruments, instruments beyond their preventative maintenance due date, and instruments under investigation and prevent their selection for use.			
					E-3-10	The system shall be able to record the multiple occurrences of next calibration date, next preventative maintenance date, and service date.			
					E-3-11	The system shall be able to calculate instrument event dates based upon a predefined interval.			
					E-3-12	The system should be able to identify uniquely each instrument or instrument part registered within the system.			

TABLE X1.1 *Continued*

SDMS	ELN	IDS	LIMS/LIS	OTHER	# (Fig. 3)	Laboratory Informatics Functions/Requirements	C	S	P
					E-3-13	The system shall have the capability to take a parent instrument offline automatically when a child instrument or part goes offline. (For example, a detector or pump represents the child instrument of a liquid chromatography system.)			
					E-3-14	The system should be capable of logging maintenance activities (scheduled or otherwise) for each instrument.			
Batch/Lot Management (E-4)									
					E-4-1	The system shall be able to group samples or processes into identifiable batches of predefined commonalities.			
					E-4-2	The system shall be capable of maintaining the linkage of the component parts and be able to track the batch throughout the system.			
					E-4-3	The system shall allow for batch review as an entity with all associated batch quality control parameters.			
					E-4-4	The system should have a process for determining the effectiveness of the process and be able to designate the disposition of the batch based on study/experiment results.			
Scheduled Event Management (E-5)									
					E-5-1	The system shall provide the ability to maintain, update, and revise schedules for various tasks and processes.			
					E-5-2	The system scheduler shall be able to handle calibrations, maintenance, stability studies, and so forth.			
					E-5-3	The system scheduler shall create an automated sample number for recording the results of calibration or validation.			
					E-5-4	The system shall be able to handle special testing characteristics such as sampling points, human body sampling location, and so forth, for specialized industries.			
					E-5-5	The system shall be able to set the retest date for a standard or reagent based upon a retest interval.			
Instrument Data Capture & Control (section E-6)									
					E-6-1	The system shall have the capability to trigger an instrument event based upon the number of uses of that instrument.			
					E-6-2	The system shall be able to take an instrument offline automatically when calibration or maintenance dates expire.			
					E-6-3	The system should provide auto sampler and robotic system control via application programming interface or web services.			
					E-6-4	The system should capture the personnel or instrument information relating to the results/ determinations entered into the system.			
					E-6-5	In cases in which instruments interface with the system, the system should accept the results uploaded from the instrument.			
					E-6-6	In cases in which a bidirectional interface is possible, the system should be capable of sending sample sequences like unknown samples, control samples, and standards to the instrument.			
					E-6-7	The system should support integration with simple laboratory instruments via RS-232, TCP/IP, and USB connection.			
					E-6-8	The system should support bidirectional interfacing with laboratory instrumentation software that are capable of communication using industry standards like Specification E1394, Health Level Seven, and other standard communication protocols.			
Standard & Reagent Management (E-7)									
					E-7-1	The system shall maintain an inventory of all standards and reagents used in the testing process and allow for linking of manufacturer documents, for example, certificates of analysis, safety data sheets, and so forth.			
					E-7-2	The inventory system shall allow for the marking of material as a controlled substance and flag material based on material stability.			

TABLE X1.1 *Continued*

SDMS	ELN	IDS	LIMS/LIS	OTHER	# (Fig. 3)	Laboratory Informatics Functions/Requirements	C	S	P
					E-7-3	The system shall allow tracking of standards and reagents that are nearing expiration date and produce a list of items that are nearing reorder levels based on amounts of material used/consumed.			
					E-7-4	System should maintain an inventory of working standards and volumetric solutions prepared in the lab. Work standard details such as concentration, purity, and so forth, and volumetric solution details such as molarity factor are to be maintained. Such details should be available during calculation of results.			
					E-7-5	The system shall have the ability to assign a review status to the standard/reagent properties and require additional review if properties are changed (expiry, vendor lot number, and so forth).			
Inventory Management (Section E-8)									
					E-8-1	The system should provide the ability to track the inventory amounts of samples received and consumed during testing.			
					E-8-2	Inventory items should have attributes like chemical name, internal name, catalog number, reorder level, consumable or reusable status, expiry date, open-expiry days, and so forth.			
					E-8-3	The system should be able to generate labels of different formats and sizes for different inventory item categories.			
					E-8-4	The system should allow for ordering from qualified vendors, with a vendor master list with approval status for order validation.			
					E-8-5	The system should allow recording of the receipt, issue, and disposal of materials.			
					E-8-6	The system should be able to quarantine certain types of materials until they are validated by the laboratory by a quality control test.			
					E-8-7	The system should allow the retiring of materials that are found to be faulty or not of good quality.			
					E-8-8	The system shall be able to designate a standard or reagent as being disposed or consumed.			
Investigation Management (E-9)									
					E-9-1	The system shall be able to record instances of identified nonconformance, planned deviations, and the actions required to restore the process to conformity.			
					E-9-2	The system should record the out-of-specification (OOS) and out-of-trend (OOT) samples as unique incidents and show the tests that are OOS or OOT.			
					E-9-3	Each OOS/OOT status should trigger an investigation to identify the root cause and document it.			
					E-9-4	The system shall provide the ability to take corrective and preventive actions, if needed, for all nonconformances.			
					E-9-5	The system should include a tracking mechanism for corrective actions to record the effectiveness of the actions and identify instances of insufficient resolution or repeat occurrences.			
					E-9-6	OOS/OOT workflows should be configurable based on each organization's standard operating procedures.			
Statistical Trending, Control Charts (E-10)									
					E-10-1	The system should provide a means to configure automatic generation of trending and control charts.			
					E-10-2	The system should allow choosing of sample types, tests, and parameters based on which statistical trending and control charts can be generated.			
Scientific Data Management (E-11)									
					E-11-1	The system shall be able to capture instrument-generated raw and metadata either as files or directly via RS-232 or TCP/IP communication ports.			
					E-11-2	The data captured by a scientific data management system (SDMS) should have checksum verification of source and destination data to verify originality and should store that information in a secure server with controlled access.			
					E-11-3	Metadata related to raw data need to be stored in a database such that it is easy to go back to the original data generated by instruments for specific samples/ tests.			

TABLE X1.1 *Continued*

SDMS	ELN	IDS	LIMS/LIS	OTHER	# (Fig. 3)	Laboratory Informatics Functions/Requirements	C	S	P
					E-11-4	The system should capture a true copy of the original data in order to maintain the integrity of data and should not obscure previous versions, if any.			
					E-11-5	The system should protect the raw data from being deleted and have version control when data is modified by any user or specific software.			
					E-11-6	The SDMS should help in achieving ALCOA principles defined by the FDA.			
					E-11-7	The system should have security and access controls for data stored.			
					E-11-8	The system should audit trail each and every record created and modified with version control.			
					E-11-9	Data within the SDMS will become the primary source of truth and hence needs to have proper systems for backup, restore, and archival for long term use.			
Instrument Data Systems Functions (E-12)									
					E-12-1	The system should be able to communicate with instrument systems via an application programming interface or web services.			
					E-12-2	The system should be able to send sample and test order information to instrument data systems.			
					E-12-3	The system should be able to receive results from instrument data systems.			
					E-12-4	The system should have generic parsing capability of instrument data to extract data of interest such as sample details and results.			
Systems Integration (E-13)									
					E-13-1	The system should provide a means to communicate status changes for dynamic entities (samples, lots, instruments, and so forth) to and from external systems.			
					E-13-2	The system should provide a means to communicate status changes to external systems.			
					E-13-3	The system should provide a mechanism to archive both data and metadata.			
					E-13-4	The system should provide a mechanism to restore archived data.			
					E-13-5	The system should be based on a reliable, effective, and supported data storage system.			
					E-13-6	The system should provide an interface to a third-party reporting tool.			
					E-13-7	The system should conform to the data storage platforms and standards currently in place in your enterprise.			
					E-13-8	The system's data storage mechanism should support the ability to modify the data structures as needed.			
					E-13-9	The system's data storage tools should allow for multiple environments (that is, development and production) and the ability to move records from one environment to another.			
					E-13-10	The system's data storage tool shall contain the ability to fine tune performance and security of data.			
					E-13-11	The system's data storage tools should support industry best practices for backup and recovery.			
					E-13-12	The system's architecture should be clearly separated into logical modules with standard interfaces between the modules.			
					E-13-13	The system should use the system and hardware platform as efficiently as possible to allow for concurrent usage and high peak usage.			
					E-13-14	The system should present a well-documented application programming interface for interfacing with the application's underlying functionality at a granular level.			
					E-13-15	The system should provide a means to integrate and exchange data based on common methods.			
					E-13-16	The system's data storage tools shall be replicable to ensure recoverability in the event of hardware failure.			
					E-13-17	The system should support interfacing with the company's enterprise resource planning systems.			
					E-13-18	The system should be able to interface with enterprise middleware.			
Capacity Planning & Laboratory Scheduling (E-14)									
					E-14-1	The system should be able to measure the test-based work capacity or throughput.			

TABLE X1.1 *Continued*

SDMS	ELN	IDS	LIMS/LIS	OTHER	# (Fig. 3)	Laboratory Informatics Functions/Requirements	C	S	P
					E-14-2	The system should be able to allocate work for personnel and instruments.			
					E-14-3	The system should allow laboratory tasks to be scheduled and allocated against available resources (people and equipment).			
Lean Laboratory/Continuous Improvement (Section E-15)									
					E-15-1	The system should allow for the automation of the controlled release of work into the laboratory based on the workload leveling strategy.			
					E-15-2	The system should allow for visual and quick assessment of the workflow processes at strategic points.			
					E-15-3	The system should provide tools that map actual workflow, identifying potential failure points.			
Artificial Intelligence (AI) – Smart Objects (Section E-16)									
					E-16-1	The system should allow for the integration of AI into the laboratory workflow, acting as a smart agent or assistant to the laboratory analyst, laboratory supervisor, or laboratory customer.			
					E-16-2	The system should have the ability to support machine learning based on unique ontologies to assist in the analysis of laboratory informatics datasets.			
					E-16-3	The system should have the ability to support smart laboratory assistance functions for voice data capture and retrieval.			
					E-16-4	The system should have the ability to respond to voice commands and take appropriate actions (for example, reschedule samples, change test assignments, and notify customers of laboratory results).			
					E-16-5	The system should have the ability to support predictive maintenance for laboratory instruments and equipment.			
					E-16-6	The system should allow AI to optimize various parameters and to set ideal models within the system.			
					E-16-7	The system should allow customization of which parameters, without explicit instruction, may be algorithmically inferred from past learning (for example, data mining, aggregation, transformation, and reporting of data).			
					E-16-8	The system's AI functions should monitor laboratory operation parameters such as capacity utilization, optimization of scheduling, QA/QC trending, and data integrity risks, and it should initiate notifications (using email, text, or social media) when necessary with recommendations for action.			
					E-16-9	The system's AI functions should allow for other types of facility monitoring (for example, alarms, lights' on/off status, temperature controls for central heat and air condition, open/closed/locked door status) and should initiate notifications (using email, text, or social media) when necessary with recommendations for action.			
					E-16-10	The system's AI functions should monitor instrument health (usage, critical parameters, maintenance) and provide the ability to send notifications with recommended actions.			
					E-16-11	The system should allow for environmental equipment control and monitoring of incubators, refrigerators, and so forth, and provide the ability to send notifications with recommended actions.			
Data Integrity (Section E-17)									
					E-17-1	The system shall have functionality to support ALCOA principles.			
					E-17-2	The system shall protect data entered by not allowing it to be obscured by a new set of data. Both the old and current data should be available for review.			
					E-17-3	The system shall maintain a true copy of the original instrument data for on-demand review.			
					E-17-4	The system shall retain data in the system for a set period and protect that data to enable their accurate and ready retrieval throughout the records retention period.			
					E-17-5	The system shall require and authenticate a user with either (1) two identifiers, a user ID and password or (2) unique biometric data.			

TABLE X1.1 *Continued*

SDMS	ELN	IDS	LIMS/LIS	OTHER	# (Fig. 3)	Laboratory Informatics Functions/Requirements	C	S	P
					E-17-6	The system's use of electronic record time stamps shall accurately reflect the system date and time.			
					E-17-7	The system shall require a unique user ID to be assigned to each user.			
					E-17-8	The audit trail shall not be able to be modified, deleted, or disabled by users with general access permissions.			
Configuration Management (Section S-1)									
					S-1-1	The system will allow for the entry and management of user configurable lookup/master data.			
					S-1-2	The system shall be able to assign specification limits for sample and instrument tests.			
					S-1-3	The system should provide for configurable review and approval of multiple tests at higher levels such as the sample, batch, project, and experiment levels.			
					S-1-4	The system should allow for the entry and configuration of warning/action and material specification limits that can be compared against entered results/determinations to determine whether the results meet specifications or other limits.			
					S-1-5	The system should allow for the entry of electronic signatures for both entered results and approved reports. This should be configurable based on sample or test, or both.			
					S-1-6	The system should provide a listing of all tests that shall be performed, the amount of material required, and to which location the samples are to be sent.			
					S-1-7	The system will allow for the configuration of the appropriate laboratory workflows.			
					S-1-8	The system will allow for the assignment of status values to track progress of samples or other entities in the laboratory workflow.			
					S-1-9	The system should allow for revision control of lookup/master data on a user-defined basis.			
					S-1-10	The system should allow for import of lookup/master data.			
					S-1-11	The system shall include the ability to define rounding rules for numeric data.			
					S-1-12	The system should allow for the creation of calculated limits based on test results and relevant metadata.			
					S-1-13	The system should provide a warning to users for entry of out-of-specification results (audible, color code, icon, screen message, email notification).			
					S-1-14	The system shall provide a means to update static and dynamic data (for example, sample information, test results, specifications, and so forth).			
					S-1-15	The system should provide a means for user-defined actions to trigger based on workflow events or status changes.			
					S-1-16	The system should secure data and operations from unintended exposure/use by unapproved personnel.			
					S-1-17	The system should provide administrative interfaces that allow approved users to configure the system without programming or direct manipulation of the data storage system.			
					S-1-18	The system should provide a programming tool to customize the applications or build calculations within the application.			
					S-1-19	The system should present multiple user interfaces that reflect the specific geographic needs of local users, including elements of language, character sets, and time zones.			
					S-1-20	In regulated environments, the system should support rules governing electronic records and electronic signatures, for example, 21 CFR Part 11 and Annex 11.			
					S-1-21	The system should provide a security interface that prevents unauthorized access to data and functions that can be conveniently administered for all modules of the system.			
					S-1-22	The system should provide for a unified authentication scheme whereby a user can sign on once and access all permitted functions and data.			

TABLE X1.1 *Continued*

SDMS	ELN	IDS	LIMS/LIS	OTHER	# (Fig. 3)	Laboratory Informatics Functions/Requirements	C	S	P
					S-1-23	The system shall meet enterprise password policy guidelines and applicable regulatory guidelines, for example, 21 CFR Part 11 and Annex 11.			
					S-1-24	The system should comply with applicable regulatory standards and company standards for electronic signatures.			
					S-1-25	The system should support the ability to assign individual users to system groups or roles, or both.			
					S-1-26	The system should provide tools for static data migration into the system.			
Validate/Commission Systems (Section S-2)									
					S-2-1	The vendor should have established software development standards, formal change control, and software revision control. Vendor staff quality and skills should be documented.			
					S-2-2	The vendor should provide access to source code through suitable escrow organization.			
					S-2-3	The system should provide functionality to summarize and evaluate performance markers and processes of the enterprise.			
					S-2-4	The system should be well documented with comprehensive manuals for all phases of operation, including administration, operation, and troubleshooting.			
System Administration (Section S-3)									
					S-3-1	The system should allow the user to interact with the system for a configurable period of time before being prompted again for authentication credentials.			
					S-3-2	The system should allow for batch modification of personnel data including groups and role assignments.			
					S-3-3	The system should support storage of a wide variety of data formats used by the organization.			
					S-3-7	The system should support the ability to assign individual users to system groups or roles, or both.			
					S-3-8	The vendor should provide maintenance agreements and support services.			
					S-3-9	The vendor shall provide help desk support, training support, installation support, and high quality documentation.			
					S-3-10	The system should integrate with an organization's enterprise personnel security directory or physical security systems, or both.			
					S-3-11	The vendor should provide upgrades and documentation for the upgrade.			
					S-3-12	The upgrade shall provide a means to migrate data to a new release.			
					S-3-13	The system should allow for fast retrieval of stored items.			
					S-3-14	The system should support interaction with mobile technologies like smartphones and tablet PCs for timely notification of imminent or actual issues.			
					S-3-15	The upgrade shall include the capability to install in parallel for testing.			
Cyber Security (Section S-4)									
					S-4-1	The system should be able to work in secure communication protocols like SSL/TLS over Secure Hypertext Transfer Protocol with 256-bit encryption.			
					S-4-2	The system should be able to support an encrypted database for cyber security.			
Privacy Information (Section S-5)									
					S-5-1	The system should be able to comply with privacy protection compliance like that found in the HIPAA provisions.			
					S-5-2	The system should have appropriate security to prevent data from being compromised for personally identifiable information.			

BIBLIOGRAPHY

The references shown here provide additional sources of information related to laboratory informatics. The landscape of laboratory informatics is changing rapidly. This list of resources is intended to inform the reader of the many organizations involved in this area.

Artificial Intelligence and Internet of Things:

- (1) Finlay, Steven, *Artificial Intelligence and Machine Learning for Business*, Relativistic, Great Britain, 2017.
- (2) Information Resources Management Association, *The Internet of Things: Breakthroughs in Research and Practice*, IGI Global, Hershey, PA, 2017.

Clinical and Health Data Standards and Networks:

- (3) The National Centers for Disease Control and Prevention (CDC) manages the Laboratory Response Network (LRN). This includes the CDC LRN-Biological (LRN-B) and CDC LRN-Chemical (LRN-C).³²
- (4) The Food Emergency Response Network (FERN) is managed by the U.S. Department of Agriculture's Food Safety and Inspection Service and the Food and Drug Administration. FERN uses the Electronic Laboratory Exchange Network (eLEXNET), which allows multiple government agencies engaged in food safety activities to compare, communicate, and coordinate findings of laboratory analyses.³³
- (5) The National Animal Health Laboratory Network's (NAHLN) purpose is to enhance the nation's early detection of, response to, and recovery from animal health emergencies, including bioterrorist incidents, newly emerging diseases, and foreign animal disease agents that threaten the nation's food supply and public health.³⁴
- (6) The National Plant Diagnostic Network (NPDN) is managed by the U.S. Department of Agriculture's National Institute of Food and Agriculture and represents an international consortium of plant diagnostic laboratories.³⁵
- (7) The Integrated Consortium of Laboratory Networks (ICLN) coordinates detection efforts and data sharing across nine U.S. federal agencies during chemical, biological, radiological, and nuclear (CBRN) incidents.³⁶

Data Exchange Standards:

- (8) NetCDF (Network Common Data Form) is an interface for array-oriented data access and a library that provides an implementation of the interface. The NetCDF library also defines a machine-independent format for representing scientific data. Together, the interface, library, and format support the creation, access, and sharing of scientific data.³⁷
- (9) The EPA's Environmental Response Laboratory Network (ERLN) "Requirements for Environmental Response Laboratory Network (ERLN) Data Submissions" is a granular environmental electronic data deliverable (EDD) with a standardized, defined list of data elements and their associated data structures and components. This EDD is agnostic with regard to method, matrix, or governmental program and is used by public and private laboratories to provide

emergency response laboratory data. The companion "WebEDR" is a web service that provides data exchange and automated data review.³⁸

- (10) The EPA's Scribe software tool is used to capture a variety of field data for their Environmental Response Team. Scribe projects are stored and shared via Scribe.NET.³⁹
- (11) The Environmental Information Exchange Network provides information on several data exchange formats including eDWR, SDWIS, AQDE, AQS-Air, ICIS-NPDES-Water, NetDMR, OWIR, ODPX, WQDE, WQX, HERE, and SRS.⁴⁰
- (12) The EPA's Staged Electronic Data Deliverable (SEDD) is a hierarchical, uniform file format used as part of its Superfund program, as well as outside the agency.³⁸
- (13) The Standardization in Lab Automation (SiLA) consortium develops standards for the integration of laboratory instrumentation and the exchange of data between laboratory systems.⁴¹

Environmental/Chemical Data Exchanges:

- (14) The Association of Public Health Laboratories' system for environmental data delivery is based on the EPA's ERLN. This EDD is agnostic to methods, matrix, and program.⁴²
- (15) The CDC's Healthy Homes and Lead Poisoning Surveillance System (HHLPS) is a web-based replacement for its former data reporting system STELLAR. It combines and tracks data on lead- and nonlead-based environmental activities.³¹
- (16) The North American Association for Central Cancer Registries (NAACCR) provides central registry structural requirements, process standards, and outcome measures for access to source data and completeness of reporting, data quality, data analysis and reporting, and data management.⁴³
- (17) The Public Health Data Standards Consortium (PHDSC) is an organization of federal, state, and local health agencies; professional associations, academia; public and private sector organizations; international members; and individuals. Its goal is to empower the healthcare and public health communities with health information technology standards to improve individual and community health.⁴⁴

³⁸ For more information, visit <https://www.epa.gov/>.

³⁹ For more information, see the EPA's Response Team website at <https://response.epa.gov>.

⁴⁰ For more information, visit <http://www.exchangenetwork.net>.

⁴¹ Additional information available from Association Consortium Standardization in Lab Automation (SiLA), Spinnerstrasse 38, CH-8645 Rapperswil-Jona, Switzerland, <http://www.sila-standard.org>.

⁴² Additional information available from Association of Public Health Laboratories, 8515 Georgia Ave., Suite 700, Silver Spring, MD 20910, <https://www.aphl.org>.

⁴³ Additional information available from North American Association of Central Cancer Registries, Inc. (NAACCR), 2050 W. Iles, Suite A, Springfield, IL 62704-4194, <http://www.naacr.org>.

⁴⁴ Additional information available from Public Health Data Standards Consortium (PHDSC), 111 South Calvert St., Suite 2700, Baltimore MD 21202, <http://www.phdsc.org/>.

³² For additional information, visit <https://emergency.cdc.gov>.

³³ For more information, visit <https://www.fernlab.org/>.

³⁴ For more information, visit <https://www.nahln.org/>.

³⁵ For more information, visit <https://www.npdn.org/>.

³⁶ For more information, visit <https://www.icln.org/>.

³⁷ Available from Unidata Program Center, UCAR Office of Programs, P.O. Box 3000, Boulder, CO 80307-3000, <https://www.unidata.ucar.edu>.

*EPA Data Standards:*⁶

- (18) 40 Code of Federal Regulations (CFR) Part 160, Good Laboratory Practice Standards, 54 FR 34067, Aug. 17, 1989.
- (19) 40 Code of Federal Regulations (CFR) Part 792, Good Laboratory Practice Standards, 54 FR 34043, Aug. 17, 1989.

*FDA Regulations:*⁷

- (20) 21 Code of Federal Regulations (CFR) Part 58, Good Laboratory Practice for Nonclinical Laboratory Studies, 43 FR 60013, Dec. 22, 1978.
- (21) 21 Code of Federal Regulations (CFR) Part 211, Current Good Manufacturing Practice for Finished Pharmaceuticals, 43 FR 45077, Sept. 29, 1978.
- (22) 21 Code of Federal Regulations (CFR) Part 820, Quality System Regulation, FR 52654, Oct. 7, 1996.
- (23) FDA Compliance Program 7346.832, Pre-approval Inspections/Investigations, published May 2010, effective May 2012.

*ICH Standards:*⁹

- (24) ICH Quality Guideline Q8, Pharmaceutical Development, 2009.
- (25) ICH Quality Guideline Q10, Pharmaceutical Quality System, 2008.

*ISO Standard:*¹²

- (26) ISO/HL7 27931 Data Exchange Standards -- Health Level Seven Version 2.5 -- An application protocol for electronic data exchange in healthcare environments.

*ISPE Guides:*¹³

- (27) *GAMP Good Practice Guide: Electronic Data Archiving*, ISPE, 2007.
- (28) *GAMP Good Practice Guide: GxP Compliant Laboratory Computerized Systems*, ISPE, 2012.
- (29) *GAMP Good Practice Guide: IT Infrastructure Control and Compliance*, ISPE, 2017.

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- (31) Ohno, Taiichi, *Toyota Production System*, Productivity Press, New York, NY, 1988.
- (32) Womack, James P. and Jones, Daniel T., *Lean Thinking*, Free Press, New York, NY, 2003.

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